# Centrifugal Compressors for Petroleum, Chemical, and Gas Service Industries

API STANDARD 617
SIXTH EDITION, FEBRUARY 1995

**American Petroleum Institute**
1220 L Street, Northwest
Washington, D.C. 20005



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**Manufacturing, Distribution and Marketing Department**

API STANDARD 617
SIXTH EDITION, FEBRUARY 1995

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# FOREWORD

This standard is based on the accumulated knowledge and experience of manufacturers and users of centrifugal compressors. The objective of this standard is to provide a purchase specification to facilitate the manufacture and procurement of centrifugal compressors for use in petroleum, chemical, and gas industry services.

The primary purpose of this standard is to establish minimum mechanical requirements. This limitation in scope is one of charter as opposed to interest and concern. Energy conservation is of concern and has become increasingly important in all aspects of equipment design, application, and operation. Thus, innovative energy-conserving approaches should be aggressively pursued by the manufacturer and the user during these steps. Alternative approaches that may result in improved energy utilization should be thoroughly investigated and brought forth. This is especially true of new equipment proposals, since the evaluation of purchase options will be based increasingly on total life costs as opposed to acquisition cost alone. Equipment manufacturers, in particular, are encouraged to suggest alternatives to those specified when such approaches achieve improved energy effectiveness and reduced total life costs without sacrifice of safety or reliability.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of this standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard, rather than by rewriting or incorporating sections thereof into another complete standard.

API standards are published as an aid to procurement of standardized equipment and materials. These standards are not intended to inhibit purchasers or producers from purchasing or producing products made to other standards.

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## IMPORTANT INFORMATION CONCERNING USE OF ASBESTOS OR ALTERNATIVE MATERIALS

Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

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# Centrifugal Compressors for Petroleum, Chemical, and Gas Service Industries

## SECTION 1—GENERAL

### 1.1 Scope

This standard covers the minimum requirements for centrifugal compressors used in petroleum, chemical, and gas industries services that handle air or gas. This standard does not apply to fans or blowers that develop less than 34 kPa (5 pounds per square inch) pressure rise above atmospheric pressure; these are covered by API Standard 673. This standard also does not apply to packaged, integrally-geared centrifugal air compressors, which are covered by API Standard 672.

Note: A bullet (●) at the beginning of a paragraph indicates that a decision is required or further information is to be provided by the purchaser. This information should be indicated on the data sheets (see Appendix A); otherwise it should be stated in the quotation request or in the order.

### 1.2 Alternative Designs

The vendor may offer alternative designs (see 5.2 for proposal requirements). Equivalent metric dimensions, fasteners, and flanges may be substituted as mutually agreed upon by the purchaser and the vendor.

### 1.3 Conflicting Requirements

In case of conflict between this standard and the inquiry or order, the information included in the order shall govern.

### 1.4 Definition of Terms

Terms used in this standard are defined in 1.4.1 through 1.4.23 (refer to Figure 1).

**1.4.1** *Actual cubic feet per minute (ACFM)* refers to the flow rate at flowing conditions of temperature and pressure at any given location. Because this term describes flow at a number of locations, it should not be used interchangeably with ICFM (see 1.4.6).

**1.4.2** The *alarm point* is a preset value of a parameter at which an alarm is actuated to warn of a condition that requires corrective action.

**1.4.3** *Axially split* refers to casing joints that are parallel to the shaft centerline.

**1.4.4** The *compressor rated point* is the intersection on the 100 percent speed curve corresponding to the highest capacity of any specified operating point.

Note: This is generally a fictitious point rather than an actual operating point (see Figure 1).

**1.4.5** The use of the word *design* in any term (such as design power, design pressure, design temperature, or design speed) should be avoided in the purchaser's specification. This terminology should be used only by the equipment designer and manufacturer.

**1.4.6** *Inlet cubic feet per minute (ICFM)* refers to the flow rate determined at the conditions of pressure, temperature, compressibility, and gas composition, including moisture, at the compressor inlet flange.

**1.4.7** *Maximum allowable temperature* is the maximum continuous temperature for which the manufacturer has designed the equipment (or any part to which the term is referred) when handling the specified fluid at the specified pressure.

**1.4.8** *Maximum allowable working pressure* is the maximum continuous pressure for which the manufacturer has designed the equipment (or any part to which the term is referred when it is operating at the maximum allowable temperature).

**1.4.9** *Maximum sealing pressure* is the highest pressure expected at the seals during any specified static or operating conditions and during startup or shutdown.

**1.4.10** *Maximum continuous speed* (in revolutions per minute) for compressors driven by variable-speed prime movers is the speed at least equal to 105 percent of the highest speed required by any of the specified operating conditions. Maximum continuous speed for constant-speed drivers shall be equal to the 100 percent speed.

**1.4.11** *Minimum allowable temperature* is the minimum continuous temperature for which the manufacturer has designed the equipment (or any part to which the term is applied).

**1.4.12** *Minimum allowable speed* (in revolutions per minute) is the lowest speed at which the manufacturer's design will permit continuous operation.

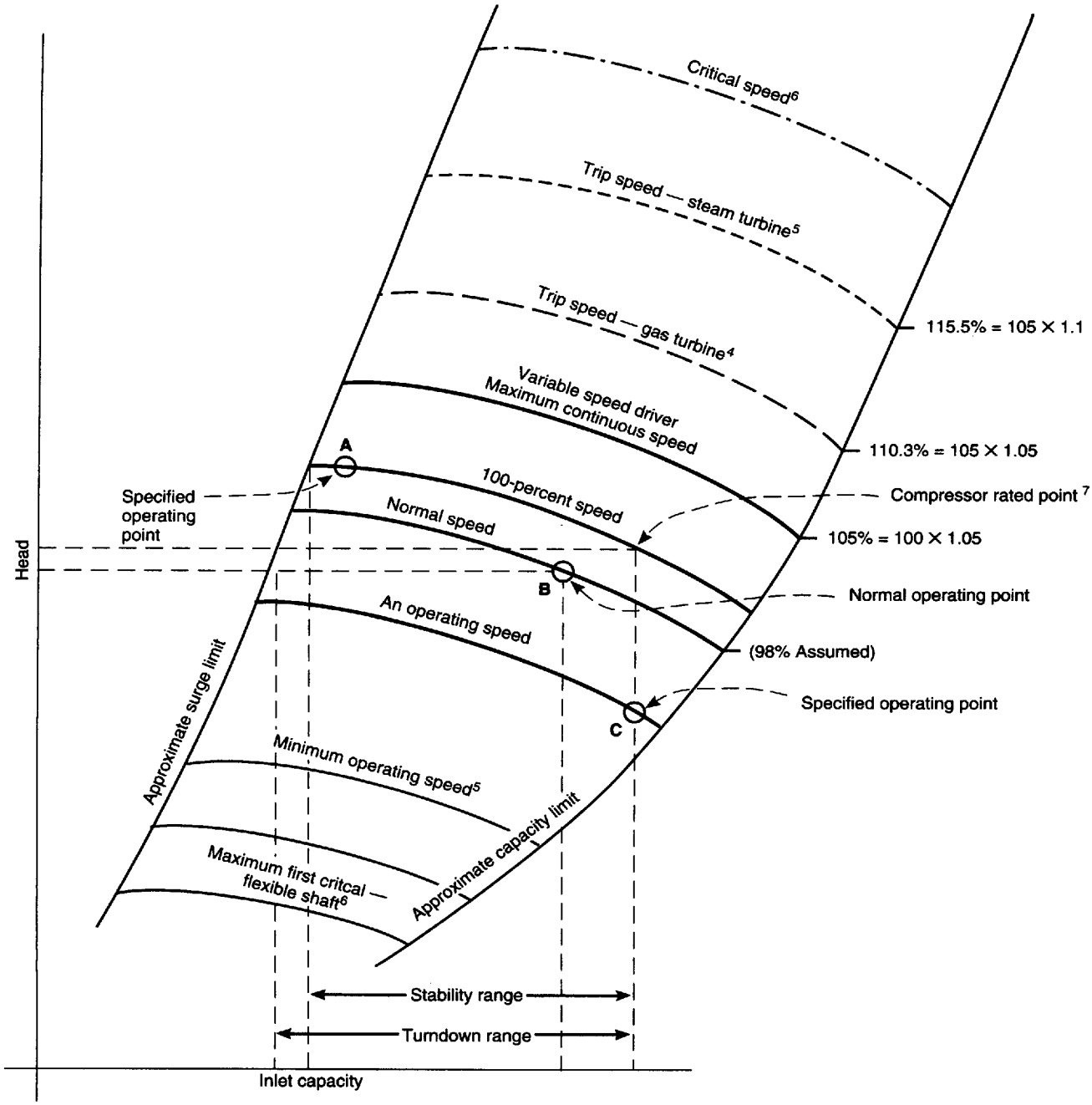
**1.4.13** The *normal operating point* is the point at which usual operation is expected and optimum efficiency is desired. This point is usually the point at which the vendor certifies that performance is within the tolerances stated in this standard.

**1.4.14** The *normal speed* is the speed corresponding to the requirements of the normal operating point.

**1.4.15** *One hundred percent speed* is the highest speed required for any specified operating point above the normal speed curve. If there are no specified operating points that require greater than normal speed, the 100 percent speed shall



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Notes:
1. Except where specific numerical relationships are stated, the relative values implied in this figure are assumed values for illustration only.
2. The 100 percent speed curve is determined from the operating point requiring the highest head; point A in the illustration.
3. The compressor rated point (CRP) is the intersection on the 100-percent speed line corresponding to the highest flow of any operating point; point C in the illustration.
4. The head-capacity curve at 100-percent speed shall extend to at least 115-percent of the capacity of the CRP. Head-capacity curves at other speeds shall be extended to equivalent capacity at each speed. For example, the head-capacity curve at 105-percent speed shall be extended to at least 1.05 times 1.15 times the capacity of the CRP; the head-capacity curve at 90-percent speed shall be extended to at least 0.9 times 1.15 times capacity of the CRP; and so on. These points define the "approximate capacity limit" curve.
5. Refer to the applicable standard for the compressor driver such as API Standard 612 (Steam Turbine) or API Standard 616 (Gas Turbine) for trip speed and minimum operating speed limits.
6. Refer to 2.9 for allowable margins of critical speeds to operating speeds.
7. See 1.1.4.

Figure 1—Illustration of Terms

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be the normal speed. For motor-driven compressors, the 100 percent speed shall be equal to the gear ratio (if any) times the full load speed of the motor being furnished.

**1.4.16** *Radially split* refers to casing joints that are transverse to the shaft centerline.

**1.4.17** *Settling out pressure* is the pressure of the compressor system when the compressor is shut down.

**1.4.18** *Shutdown point* is a preset operational value of a parameter at which automatic or manual shutdown of the system is required.

**1.4.19** *Stability* is the difference in capacity (in percent of rated capacity) between the rated capacity and the surge point at rated speed (and rated gas properties).

**1.4.20** *Standard cubic feet per minute (SCFM)* refers to the flow rate at any location corrected to a pressure at 14.7 pounds per square inch absolute (1.01 bar) and a temperature of 60°F (15.56°C) with a compressibility factor of 1.0 and in a dry condition.

**1.4.21** *Trip speed* (in revolutions per minute) is the speed at which the independent emergency overspeed device operates to shut down a prime mover.

**1.4.22** *Turndown* is the percentage of change in capacity (referred to rated capacity) between the rated capacity and the surge point capacity at the rated head when the unit is operating at rated suction temperature and gas composition.

**1.4.23** *Unit responsibility* refers to the responsibility for coordinating the technical aspects of the equipment and all auxiliary systems included in the scope of the order. Responsibility for such factors as the power requirements, speed, rotation, piping, and testing of components shall be reviewed.

## 1.5 Referenced Publications

**1.5.1** The editions of the following standards, codes, and specifications that are in effect at the time of publication of this standard shall, to the extent specified herein, form a part of this standard. The applicability of changes in standards, codes, and specifications that occur after the inquiry shall be mutually agreed upon by the purchaser and the vendor.

ANSI[1]

B1.20.3 *Dryseal Pipe Threads (Inch)*
B16.1 *Cast Iron Pipe Flanges and Flanged Fittings, Class 25, 125, 250, and 800*
B16.5 *Pipe Flanges and Flanged Fittings, Steel Nickel Alloy and Other Special Alloys*
B16.11 *Forged Steel Fittings, Socket-Welding and Threaded*
B16.42 *Ductile Iron Pipe Flanges and Flanged Fittings, Class 150 and 300*
B19.3 *Safety Standards for Compressors for General Refinery Services*
S18.1 *Annunciator Sequences and Specifications*

API

Std 541 *Form-Wound Squirrel-Cage Induction Motors—250 Horsepower and Larger*
Std 546 *Form-Wound Brushless Synchronous Motors—500 Horsepower and Larger*
Std 612 *Special-Purpose Steam Turbines for Refinery Services*
Std 613 *Special-Purpose Gear Units for Refinery Service*
Std 614 *Lubrication, Shaft-Sealing, and Control-Oil Systems for Special-Purpose Applications*
Std 616 *Gas Turbines for Refinery Services*
Std 670 *Vibration, Axial-Position, and Bearing-Temperature Monitoring Systems*
Std 671 *Special-Purpose Couplings for Refinery Service*
Std 672 *Packaged, Integrally Geared, Centrifugal Air Compressors for General Refinery Service*
Std 673 *Special-Purpose Centrifugal Fans for General Refinery Service* (under development)
RP 683 *Quality Improvement Manual for Mechanical Equipment in Petroleum, Chemical, and Gas Industries*

ASME[2]

B1.1 *Unified Inch Screw Threads (UN and UNR Thread Form)*
B1.20.1 *General Purpose (Inch) Pipe Threads*
B16.47 *Large-Diameter Carbon Steel Flanges (Nominal Pipe Sizes 26 Through 60; Classes 75, 150, 300, 400, 600, and 900)*
B31.3 *Chemical Plant and Petroleum Refinery Piping*
PTC 10 *Compressors and Exhausters*
PTC 1 *General Instructions*
*Boiler and Pressure Vessel Code*, Section VIII, "Rules for Construction of Pressure Vessels," and Section IX, "Welding and Brazing Qualifications"

ASTM[3]

A 48 *Gray Iron Castings*
A 148 *High-Strength Steel Castings for Structural Purposes*

[1]American National Standards Institute, 11 West 42nd Street, New York, New York 10036.

[2]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

[3]American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103-1187.

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A 193 *Alloy Steel and Stainless Steel Bolting Materials for High-Temperature Service*
A 194 *Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service*
A 203 *Pressure Vessel Plates, Alloy Steel, Nickel*
A 216 *Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service*
A 217 *Martensitic Stainless Steel and Alloy Steel Castings for Pressure-Containing Parts Suitable for High-Temperature Service*
A 240 *Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels*
A 266 *Carbon Steel Forgings for Pressure Vessels Components*
A 278 *Gray Iron Casting for Pressure-Containing Parts for Temperatures Up to 650°F (345°C)*
A 283 *Low and Intermediate Tensile Strength Carbon Steel Plates, Shapes, and Bars*
A 284 *Low and Intermediate Tensile Strength Carbon Silicon Steel Plates for Machine Parts and General Construction*
A 285 *Low-and Intermediate-Tensile Strength Carbon Steel Pressure Vessel Plates*
A 307 *Carbon Steel Externally Threaded Standard Fasteners*
A 320 *Alloy-Steel Bolting Materials for Low-Temperature Service*
A 336 *Alloy Steel Forgings for Pressure and High Temperature Parts*
A 351 *Austenitic Steel Castings for High Temperature Service*
A 352 *Ferritic and Martensitic Steel Castings for Pressure-Containing Parts Suitable for Low Temperature Service*
A 353 *Double-Normalized and Tempered 9 Percent Nickel Alloy Steel Pressure Vessel Plates*
A 367 *Chill Testing of Cast Iron*
A 370 *Methods and Definitions for Mechanical Testing of Steel Products*
A 388 *Practice for Ultrasonic Examination of Heavy Steel Forgings*
A 395 *Ferritic Ductile Iron Pressure Retaining Castings for Use at Elevated Temperatures*
A 414 *Carbon Steel Sheet for Pressure Vessels*
A 436 *Austenitic Gray Iron Castings*
A 470 *Vacuum-Treated Carbon and Alloy Steel Forgings for Turbine Rotors and Shafts*
A 473 *Stainless and Heat-Resisting Steel Forgings*
A 494 *Nickel and Nickel Alloy Castings*
A 515 *Carbon Steel Pressure Vessel Plates for Moderate and Lower Temperature Service*
A 522 *Forged or Rolled 8 and 9 Percent Nickel Alloy Steel Flanges, Fittings, Valves, and Parts for Low-Temperature Service*
A 536 *Ductile Iron Castings*
A 537 *Carbon Manganese-Silicon Heat Treated Pressure Vessel Plates*
A 543 *Quenched and Tempered Nickel-Chromium-Molybdenum Alloy Steel Pressure Vessel Plates*
A 553 *Quenched and Tempered 8 and 9 Percent Nickel Alloy Steel Pressure Vessel Plates*
A 571 *Austenitic Ductile Iron Casting for Pressure-Containing Parts Suitable for Low Temperature Service*
A 578 *Specification for Ultrasonic Angle-beam Examination of Steel Plates*
A 609 *Practice for Castings, Carbon, Low-Alloy, and Martensitic Stainless Steel, Ultrasonic Examination Thereof*
A 693 *Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip*
A 705 *Age-Hardening Stainless and Heat-Resisting Steel Forgings*
A 709 *Structural Steel for Bridges*
A 743 *Iron-Chromium, Iron-Chromium-Nickel, and Nickel-Base Corrosion-Resistant Castings for General Application*
A 744 *Iron-Chromium-Nickel and Nickel-Base Corrosion-Resistant Castings for Severe Service*
A 747 *Precipitation Hardening Stainless Steel Castings*
B 23 *White Metal Bearing Alloys*
B 26 *Aluminum-Alloy Sand Castings*
B 29 *Pig Lead*
B 127 *Nickel-Copper Alloy Plate, Sheet, and Strip*
B 164 *Nickel-Copper Alloy Rod, Bar, and Wire*
B 367 *Titanium and Titanium Alloy Castings*
B 462 *Forged or Rolled Chromium-Nickel-Iron Molybdenum-Copper-Columbian Stabilized Alloy Pipe Flanges, Forged Fittings, and Valves and Parts for Corrosive High Temperature Service*
B 574 *Low-Carbon Nickel-Molybdenum-Chromium Alloy Rod*
DS56C *Metals and Alloys in the Unified Numbering System*

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E 94 *Guide for Radiographic Testing*
E 125 *Reference Photographs for Magnetic Particle Indications on Ferrous Castings*
E 165 *Practice for Liquid Penetrant Inspection Method*
E 709 *Practice for Magnetic Particle Examination*

AWS[4]
*Structural Welding Code - Steel*

ISA[5]
S18.1 *Annunciator Sequences and Specifications*

MSS[6]
SP-44 *Steel Pipe Line Flanges*

NACE[7]
MRO175-93 *Sulfide Stress Cracking Resistant Metallic Material for Oil Field Equipment*

NFPA[8]
70 *National Electrical Code, Articles 500, 501, and 502*
496 *Purged and Pressurized Enclosures for Electrical Equipment*

SAE[9]
AMS 4676 *Bars and Forgings*

SSPC[10]
SP6 *Commercial Blast Cleaning*

U.S. Government Specifications
QQ-N-286 *Wrought Nickel-Copper-Aluminum Alloy* (UNS N0550)

**1.5.2** The purchaser and the vendor shall mutually determine the measures that must be taken to comply with any governmental regulations, ordinances, or rules that are applicable to the equipment.

[4]American Welding Society, 550 N.W. LeJeune Road, Miami, Florida 33135.
[5]Instrument Society of America, P.O. Box 12277, Research Triangle Park, North Carolina, 27709.
[6]Manufacturers Standardization Society of the Valve and Fittings Industry, 127 Park Street, N.E., Vienna, Virginia 22180.
[7]National Association of Corrosion Engineers, P.O. Box 218340, Houston, Texas 77218.
[8]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269-9101.
[9]Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096-0001.
[10]Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, Pennsylvania 15213-2683.

# SECTION 2—BASIC DESIGN

## 2.1 General

**2.1.1** The equipment (including auxiliaries) covered by this standard shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation.

Note: It is recognized that this is a design criterion.

**2.1.2** Unless otherwise specified, the compressor vendor shall assume unit responsibility for all equipment and all auxiliary systems included in the scope of the order.

**2.1.3** Unless otherwise specified, the compressor shall be designed to deliver normal head and normal capacity without negative tolerance. The horsepower at the above condition shall not exceed 104 percent of the normal value.

Note: See the optional performance test criteria in 4.3.6.1, and handling of excess head for constant speed drivers.

**2.1.4** The head-capacity characteristic curve shall rise continuously from the rated point to predicted surge. The compressor, without the use of a bypass, shall be suitable for continuous operation at any capacity at least 10 percent greater than the predicted approximate surge capacity shown in the proposal.

**2.1.5** Unless otherwise specified, cooling water systems shall be designed for the conditions specified in Table 1. Provision shall be made for complete venting and draining of the system.

Note: The vendor shall notify the purchaser if the criteria for minimum temperature rise and velocity over heat exchange surfaces result in a conflict. The criterion for velocity over heat exchanger surfaces is intended to minimize water-side fouling; the criterion for minimum temperature rise is intended to minimize the use of cooling water. The purchaser will approve the final selection.

**2.1.6** The arrangement of the equipment, including piping and auxiliaries, shall be developed jointly by the purchaser and the vendor. The arrangement shall provide adequate clearance areas and safe access for operation and maintenance.

Table 1—Design Criteria and Specifications for Cooling Water Systems

| Criteria | Specifications | |
|---|---|---|
| | Metric | U.S. Customary |
| Velocity over heat exchanger surfaces | 1.5–2.5 m/s | 5–8 fps |
| Maximum allowable working pressure | ≥520 kPa(ga) | ≥75 psig |
| Test pressure | ≥790 kPa(ga) | ≥115 psig |
| Maximum pressure drop | 100 kPa | 15 psi |
| Maximum inlet temperature | 32°C | 90°F |
| Maximum outlet temperature | 49°C | 120°F |
| Maximum temperature rise | 17°C | 30°F |
| Minimum temperature rise | 11°C | 20°F |
| Fouling factor on water side | 0.35$m^2$×K/kW | 0.002 hr-$ft^2$-°F/Btu |
| Shell Corrosion Allowance | 3.2 mm | 0.125 in. |

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**2.1.7** All equipment shall be designed to permit rapid and economical maintenance. Major parts such as casing components and bearing housings shall be designed (shouldered or cylindrically doweled) and manufactured to ensure accurate alignment on reassembly.

**2.1.8** The inner casing of radially split barrel compressors shall be designed for easy withdrawal from the outer shell and easy disassembly for inspection or replacement of parts.

● **2.1.9** The purchaser will specify whether the installation is indoors (heated or unheated) or outdoors (with or without a roof), as well as the weather and environmental conditions in which the equipment must operate (including maximum and minimum temperatures, unusual humidity, and dusty or corrosive conditions).

● **2.1.10** Control of the sound pressure level (SPL) of all equipment furnished shall be a joint effort of the purchaser and the vendor. The equipment furnished by the vendor shall conform to the maximum allowable sound pressure level specified.

● **2.1.11** The purchaser will advise the vendor in the inquiry of any specific requirements for liquid injection.

**2.1.12** Equipment shall be designed to run without damage to the trip speed and the maximum allowable working pressure.

**2.1.13** The machine and its driver shall perform on the test stand and on their permanent foundation within the specified acceptance criteria. After installation, the performance of the combined units shall be the joint responsibility of the purchaser and the vendor who has unit responsibility.

● **2.1.14** Many factors (such as piping loads, alignment at operating conditions, supporting structure, handling during shipment, and handling and assembly at the site) may adversely affect site performance. To minimize the influence of these factors, the vendor shall review and comment on the purchaser's piping and foundation drawings. When specified, the vendor's representative shall:

a. Observe a check of the piping performed by parting the flanges.
b. Check alignment at the operating temperature.
c. Be present during the initial alignment check.

● **2.1.15** Motors, electrical components, and electrical installations shall be suitable for the area classification (class, group, and division) specified by the purchaser on the data sheets, and shall meet the requirements of NFPA 70, Articles 500, 501, and 502, as well as local codes specified and furnished by the purchaser.

**2.1.16** Spare parts for the compressor and all furnished auxiliaries shall meet all the criteria of this standard.

● **2.1.17** When specified, the compressor or compressors shall be suitable for field run on air. Performance parameters, including any required precautions, shall be mutually agreed upon by the purchaser and the vendor.

**2.1.18** A guide to centrifugal compressor nomenclature will be found in Appendix F.

## 2.2 Casings

**2.2.1** The thickness of the casing shall be suitable for the maximum allowable working and test pressures and shall include at least a 3.175 millimeters ($^1/_8$-inch) corrosion allowance. The hoop-stress values used in the design of the casing shall not exceed the maximum allowable stress values in tension specified in Section VIII, Division 1 or 2, of the ASME Code at the maximum operating temperature of the material used.

**2.2.2** The equipment feet shall be provided with vertical jack-screws and shall be drilled with pilot holes that are accessible for use in final doweling.

**2.2.3** Supports and alignment bolts shall be rigid enough to permit the machine to be moved by the use of its lateral and axial jackscrews.

● **2.2.4** The maximum allowable working pressure of the casing shall be at least equal to the specified relief valve setting; if a relief valve setting is not specified, the maximum allowable working pressure shall be at least 1.25 times the maximum specified discharge pressure (gauge). System protection will be furnished by the purchaser.

**2.2.5** Casings designed for more than one maximum allowable pressure level (split pressure-level casings) are not permitted unless specifically approved by the purchaser. If approved, the vendor shall define the physical limits and the maximum allowable working pressure of each section of the casing.

**2.2.6** Each axially split casing shall be sufficiently rigid to allow removal and replacement of its upper half without disturbing rotor-to-casing running clearances and bearing alignment.

**2.2.7** Casings shall be made of steel for the following:

a. Air or nonflammable gas at a maximum allowable working pressure that is over 2760 kPa gauge (400 pounds per square inch gauge).
b. Air or nonflammable gas at a calculated discharge temperature that is over 260°C (500°F) at maximum continuous speed at any point within the operating range. (The maximum temperature usually will occur near the surge point.)
c. Flammable or toxic gas.

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**2.2.8** Cast iron or other materials of construction may be offered for operating conditions other than those specified in 2.2.7.

**2.2.9** Unless otherwise specified, casings shall be radially split when the partial pressure of hydrogen (at maximum allowable working pressure) exceeds 1380 kPa gauge (200 pounds per square inch gauge).

Note: The partial pressure of hydrogen is calculated by multiplying the highest specified mole (volume) percent of hydrogen by the maximum allowable working pressure.

**2.2.10** Axially split casings shall use a metal-to-metal joint (with a suitable joint compound) that is tightly maintained by suitable bolting. Gaskets (including string type) shall not be used on the axial joint. O-rings or ring grooves machined into the flange facing of an axially split casing joint may be used with purchaser's approval. When gasketed joints are used between the end covers and the cylinder of radially split casings, they shall be securely maintained by confining the gaskets. Gasket materials shall suitable be for all specified service conditions.

**2.2.11** Jackscrews, guide rods, and casing alignment dowels shall be provided to facilitate disassembly and reassembly. When jackscrews are used as a means of parting contacting faces, one of the faces shall be relieved (counterbored or recessed) to prevent a leaking joint or improper fit caused by marring of the face. Guide rods shall be of sufficient length to prevent damage to the internals or casing studs by the casing during disassembly and re-assembly. Lifting lugs or eyebolts shall be provided for lifting only the top half of the casing. Methods of lifting the assembled machine shall be specified by the vendor.

**2.2.12** The use of tapped holes in pressure parts shall be minimized. To prevent leakage in pressure sections of casings, sufficient metal in addition to the allowance for corrosion shall be left around and below the bottom of drilled and tapped holes. The depth of the tapped holes shall be at least 1.5 times the stud diameter.

**2.2.13** The sealing of stud clearance holes to prevent leakage is not permitted.

**2.2.14** Compressor mounting surfaces shall be parallel and shall have a machined finish of 3.2 micrometers to 6.4 micrometers (125 microinches to 250 microinches) arithmetic average roughness ($R_a$). Hold-down or foundation bolt holes shall be drilled perpendicular to the compressor mounting surface or surfaces and spot faced to a diameter three times that of the hole.

**2.2.15** Studded connections shall be furnished with studs installed. Blind stud holes should be drilled only deep enough to allow a preferred tap depth of 1.5 times the major diameter of the stud; the first 1.5 threads at both ends of each stud shall be removed.

**2.2.16** Bolting external and internal to the casing shall be furnished as specified in 2.2.16.1 through 2.2.16.4.

**2.2.16.1** The details of threading shall conform to ASME B1.1.

**2.2.16.2** Studs are preferred to cap screws (external only).

**2.2.16.3** Adequate clearance shall be provided at bolting locations to permit the use of socket or box wrenches (external only).

**2.2.16.4** Socket-head, slotted-nut, or spanner-type bolting shall not be used unless specifically approved by the purchaser (external only).

## 2.3 Interstage Diaphragms and Inlet Guide Vanes

● **2.3.1** Interstage diaphragms and inlet guide vanes shall be suitable for all specified operating conditions, including startup, shutdown, trip-out, settling-out, and momentary surge. When intermediate main process connections are used, the purchaser will specify the maximum and minimum pressure at each connection. The vendor shall confirm that the diaphragms furnished are suitable for the maximum differential pressure.

**2.3.2** Internal joints shall be designed to minimize leakage and permit easy disassembly.

**2.3.3** Renewable labyrinths shall be provided at all internal close clearance points to minimize internal leakage. These labyrinths shall be stationary and easily replaceable unless otherwise approved by the purchaser.

**2.3.4** Diaphragms shall be axially split unless otherwise approved by the purchaser. The diaphragms shall be furnished with threaded holes for eyebolts or with another means to facilitate removal.

**2.3.5** If diaphragm cooling is specified, the top and bottom halves of axially split diaphragms shall have independent cooling passages to prevent leaks across the horizontal split line. Each coolant inlet and outlet connection shall be manifolded at both the top and at the bottom of each casing.

## 2.4 Casing Connections

### 2.4.1 GENERAL

**2.4.1.1** All process gas connections to the casing shall be suitable for the maximum allowable working pressure of the casing as defined in 2.2.4.

**2.4.1.2** All of the purchaser's connections shall be accessible for maintenance without the machine being moved.

**2.4.1.3** Openings for NPS $1^1/_4$, $2^1/_2$, $3^1/_2$, 5, 7, and 9 shall not be used.

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**2.4.1.4** Connections welded to the casing shall meet the material requirements of the casing, including impact values, rather than the requirements of the connected piping.

**2.4.1.5** All welding of connections shall be done before hydrostatic testing (see 4.3.2).

### 2.4.2 MAIN PROCESS CONNECTIONS

● **2.4.2.1** Inlet and outlet connections shall be flanged or machined and studded, and oriented as specified in the data sheets. Inlet and outlet connections for barrel compressors shall be located in the outer casing, not in the end covers. On radially split overhung design compressors, the process inlet connection may be in the end cover.

**2.4.2.2** Flanges shall conform to ANSI B16.1, B16.5, or B16.42. Steel flanges larger than 24 inches shall conform to API Standard 605 or to MSS SP-44.

**2.4.2.2.1** Flat-faced flanges with full raised-face thickness are acceptable on cases other than cast iron.

**2.4.2.2.2** Flanges that are thicker or have a larger outside diameter than that required by ANSI B16.5, API Standards 605,or MSS SP-44 are acceptable.

● **2.4.2.3** Connections other than those covered by ANSI B16.5, API Standards 605, or MSS SP-44 require the purchaser's approval. When specified, the vendor shall supply all mating flanges, including studs and nuts.

**2.4.2.4** Cast iron flanges shall be flat faced and shall have a minimum thickness of Class 250 per ANSI B16.1 for sizes 8 inches and smaller.

**2.4.2.5** The concentricity of the bolt circle and the bore of all casing flanges shall be such that the area of the machined gasket-seating surface is adequate to accommodate a complete standard gasket without protrusion of the gasket into the fluid flow.

**2.4.2.6** The finish of all flanges and nozzles shall conform to ANSI B16.1, B16.5, or B16.42 as applicable to the material furnished, including flange finish roughness requirements.

### 2.4.3 AUXILIARY CONNECTIONS

**2.4.3.1** Auxiliary connections may include but are not limited to those for vents, liquid injection, drains (see 2.4.3.2), water cooling, lube and seal oil, flushing, buffer gas, and balance piston cavity.

● **2.4.3.2** For axially split casings, the vendor shall provide connections for complete drainage of all gas passages. For radially split casings, the drains shall be located at the lowest point of each inlet section, the lowest point of the section between the inner and outer casings, and the lowest point of each discharge section. When specified, individual stage drains, including a drain for the balance piston cavity, shall be provided.

**2.4.3.3** Flanges shall conform to ANSI B16.5.

**2.4.3.4** Auxiliary connections shall be at least $^3/_4$-inch nominal pipe size (see 2.4.1.3) and shall be socket welded and flanged, or machined and studded. For socket welded construction, a 1.5-millimeter ($^1/_{16}$-inch) gap, as measured prior to welding, shall be left between the pipe end and the bottom of the socket in the casing.

**2.4.3.5** Where socket-welded and flanged or machined and studded openings cannot be provided, threaded openings in sizes $^3/_4$-inch through $1^1/_2$ inches nominal pipe size are permissible with the approval of the purchaser. These threaded openings shall be installed as specified in 2.4.3.5.1 through 2.4.3.5.3

**2.4.3.5.1** Tapped openings and bosses for pipe threads shall conform to ANSI B1.20.3.

**2.4.3.5.2** Pipe threads shall be taper thread conforming to ANSI B1.20.3.

**2.4.3.5.3** The threaded connection shall be seal welded; however, seal welding is not permitted on cast iron equipment, for instrument connections, or where disassembly is required for maintenance. Seal-welded joints shall be in accordance with ASME B31.3.

**2.4.3.6** A pipe nipple, preferably not more than 150 millimeters (6 inches) long, shall be installed in a threaded and socket-weld opening. Pipe nipples shall be a minimum of Schedule 160 seamless for threaded connections and Schedule 80 for socketwelded connections. Each pipe nipple shall be provided with a welding-neck, socket-weld, or slip-on flange.

**2.4.3.7** Tapped openings not connected to piping shall be plugged with solid round-head steel plugs furnished in accordance with ANSI B16.11. As a minimum plugs shall meet the material requirements of the casing. Plugs that may later require removal shall be of corrosion-resistant material. Threads shall be lubricated. Tape shall not be applied to threads of plugs inserted into oil passages. Plastic plugs are not permitted.

## 2.5 External Forces and Moments

**2.5.1** The compressor shall be designed to withstand external forces and moments on each nozzle calculated per Equations G-1 and G-2 of Appendix G. The vendor shall furnish the allowable forces and moments for each nozzle in tabular form.

**2.5.2** Casing and supports shall be designed to have sufficient strength and rigidity to limit coupling misalignment

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caused by imposing allowable forces and moments to 50 micrometers (0.002 inch).

## 2.6 Rotating Elements

**2.6.1** Shafts shall be made of one-piece, heat treated steel that is suitably machined. Shafts that have a finished diameter larger than 203 millimeters (8 inches) shall be forged steel. Shafts that have a finished diameter of 203 millimeters (8 inches) or less shall be forged steel or, with the purchaser's approval, hot rolled barstock, providing such barstock meets all quality and heat treatment criteria established for shaft forgings.

**2.6.2** Shaft ends for couplings shall conform to API Standard 671.

**2.6.3** Unless other shaft protection is approved by the purchaser, renewable shaft sleeves shall be furnished at interstage close-clearance points, under all carbon-ring packing, and at shaft seals (for gas service). These sleeves shall be made of a material that is corrosion-resistant in the specified service. The sleeves under close-clearance bushing end seals shall be suitably treated to resist wear and sealed to prevent leakage between the shaft and sleeve (see 2.11.1.7 for limitations).

**2.6.4** The design of shaft-sleeve-impeller assemblies shall not create temporary nor permanent distortions of the rotor assembly. The method of attaching the impeller shall adequately maintain concentricity and balance under all specified operating conditions, including overspeed to trip speed.

**2.6.5** The rotor shaft sensing areas to be observed by radial-vibration probes shall be concentric with the bearing journals. All shaft sensing areas (both radial vibration and axial position) shall be free from stencil and scribe marks or any other surface discontinuity, such as an oil hole or a keyway, for a minimum of one probe-tip diameter on each side of the probe. These areas shall not be metalized, sleeved, or plated. The final surface finish shall be from 0.4 to 0.8 micrometer (16 to 32 microinches) $R_a$, preferably obtained by honing or burnishing. These areas shall be properly demagnetized or otherwise treated so that the combined total electrical and mechanical runout does not exceed 25 percent of the maximum allowed peak-to-peak vibration amplitude or the following value, whichever is greater:

a. For areas to be observed by radial vibration probes, 6 micrometers (0.25 mil).
b. For areas to be observed by axial position probes, 13 micrometers (0.5 mil).

Note: If all reasonable efforts fail to achieve the limits noted in 2.6.5, the vendor and the purchaser shall mutually agree on alternate acceptance criteria.

**2.6.6** Each rotor shall be clearly marked with a unique identification number. This number shall be on the end of the shaft opposite the coupling or in an accessible area that is not prone to maintenance damage.

**2.6.7** Impellers may be closed, consisting of a disk, vanes, and a cover; or semi-open, consisting of disk and vanes. Impellers shall be of welded, riveted, milled, or case construction. Other manufacturing methods, such as electroerosion and brazing, may be permitted if approved by the purchaser. Each impeller shall be marked with a unique identification number.

**2.6.8** Welded and riveted impellers may consist of forged and cast components. Welds in the gas passageway shall be smooth and free of weld spatter. Impellers shall be heat treated and stress relieved after welding. Vane entrances and exits shall not have knife edges.

**2.6.9** Upgrade or repair welding of cast impellers may be permitted only with the purchaser's approval.

**2.6.10** Welding as a means of balancing an impeller is not permitted.

**2.6.11** The design of stressed parts shall include proper evaluation of the stress concentration factor (SCF) for the geometry. The design of stressed rotating parts shall include fillets that will limit the SCF. (Areas of concern include the impeller, vane-to-disk intersections, keyways, and shaft section changes.)

**2.6.12** Integral thrust collars are preferred. Replaceable thrust collars shall be furnished when they are required for removal of liquid-film, mechanical-contact, or gas shaft seals. When integral collars are furnished, they shall be provided with at least 3.2 millimeters ($^1/_8$-inch) of additional stock to enable refinishing if the collar is damaged. When replaceable collars are furnished (for assembly and maintenance purposes), they shall be positively locked to the shaft to prevent fretting.

**2.6.13** Both faces of thrust collars shall have a surface finish of not more than 0.4 micrometers (16 microinches) $R_a$, and the axial total indicator runout of either face shall not exceed 12.7 micrometers (0.0005 inches).

**2.6.14** Compressor designs that do not require a balance drum are acceptable.

**2.6.15** A balance drum, line, and porting shall be provided if required to limit axial loads on the thrust bearings. A separate pressure-tap connection or connections shall be provided to indicate the pressure in the balancing chamber, not in the balance line.

**2.6.16** The balance line shall be flanged and sized to handle balance drum gas leakage at twice the initial design labyrinth clearance without exceeding the load rating of the

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thrust bearings (see 2.7.3.3). If the balance line involves a purchaser's connection to his piping, then the connection sizes shall be indicated on the data sheets.

**2.6.17** To prevent the buildup of potential voltages in the shaft, residual magnetism of the rotating element shall not exceed 0.0005 tesla (5 gauss).

## 2.7 Bearings and Bearing Housings

### 2.7.1 GENERAL

**2.7.1.1** Hydrodynamic radial and thrust bearings shall be provided unless specific approval to the contrary is obtained from the purchaser.

Note: Appendix J gives application considerations for use of active magnetic bearings. These bearings are new technology at this time, and are not specifically being recommended in this standard, although some users may decide to incorporate this new technology into their units.

**2.7.1.2** Unless otherwise specified, thrust bearings and radial bearings shall be fitted with bearing-metal temperature sensors installed in accordance with API Standard 670.

### 2.7.2 RADIAL BEARINGS

**2.7.2.1** Sleeve or pad radial bearings shall be used and shall be split for ease of assembly. The use of non-split designs requires the purchaser's approval. The bearings shall be precision bored with steel-backed babbitted replaceable liners, pads, or shells. The bearings shall be equipped with anti-rotation pins and shall be positively secured in the axial direction.

**2.7.2.2** The bearing design shall suppress hydrodynamic instabilities and provide sufficient damping over the entire range of allowable bearing clearances to limit rotor vibration to the maximum specified amplitudes (see 2.9.5.5) while the equipment is operating loaded or unloaded, including operation at any critical frequency (see 2.9.1.3) within the specified operating speed range.

**2.7.2.3** The liners, pads, or shells shall be in axially split housings and shall be replaceable. The removal of the top half of the casing of an axially split machine or the head of a radially split unit shall not be required for replacement of these elements. The bearing design shall not require removal of the coupling hub to permit replacement of the bearing liners, pads, or shells unless approved by purchaser.

**2.7.2.4** Compressors equipped with sleeve journal bearings shall be designed for field installation of tilting-pad radial bearings without remachining of the bearing bracket.

### 2.7.3 THRUST BEARINGS

**2.7.3.1** Hydrodynamic thrust bearings shall be steel-backed, babbitted multiple segments designed for equal thrust capacity in both directions and arranged for continuous pressurized lubrication to each side. Both sides shall be tilting pads, incorporating a self-leveling feature which ensures that each pad carries an equal share of the thrust load even with minor variation in pad thickness.

**2.7.3.2** Each pad shall be designed and manufactured with dimensional precision thickness variation) that will allow interchange or replacement of the individual pads.

**2.7.3.3** Thrust bearings shall be sized for continuous operation under the most adverse specified operating conditions. Calculations of the thrust forces shall include but shall not be limited to the following factors:

a. Seal maximum design internal clearances and twice the maximum design internal clearances.
b. Pressurized rotor diameter step changes.
c. Stage maximum differential pressures.
d. Specified extreme variations in inlet, interstage, and discharge pressures.
e. External thrust forces transmitted through the couplings.
f. The maximum thrust force from the sleeve bearing type drive if the motor is directly connected.

**2.7.3.4** For gear couplings, the external thrust force shall be calculated from Equation 1.

$$F = \frac{(0.25)(9550)P_r}{(N_r D)} \qquad (1)$$

In U.S. Customary units:

$$F = \frac{(0.25)(63{,}300)P_r}{(N_r D)}$$

*Where:*

$F$ = external thrust force, in kilonewtons (pounds).
$P_r$ = rated power, in kilowatts (horsepower).
$N_r$ = rated speed, in revolutions per minute.
$D$ = shaft diameter at the coupling, in millimeters (inches).

**2.7.3.5** Thrust forces for flexible-element couplings shall be calculated on the basis of the maximum allowable deflection permitted by the coupling manufacturer.

**2.7.3.6** If two or more rotor thrust forces are to be carried by one thrust bearing (such as in a gear box), the resultant of the forces shall be used, provided the directions of the forces make them numerically additive; otherwise, the largest of the forces shall be used.

**2.7.3.7** Hydrodynamic thrust bearings shall be selected at no more than 50 percent of the bearing manufacturer's ultimate load rating. The ultimate load rating is the load that will produce the minimum acceptable oil film thickness without inducing failure during continuous service, or the load that will not exceed the creep initiation or yield strength

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of the babbitt at the location of maximum temperature on the pad, whichever load is less. In sizing thrust bearings, consider the following for each specified application:

a. The shaft speed.
b. The temperature of the bearing babbitt.
c. The deflection of the bearing pad.
d. The minimum oil film thickness.
e. The feed rate, viscosity, and supply temperature of the oil.
f. The design configuration of the bearing.
g. The babbitt alloy and pad material.
h. The turbulence of the oil film.

The basis for the sizing of thrust bearings shall be reviewed and approved by the purchaser.

**2.7.3.8** Thrust bearings shall be arranged to allow axial positioning of each rotor relative to the casing and setting of the thrust bearings' clearance.

### 2.7.4 BEARING HOUSINGS

**2.7.4.1** Rotor support system parts (bearings, bearing housings, bearing shells, and bearing brackets) shall be separable from the casing, axially split, non-pressurized (vented to atmosphere), and furnished with plugged connections for dry air or inert gas purge to any atmospheric labyrinth seals. Axially split bearing housings shall have a metal-to-metal split joint whose halves are located by means of cylindrical dowels.

**2.7.4.2** Compressors using semi-enclosed coupling guards shall have bearing housings equipped with replaceable labyrinth-end seals and deflectors where the shaft passes through the housing; lip seals shall not be used. The seals and deflectors shall be made of nonsparking materials. The design of the seals and deflectors shall effectively retain oil in the housing and prevent entry of foreign material into the housing.

**2.7.4.3** Bearing housings for pressure-lubricated hydrodynamic bearings shall be arranged to minimize foaming. The drain system shall be adequate to maintain the oil and foam level below shaft seals. Oil outlets from thrust bearings shall be tangential and in the upper half of the control ring or, if control rings are not used, in the thrust bearing cartridge.

**2.7.4.4** The rise in oil temperature through the bearing and housings shall not exceed 28°C (50°F) under the most adverse specified operating conditions. The bearing outlet oil temperature shall not exceed 82°C (180°F). When the inlet oil temperature exceeds 49°C (120°F), special consideration shall be given to bearing design, oil flow, and allowable temperature rise.

**2.7.4.5** Shaft support structures bolted to casings shall be steel.

**2.7.4.6** Oil connections on bearing housings shall be in accordance with 2.4.3.

**2.7.4.7** Mount two radial-vibration probes in each bearing housing, two axial-position probes at the thrust end of each machine, and a one-event-per-revolution probe in each machine. The probe installation shall be as specified in API Standard 670.

## 2.8 Shaft Seals

● **2.8.1** Shaft seals shall be provided to restrict or prevent process gas leaks to the atmosphere or seal fluid leaks into the process gas stream over the range of specified operating conditions, including startup and shutdown. Seal operation shall be suitable for specified variations in suction conditions that may prevail during startup, shutdown, or settling out, and during any other special operation specified by the purchaser. The maximum sealing pressure shall be at least equal to the settling-out pressure. The shaft seals and seal system shall be designed to permit safe compressor pressurization with the seal system in operation prior to process startup.

Note: The purchaser should establish a realistic value for settling-out pressure. The value should be shown on the data sheets.

● **2.8.2** Shaft seals and, when specified, shaft sleeves shall be accessible for inspection and for replacement without removing the top half of the casing of an axially split compressor or the heads of a radially split unit.

Note: This requirement may not be feasible for overhung designs.

● **2.8.3** Shaft seals may be one or a combination of the types described in 2.8.3.1 through 2.8.3.5, as specified by the purchaser on the data sheets. The materials for component parts shall be suitable for the service.

**2.8.3.1** The labyrinth seal (a typical seal is shown in Figure 2) may include carbon rings in addition to the labyrinths if approved by the purchaser. Labyrinths may be stationary or rotating. Eductors or injection systems, when used, shall be furnished complete with piping, regulating and control valves, pressure gauges, and strainers. Each item shall be piped and valved to permit its removal during operation of the compressor. Where gas from the compressor discharge is used for the motivating power of the eductor, provisions must be made for sealing during startup and shutdown.

● **2.8.3.2** The mechanical (contact) seal (a typical seal is shown in Figure 3) shall be provided with labyrinths and slingers. Oil or other suitable liquid furnished under pressure to the rotating seal faces may be supplied from the lube-oil system or from an independent seal system. Mechanical seals shall be designed to prevent gas leaks while the compressor is pressurized and being shut down and after it is stopped in the event of seal-oil failure. Various supplemental devices may be provided to ensure sealing when

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the compressor is pressurized but not running and the seal-oil system is shut down. The purchaser will specify whether such a device is to be provided. The final design shall be mutually agreed upon by the purchaser and the vendor.

**2.8.3.3** The restrictive-ring seal (a typical seal is shown in Figure 4) shall include rings of carbon or other suitable material mounted in retainers or in spacers. The seal may be operated dry, as in a labyrinth seal, or with a sealing liquid, as in a mechanical seal, or with a buffer gas.

**2.8.3.4** The liquid-film seal (typical seals are shown in Figures 5 and 6) shall be provided with sealing rings or bushings and labyrinths. A sealing liquid shall be supplied as in a mechanical seal. Liquid-film seals may be cylindrical-bushing seal as shown in Figure 5 or pumping seals as shown in Figure 6. Provide an elevated tank to maintain a static head in excess of compressor sealing pressure. The vendor shall state the height of the tank above the compressor centerline. Other means to maintain this differential pressure and positive seal may be used with the purchaser's approval.

**2.8.3.5** The self-acting gas seal may require external seal gas but does not require any liquid for lubrication or cooling. A typical configuration is shown in Figure 7. Where toxic or flammable seal gases are used, an isolating seal is required to

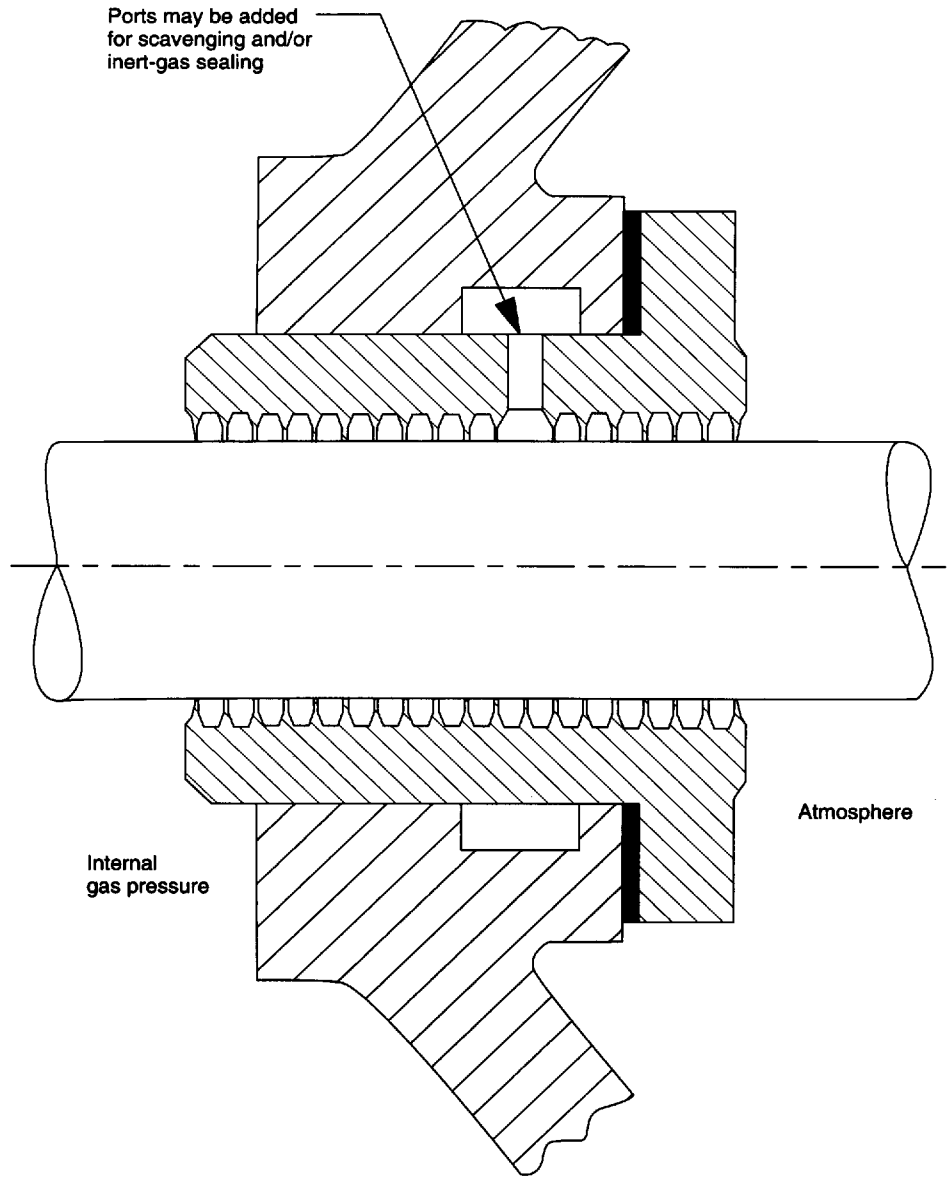



Figure 2—Labyrinth Shaft Seal

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prevent uncontrolled leaks to the atmosphere or to the bearing housing. This isolating seal shall preferably be capable of acting as a backup seal should the primary seal fail during operation. The seal gas shall be filtered and shall be free of any contaminants that form residues. The seal gas source may be taken from the compressor discharge or interstage point. An alternate seal gas source may be used, and may be required during startup or shutdown.

Note: Other variations are commonly used depending on the particular application. The seal will leak a small amount of seal gas, and may be unidirectional in operation. For testing considerations at the seal manufacturer's shop for this type of seal, see Appendix I.

**2.8.4** For any shaft end seals using sealing liquid, the inward leakage from each seal shall be piped to an independent drain pot. No individual shaft end seal shall have a leakage rate greater than 70 percent of the total expected leakage from all shaft seals in a single machine.

**2.8.5** Oil contaminated by the process gas that would damage components such as bearings, seal rings, O-rings, and couplings shall be piped away separately to allow disposal or reconditioning.

**2.8.6** Seal pressure equalizing lines and associated gas passages (including those for reference gas and axial thrust force balancing) shall be sized to maintain design shaft-end seal performance at twice the maximum initial design clearances. The lines and passages shall also be sized to maintain substantially equal pressures at both shaft-end seals during acceleration.

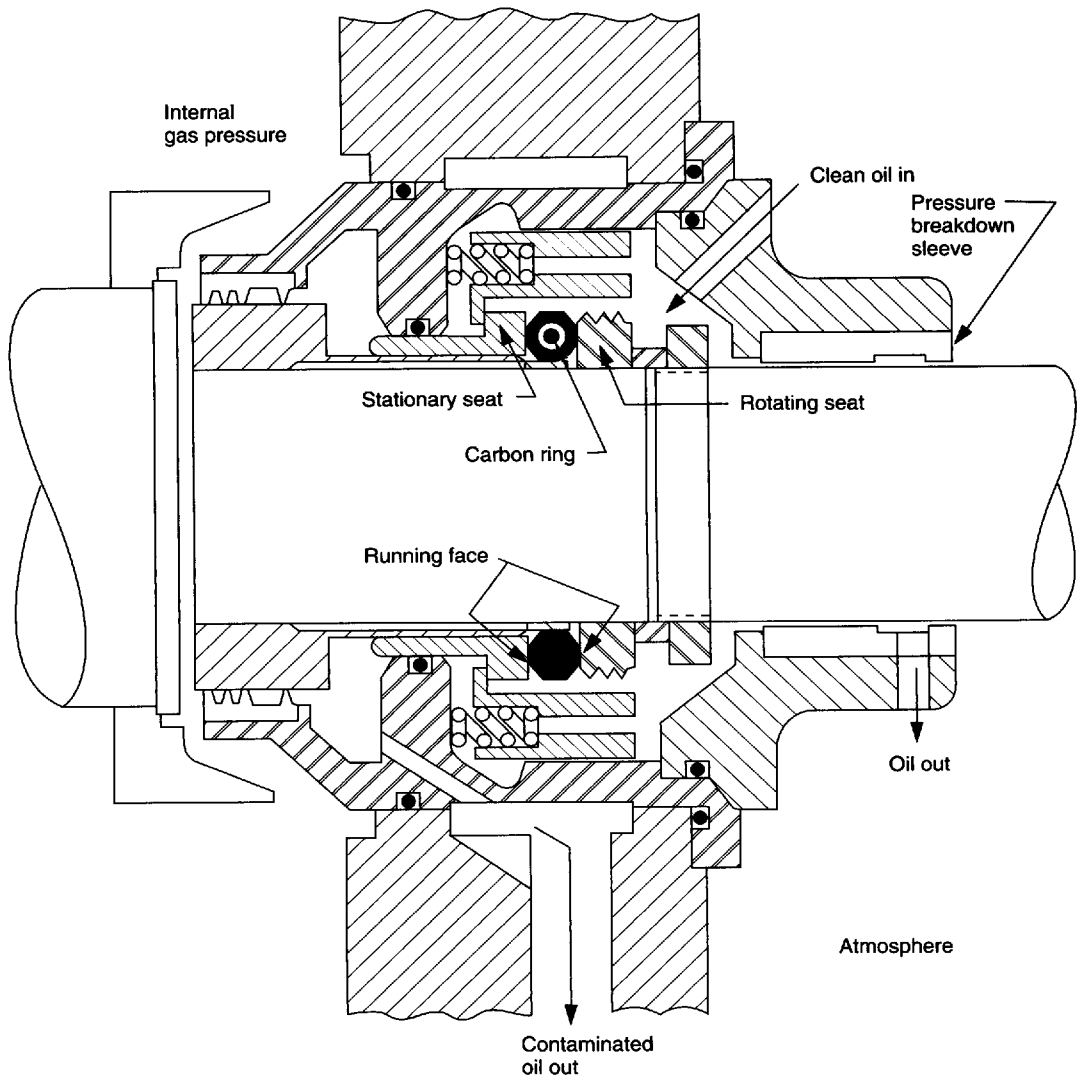



Figure 3—Mechanical (Contact) Shaft Seal

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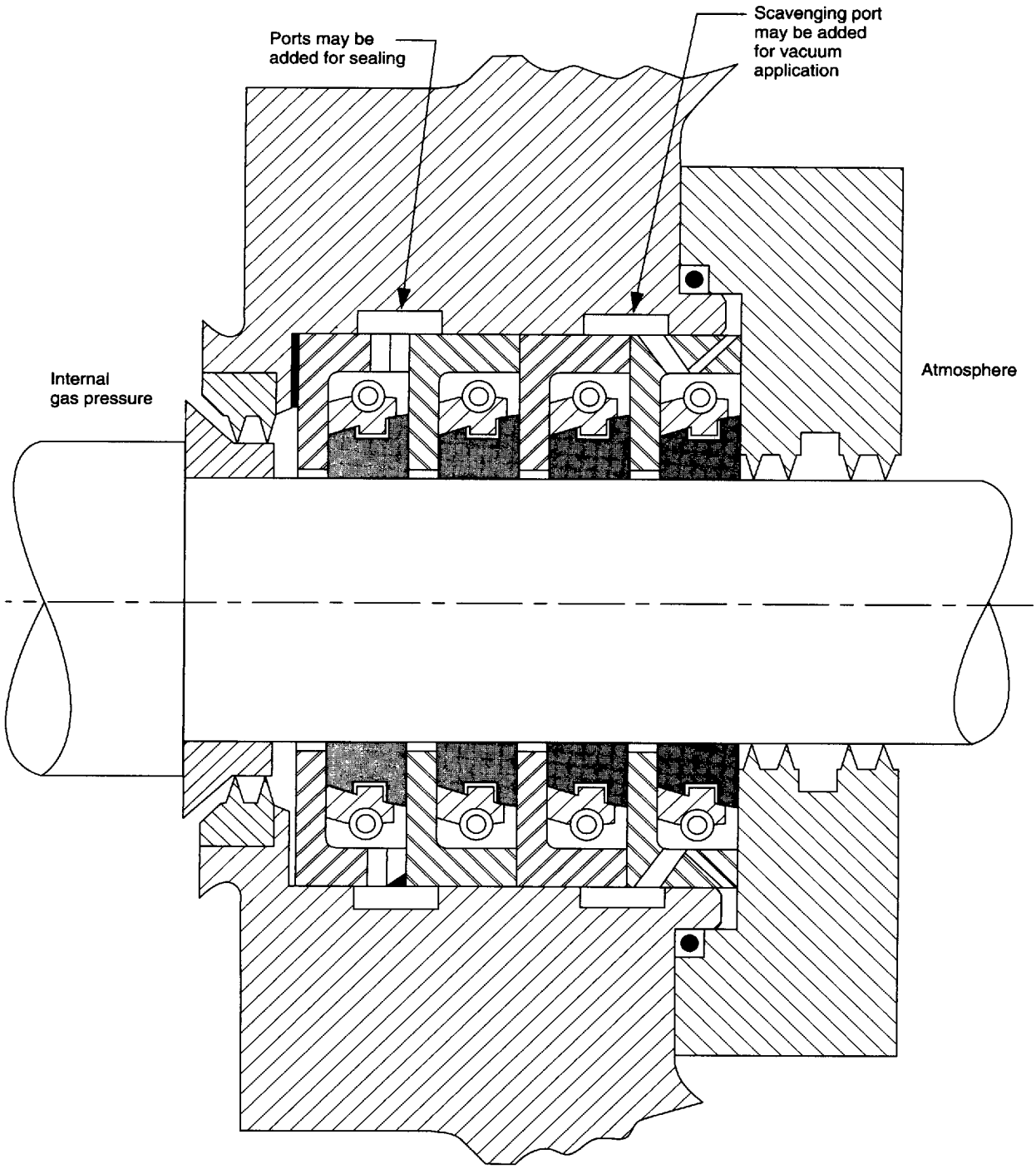



Figure 4—Restrictive-Ring Shaft Seal

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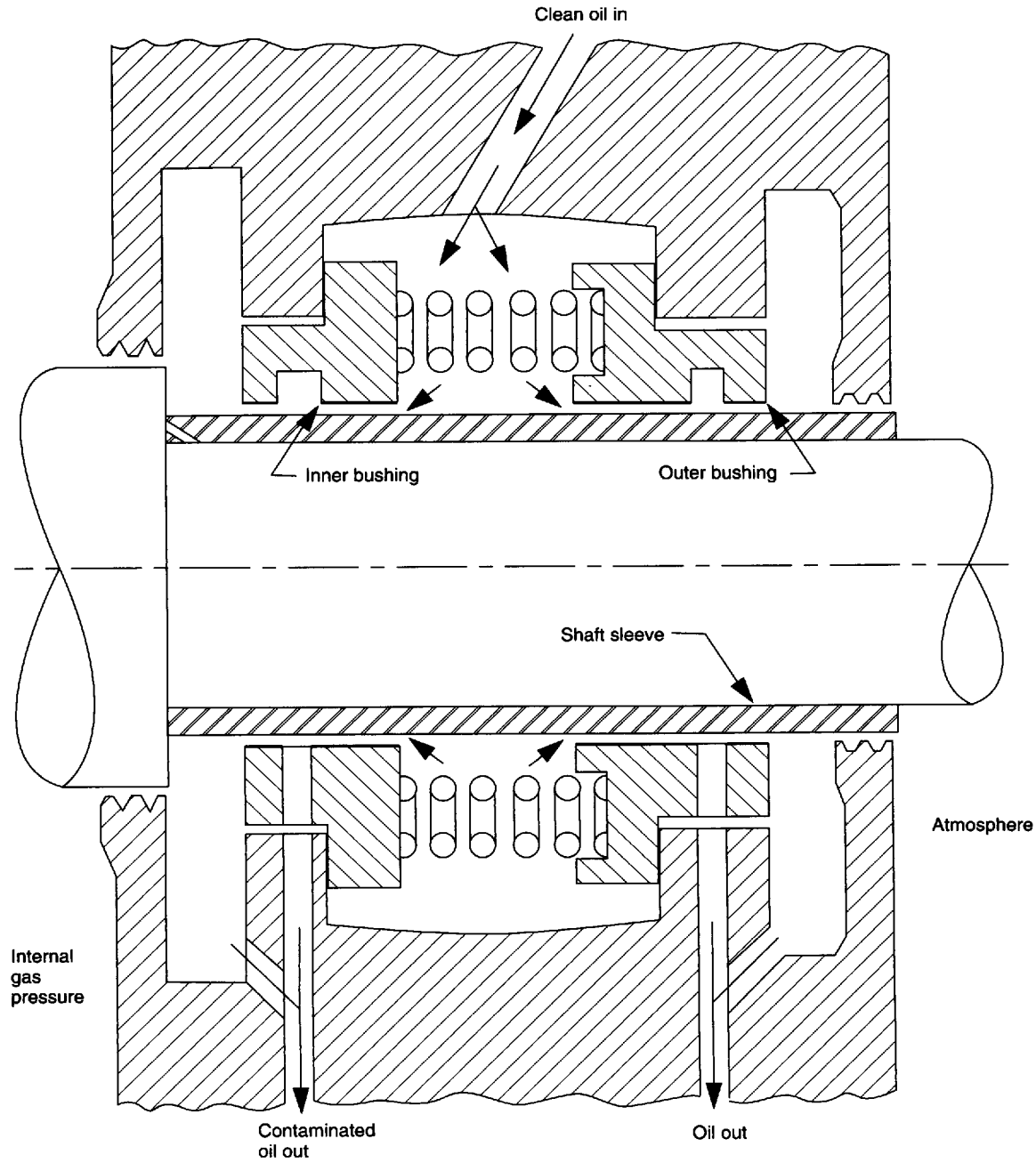



Figure 5—Liquid-Film Shaft Seal with Cylindrical Bushing

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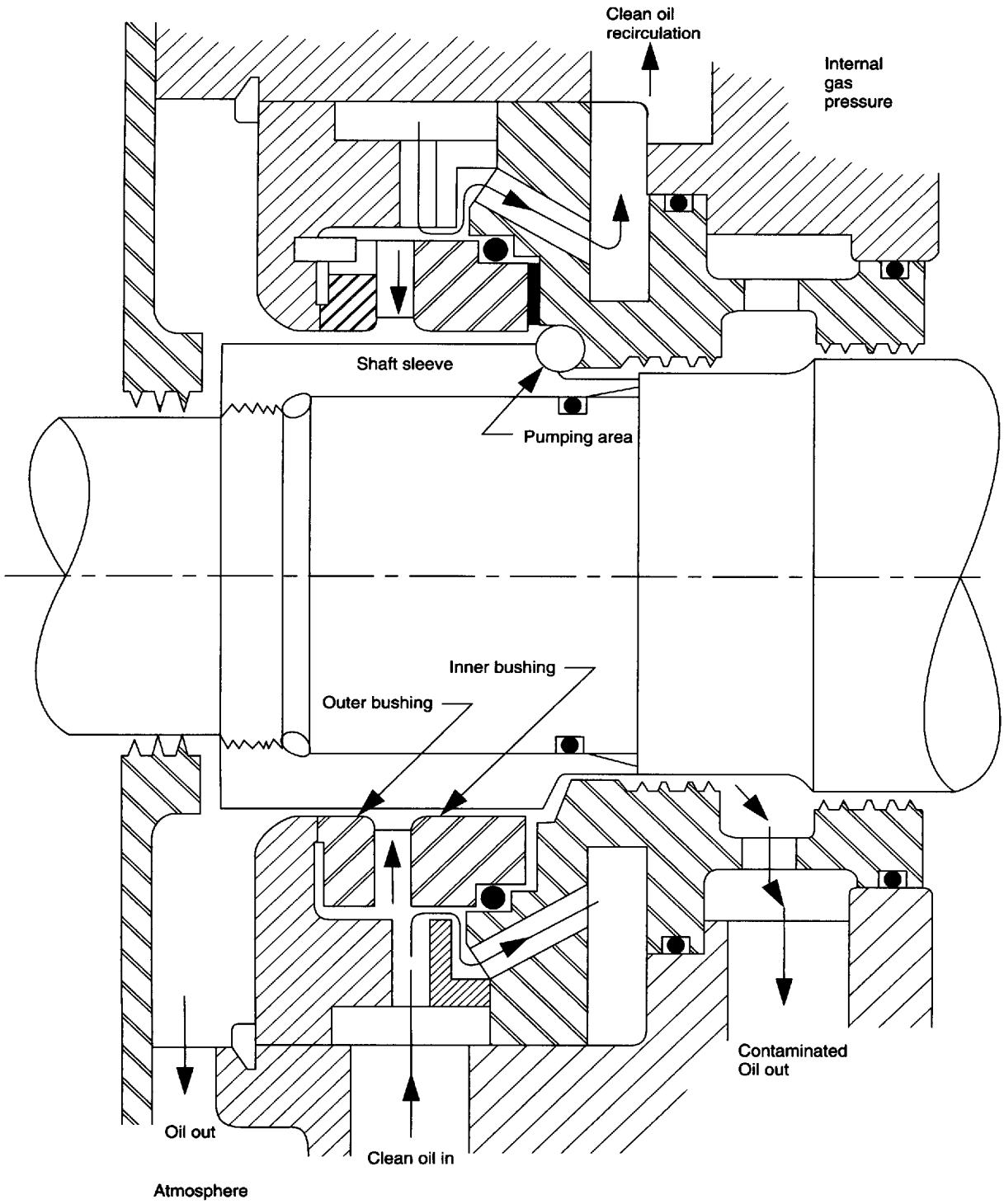



Figure 6—Liquid-Film Shaft Seal with Pumping Bushing

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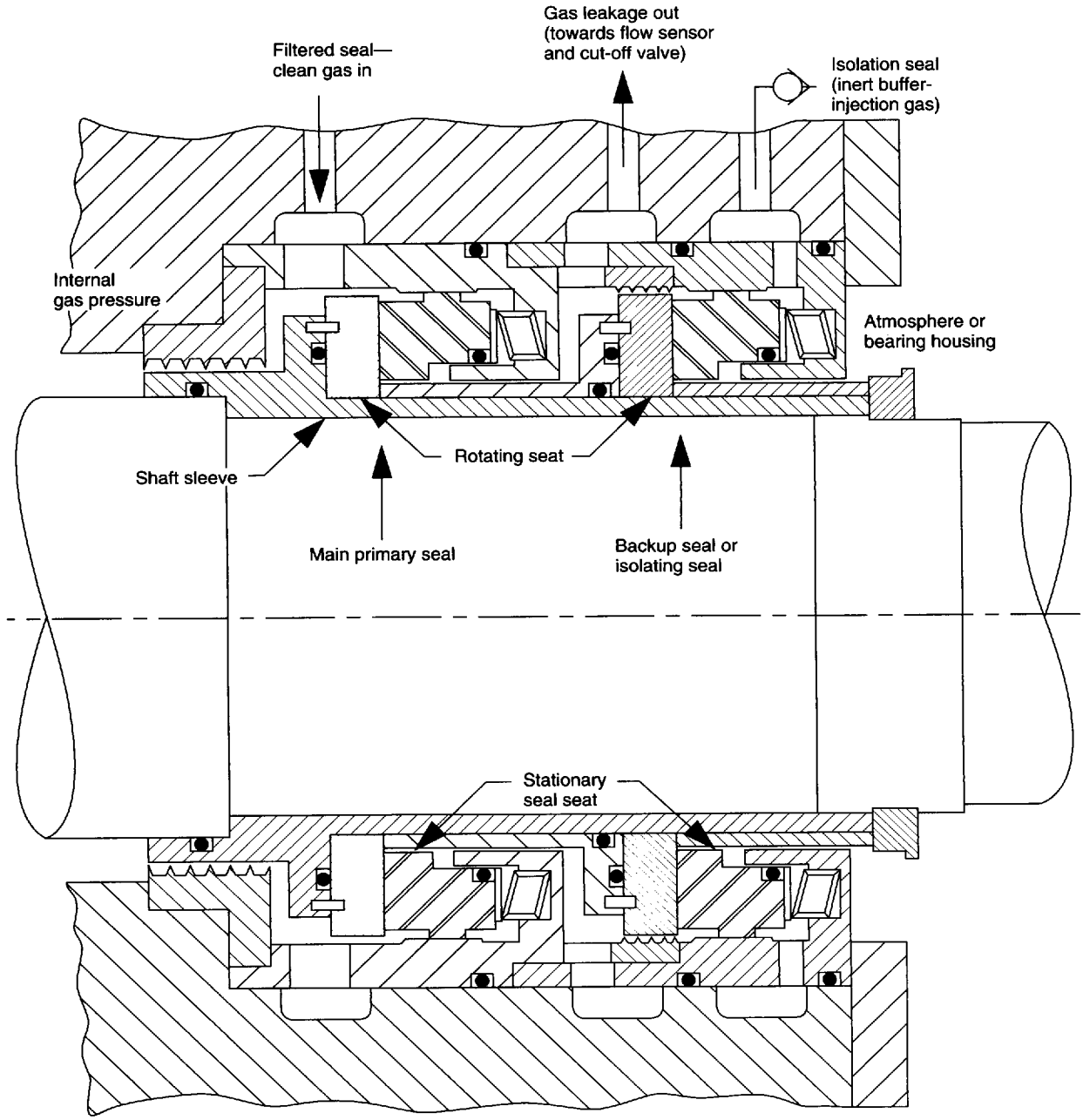



Figure 7—Self-Acting Gas Seal

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● **2.8.7** Unless otherwise specified, the seal design shall have provisions for buffer gas injection to each seal. The purchaser will specify whether buffer gas injection is to be used and, if so, the composition of that gas. In addition, the vendor shall state whether buffer gas injection is required for any specified operating conditions. When buffer gas injection is required, the vendor shall state the gas requirements including pressures, flowrates, and filtration, and, when specified, furnish the complete control system schematic and bill of material. The method of control will be specified by the purchaser.

● **2.8.8** When specified, for compressors with sub atmospheric pressure at the shaft end seals, provision shall be made to pressurize these seals with gas at a pressure that is higher than atmospheric.

## 2.9 Dynamics

Note: API Technical Publication 684 (TP 684) presents tutorial background pertinent to this section.

### 2.9.1 CRITICAL SPEEDS

**2.9.1.1** When the frequency of a periodic forcing phenomenon (exciting frequency) applied to a rotor-bearing support system corresponds to a natural frequency of that system, the system may be in a state of resonance.

**2.9.1.2** A rotor-bearing support system in resonance will have its normal vibration displacement amplified. The magnitude of amplification and the rate of phase-angle change are related to the amount of damping in the system and the mode shape taken by the rotor.

Note: The mode shapes are commonly referred to as the first rigid (translatory or bouncing) mode, the second rigid (conical or rocking) mode, and the (first, second, third, ..., *n*th) bending mode.

**2.9.1.3** When the rotor amplification factor (see Figure 8), as measured at the shaft radial vibration probes, is greater than or equal to 2.5, the corresponding frequency is called a critical speed, and the corresponding shaft rotational frequency is also called a critical speed. For the purposes of this standard, a critically damped system is one in which the amplification factor is less than 2.5.

**2.9.1.4** Critical speeds and their associated amplification factors shall be determined analytically by means of a damped unbalanced rotor response analysis and shall be confirmed during the running test and any specified optional tests.

**2.9.1.5** An exciting frequency may be less than, equal to, or greater than the rotational speed of the rotor. Potential exciting frequencies that are to be considered in the design of rotor-bearing systems shall include, but are not limited to, the following sources:

a. Unbalance in the rotor system.
b. Oil-film instabilities (whirl).
c. Internal rubs.
d. Blade, vane, nozzle, and diffuser passing frequencies.
e. Gear-tooth meshing and side bands.
f. Coupling misalignment.
g. Loose rotor-system components.
h. Hysteretic and friction whirl.
i. Boundary-layer flow separation.
j. Acoustic and aerodynamic crosscoupling forces.
k. Asynchronous whirl.

**2.9.1.6** Resonances of structural support systems may adversely affect the rotor vibration amplitude. Therefore, resonances of structural support systems that are within the vendor's scope of supply and that affect the rotor vibration amplitude shall not occur within the specified operating speed range or the specified separation margins (see 2.8.2.5) unless the resonances are critically damped.

**2.9.1.7** The vendor who is specified to have unit responsibility shall determine that the drive-train (turbine, gear, and motor) critical speeds (rotor lateral, system torsional, and blading modes) will not excite any critical speed of the machinery being supplied. The vendor also shall determine that the entire drive train is suitable for the specified operating speed range, including any starting-speed detent (hold-point) requirements. A list of all undesirable speeds from zero to trip shall be submitted to the purchaser for his review and included in the instruction manual for his guidance (see Appendix C, Item 37d).

### 2.9.2 LATERAL ANALYSIS

**2.9.2.1** The vendor shall provide a damped unbalanced response analysis for each machine to ensure acceptable amplitudes of vibration at any speed from zero to trip.

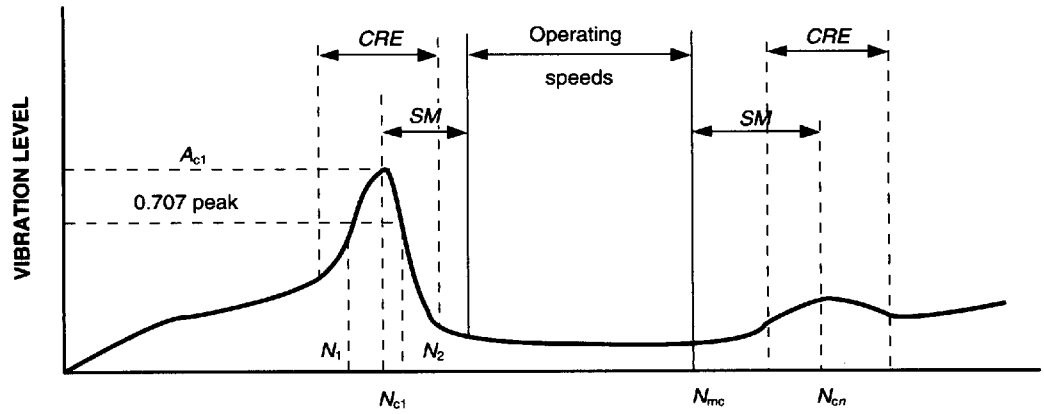
**2.9.2.2** The damped unbalanced response analysis shall include, but shall not be limited to, the following considerations:

a. Support (base, frame, and bearing-housing) stiffness, mass and damping characteristics, including effects of rotational speed variation. The vendor shall state the assumed support system values and the basis for these values (for example, tests of identical rotor support systems, assumed values).
b. Bearing lubricant-film stiffness and damping changes due to speed, load, preload, oil temperatures, accumulated assembly tolerances, and maximum to minimum clearances.
c. Rotational speed, including the various starting-speed detents, operating speed and load ranges (including agreed-upon test conditions if different from those specified), trip speed, and coast-down conditions.

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$N_{c1}$ = Rotor first critical, center frequency, cycles per minute.
$N_{cn}$ = Critical speed, $n$th.
$N_{mc}$ = Maximum continuous speed, 105 percent.
$N_1$ = Initial (lesser) speed at 0.707 × peak amplitude (critical).
$N_2$ = Final (greater) speed at 0.707 × peak amplitude (critical).
$N_2 - N_1$ = Peak width at the half-power point.
$AF$ = Amplification factor

$$= \frac{N_{c1}}{N_2 - N_1}$$

$SM$ = Separation margin.
$CRE$ = Critical response envelope.
$A_{c1}$ = Amplitude at $N_{c1}$.
$A_{cn}$ = Amplitude at $N_{cn}$.

Note: The shape of the curve is for illustration only and does not necessarily represent any actual rotor response plot.

Figure 8—Rotor Response Plot

d. Rotor masses, including the mass moment of coupling halves, stiffness, and damping effects (for example, accumulated fit tolerances, fluid stiffening and damping, and frame and casing effects).

e. Asymmetrical loading (for example, partial arc admission, gear forces, side streams, and eccentric clearances).

f. The influence, over the operating range, of the calculated values for hydrodynamic stiffness and damping generated by the casing end seals.

● **2.9.2.3** When specified, the effects of other equipment in the train shall be included in the damped unbalanced response analysis (that is, a train lateral analysis shall be performed).

Note: This analysis should be considered for machinery trains with coupling spacers greater than 1 meter (36 inches), rigid couplings, or both.

**2.9.2.4** As a minimum, the damped unbalanced response analysis shall include the following:

a. A plot and identification of the mode shape at each resonant speed (critically damped or not) from zero to trip, as well as the next mode occurring above the trip speed.

b. Frequency, phase, and response amplitude data (Bode plots) at the vibration probe locations through the range of each critical speed, using the arrangement of unbalance in Equation 2 for the particular mode. This unbalance shall be sufficient to raise the displacement of the rotor at the probe locations to the vibration limit defined by Equation 2:

$$L_v = 25.4\sqrt{\frac{12{,}000}{N}} \qquad (2)$$

In U.S. Customary units,

$$L_v = \sqrt{\frac{12{,}000}{N}}$$

*Where:*

$L_v$ = vibration limit (amplitude of unfiltered vibration), in micrometers (mils) peak to peak.
$N$ = operating speed nearest the critical of concern, in revolutions per minute.

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This unbalance shall be no less than two times the unbalance defined by the Equation 3:

$$U_{max} = 6350W/N \quad (3)$$

In U.S. Customary units,

$$U_{max} = 4W/N$$

*Where:*

$U$ = input unbalance from the rotor dynamic response analysis in gram-millimeters (ounce-inches).
$W$ = journal static weight load, in kilograms (pounds); or for bending modes where the maximum deflection occurs at the shaft ends, the overhung weight load (that is, the weight outboard of the bearing), in kilograms (pounds).
$N$ = operating speed nearest the critical of concern, in revolutions per minute.

The unbalance weight or weights shall be placed at the locations that have been analytically determined to affect the particular mode most adversely. For translatory modes, the unbalance shall be based on both journal static weights and shall be applied at the locations of maximum displacement. For conical modes, each unbalance shall be based on the journal weight and shall be applied at the location of maximum displacement of the mode nearest the journal used for the unbalance calculation, 180 degrees out of phase. Figure 2 shows the typical mode shapes and indicates the location and definition of $U$ for each of the shapes.

c. Modal diagrams for each response in 2.9.2.4, Item b, shall indicate the phase and major-axis amplitude at each coupling engagement plane, the centerlines of the bearings, the locations of the vibration probes, and each seal area throughout the machine. The minimum design diametral running clearance of the seals shall also be indicated.

d. To establish the validity of the analytical model, a verification test of the rotor unbalance is required at the completion of the mechanical running test. Therefore, additional plots based on the actual unbalance to be used during the test shall be provided. For machines that meet the requirements of 2.9.2.4, Item b, the location of the test unbalance shall be determined by the vendor. The amount of unbalance shall be sufficient to raise the vibration levels, as measured at the vibration probes, to those specified in 2.9.2.4, Item b. In all cases, the unbalance plots shall include the effects of any test-stand conditions (including the effects of test seals) that may be used during the verification test of the rotor unbalance (see 2.9.3).

● e. Unless otherwise specified, a stiffness map shall be provided of the undamped rotor response from which the damped unbalanced response analysis specified in 2.9.2.4, Item c, was derived. This plot shall show frequency versus support system stiffness, with the calculated support system stiffness curves superimposed.

f. For machines whose bearing support system stiffness values are less than or equal to 3.5 times the bearing stiffness values, use either the calculated frequency-dependent support stiffness and damping values (impedances) or the values derived from modal testing. The results of the damped unbalanced response analysis shall include Bode plots that compare absolute shaft motion with shaft motion relative to the bearing housing.

**2.9.2.5** The damped unbalance response analysis shall indicate that the machine in the unbalanced condition described in 2.9.2.4, Item b, will meet the following acceptance criteria (see Figure 8):

a. If the amplification factor is less than 2.5, the response is considered critically damped and no separation margin is required.

b. If the amplification factor is 2.5 to 3.55, a separation margin of 15 percent above the maximum continuous speed and 5 percent below the minimum operating speed is required.

c. If the amplification factor is greater than 3.55 and the critical response peak is below the minimum operating speed, the required separation margin (a percentage of minimum speed) is equal to the following:

$$SM = 100 - \left(84 + \frac{6}{AF - 3}\right) \quad (4)$$

*Where:*

$SM$ = Separation margin.
$AF$ = Amplification factor.

d. If the amplification factor is greater than 3.55 and the critical response peak is above the trip speed, the required separation margin (a percentage of maximum continuous speed) is equal to the following:

$$SM = \left(126 - \frac{6}{AF - 3}\right) - 100$$

*Where:*

$SM$ = Separation margin.
$AF$ = Amplification factor.

**2.9.2.6** The calculated unbalanced peak-to-peak rotor amplitudes (see 2.9.2.4, Item b) at any speed from zero to trip shall not exceed 75 percent of the minimum design diametral running clearances throughout the machine (with the exception of floating-ring seal locations).

**2.9.2.7** If, after the purchaser and the vendor have agreed that all practical design efforts have been exhausted, the analysis indicates that the separation margins still cannot be met or that a critical response peak falls within the operating speed range, acceptable amplitudes shall be mutually agreed upon by the purchaser and the vendor, subject to the requirements of 2.9.2.6.

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### 2.9.3 SHOP VERIFICATION OF UNBALANCED RESPONSE ANALYSIS

**2.9.3.1** A demonstration of rotor response at future unbalanced conditions is necessary because a well-balanced rotor may not be representative of future operating conditions (see 2.9.2.4, Item d). This test shall be performed as part of the mechanical running test (see 4.3.3) and the results shall be used to verify the analytical model. Unless otherwise specified, the verification test of the rotor unbalance shall be performed only on the first rotor (normally the spare rotor, if two rotors are purchased). The actual response of the rotor on the test stand to the same unbalance weight as was used to develop the Bode plots specified in 2.9.2.4 shall be the criterion for determining the validity of the damped unbalanced response analysis. To accomplish this, the following procedure shall be followed:

a. During the mechanical running test (see 4.3.4), the amplitudes and phase angle of the indicated vibration at the speed nearest the critical or criticals of concern shall be determined.
b. A trial weight, not more than one-half the amount calculated in 2.9.2.4, Item b, shall be added to the rotor at the location specified in 2.9.2.4, Item d; 90 degrees away from the phase of the indicated vibration at the speed or speeds closest to the critical or criticals of concern.
c. The machine shall then be brought up to the operating speed nearest the critical of concern, and the indicated vibration amplitudes and phase shall be measured. The results of this test and the corresponding indicated vibration data from 2.9.3.1, Item a, shall be vectorially added to determine the magnitude and phase location of the final test weight required to produce the required test vibration amplitudes.
d. The final test weight described in 2.9.3.1, Item c, shall be added to the rotor and the machine shall be brought up to the operating speed nearest the critical of concern. When more than one critical of concern exists, additional test runs shall be performed for each, using the highest speed for the initial test run.

Note: The dynamic response of the machine on the test stand will be a function of the agreed-upon test conditions and, unless the test-stand results are obtained at the conditions of pressure, temperature, speed, and load expected in the field, they may not be the same as the results expected in the field.

**2.9.3.2** The parameters to be measured during the test shall be speed and shaft synchronous (1×) vibration amplitudes with corresponding phase. The vibration amplitudes and phase from each pair of x-y vibration probes shall be vectorially summed at each response peak to determine the maximum amplitude of vibration. The major-axis amplitude of each response peak shall not exceed the limits specified in 2.9.2.6. (More than one application of the unbalance weight and test run may be required to satisfy these criteria.)

The gain of the recording instruments used shall be predetermined and preset before the test so that the highest response peak is within 60–100 percent of the recorder's full scale on the test-unit coast-down (deceleration). The major-axis amplitudes at the operating speed nearest the critical or criticals of concern shall not exceed the values predicted in accordance with 2.9.2.4, Item d, before coastdown through the critical of concern.

**2.9.3.2.1** Vectorial addition of slow-roll (300–600 revolutions per minute) electrical and mechanical runout is required to determine the actual vibration amplitudes and phase during the verification tests. Vectorial addition of the bearing-housing motion is required for machines that have flexible rotor supports (see 2.9.2.4, Item f).

Note 1: The phase on each vibration signal, x or y, is the angular measure, in degrees, of the phase difference (lag) between a phase reference signal (from a phase transducer sensing a once-per-revolution mark on the rotor, as described in API Standard 670) and the next positive peak, in time, of the synchronous (1×) vibration signal. (A phase change will occur through a critical or if a change in a rotor's balance condition occurs because of shifting or looseness in the assembly.)

Note 2: The major-axis amplitude is properly determined from a lissajous (orbit) display on an oscilloscope, or equivalent instrument. When the phase angle between the x and y signals is not 90 degrees, the major axis amplitude can be approximated by $(x^2 + y^2)^{1/2}$. When the phase angle between the x and y signals is 90 degrees, the major-axis value is the greater of the two vibration signals.

**2.9.3.2.2** The results of the verification test shall be compared with those from the original analytical model. The vendor shall correct the model if it fails to meet any of the following criteria:

a. The actual critical speeds shall not deviate from the predicted speeds by more than ±5 percent.
b. The predicted amplification factors shall not deviate from the actual test-stand values by more than ±20 percent.
c. The actual response peak amplitudes, including those that are critically damped, shall be within ±50 percent of the predicted amplitudes.

**2.9.3.3** Additional testing is required if, from the test data described or from the damped, corrected unbalanced response analysis (see 2.9.3.2.2), it appears that either of the following conditions exists:

a. Any critical response will fail to meet the separation margin requirements (see 2.9.2.5) or will fall within the operating speed range.
b. The requirements of 2.9.2.6 have not been met.

**2.9.3.4** Rotors requiring additional testing per 2.9.3.3 shall receive additional testing as follows: Unbalance weights shall be placed as described in 2.9.2.4, Item b; this may require disassembly of the machine. Unbalance magnitudes shall be achieved by adjusting the indicated unbalance that exists in the rotor from the initial run to raise the displacement of the rotor at the probe locations to the vibration limit defined by Equation 2 (see 2.9.2.4, Item b) at the maximum continuous speed. However, the unbalance used

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shall be no less than twice the unbalance limit specified in 2.9.5.2. The measurements from this test, taken in accordance with 2.9.3.2, shall meet the following criteria:

a. At no speed outside the operating speed range, including the separation margins, shall the shaft deflections exceed 90 percent of the minimum design running clearances.
b. At no speed within the operating speed range, including the separation margins, shall the shaft deflections exceed 55 percent of the minimum design running clearances or 150 percent of the allowable vibration limit at the probes (see 2.9.2.4, Item b).

The internal deflection limits specified in this section, Items a and b, shall be based on the calculated displacement ratios between the probe locations and the areas of concern identified in 2.9.2.4, Item c. Actual internal displacements for these tests shall be calculated by multiplying these ratios by the peak readings from the probes. Acceptance will be based on these calculated displacements or inspection of the seals if the machine is opened. Damage to any portion of the machine as a result of this testing shall constitute failure of the test. Minor internal seal rubs that do not cause clearance changes outside the vendor's new-part tolerance do not constitute damage.

### 2.9.4 TORSIONAL ANALYSIS

**2.9.4.1** Excitations of undamped torsional natural frequencies may come from many sources, which should be considered in the analysis. These sources may include, but are not limited to, the following:

a. Gear problems such as unbalance and pitch line runout.
b. Startup conditions such as speed detents and other torsional oscillations.
c. Torsional transients such as startups of synchronous electric motors and transients due to generator phase to phase fault or phase to ground fault.
d. Torsional excitation resulting from drivers such as electric motors and reciprocating engines.
e. Hydraulic governors and electronic feedback and control loop resonances from variable-frequency motors.
f. One- and two-time line frequency.
g. Running speed or speeds.

**2.9.4.2** The undamped torsional natural frequencies of the complete train shall be at least 10 percent above or 10 percent below any possible excitation frequency within the specified operating speed range (from minimum to maximum continuous speed).

**2.9.4.3** Torsional criticals at two or more times running speeds shall preferably be avoided or, in systems in which corresponding excitation frequencies occur, shall be shown to have no adverse effect. In addition to multiples of running speeds, torsional excitations that are not a function of operating speeds or that are nonsynchronous in nature shall be considered in the torsional analysis when applicable and shall be shown to have no adverse effect. Identification of these frequencies shall be the mutual responsibility of the purchaser and the vendor.

**2.9.4.4** When torsional resonances are calculated to fall within the margin specified in 2.9.4.2 and 2.9.4.3 (and the purchaser and the vendor have agreed that all efforts to remove the critical from within the limiting frequency range have been exhausted), a stress analysis shall be performed to demonstrate that the resonances have no adverse effect on the complete train. The acceptance criteria for this analysis shall be mutually agreed upon by the purchaser and the vendor.

**2.9.4.5** For motor-driven units and units including gears, or when specified for turbine-driven units, the vendor shall perform a torsional vibration analysis of the complete coupled train and shall be responsible for directing the modifications necessary to meet the requirements of 2.9.4.1 through 2.9.4.4.

**2.9.4.6** In addition to the torsional analyses required in 2.9.4.2 through 2.9.4.5, the vendor shall perform a transient torsional vibration analysis for synchronous-motor-driven units or variable speed motors. The acceptance criteria for this analysis shall be mutually agreed upon by the purchaser and the vendor.

### 2.9.5 VIBRATION AND BALANCING

**2.9.5.1** Major parts of the rotating element, such as the shaft, balancing drum, and impellers, shall be dynamically balanced. When a bare shaft with a single keyway is dynamically balanced, the keyway shall be filled with a fully crowned half-key. The initial balance correction to the bare shaft shall be recorded. A shaft with keyways 180 degrees apart but not in the same transverse plane shall also be filled as described above.

**2.9.5.2** The rotating element shall be multiplane dynamically balanced during assembly. This shall be accomplished after the addition of no more than two major components. Balancing correction shall only be applied to the elements added. Minor correction of other components may be required during the final trim balancing of the completely assembled element. In the sequential balancing process, any half-keys used in the balancing of the bare shaft (see 2.9.5.1) shall continue to be used until they are replaced with the final key and mating element. On rotors with single keyways, the keyway shall be filled with a fully crowned half-key. The weight of all half-keys used during final balancing of the assembled element shall be recorded on the residual unbalance work sheet (see Appendix D). The maximum

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allowable residual unbalance per plane (journal) shall be calculated as follows:

$$U_{max} = 6350W/N \quad (4)$$

In U.S. Customary units,

$$U_{max} = 4W/N$$

*Where:*

$U_{max}$ = residual unbalance, in gram-millimeters (ounce-inches).
$W$ = journal static weight load, in kilograms (pounds).
$N$ = maximum continuous speed, in revolutions per minute.

When spare rotors are supplied, they shall be dynamically balanced to the same tolerances as the main rotor.

**2.9.5.3** After the final balancing of each assembled rotating element has been completed, a residual unbalance check shall be performed and recorded as described in Appendix D.

**2.9.5.4** High-speed balancing (balancing in a high-speed balancing machine at the operating speed) shall be done only with the purchaser's specific approval. The acceptance criteria for this balancing shall be mutually agreed upon by the purchaser and the vendor.

**2.9.5.5** During the shop test of the machine—assembled with the balanced rotor, operating at its maximum continuous speed or at any other speed within the specified operating speed range—the peak-to-peak amplitude of unfiltered vibration in any plane, measured on the shaft adjacent and relative to each radial bearing, shall not exceed the value in Equation 5 or 25 micrometers (1.0 mils), whichever is less:

$$A = 25.4\sqrt{\frac{12,000}{N}} \quad (5)$$

In U.S. Customary units,

$$A = \sqrt{\frac{12,000}{N}}$$

*Where:*

$A$ = amplitude of unfiltered vibration, in micrometers (mils) peak to peak.
$N$ = maximum continuous speed, in revolutions per minute.

At any speed greater than the maximum continuous speed, up to and including the trip speed of the driver, the vibration shall not exceed 150 percent of the maximum value recorded at the maximum continuous speed.

Note: These limits are not to be confused with the limits specified in 2.9.3 for shop verification of unbalanced response.

**2.9.5.6** Electrical and mechanical runout shall be determined and recorded by rolling the rotor in V blocks while measuring runout with a noncontacting vibration probe and a dial indicator at the same shaft location.

**2.9.5.7** If the vendor can demonstrate that electrical or mechanical runout is present, a maximum of 25 percent of the test level calculated from Equation 5, or 6.4 micrometers (0.25 mil), whichever is greater, may be vectorially subtracted from the vibration signal measured during the factory test.

## 2.10 Lube Oil and Seal Oil Systems

**2.10.1** Unless otherwise specified, a pressurized oil system or systems shall be furnished to supply oil at a suitable pressure or pressures, as applicable, to the following:

a. The bearings of the driver and of the driven equipment (including any gear).
b. The continuously lubricated couplings.
c. The governing and control-oil system.
d. The shaft seal-oil system.
e. The purchaser's control system (if hydraulic).

**2.10.2** Oil reservoirs and housings that enclose moving lubricated parts (such as bearings, shaft seals, highly polished parts, instruments, and control elements) shall be designed to minimize contamination by moisture, dust, and other foreign matter during periods of operation or idleness.

● **2.10.3** The purchaser will specify on the data sheets whether the seal-oil and lube-oil systems are to be separate or combined. If separate systems are specified, the means of preventing interchange of oil between the two systems shall be described in the vendor's proposal.

**2.10.4** Unless otherwise specified, bearings and bearing housings shall be arranged for hydrocarbon oil lubrication.

**2.10.5** Unless otherwise specified, pressurized oil systems shall conform to the requirements of API Standard 614. When approved by the purchaser, a pressurized integral oil system may be provided for a closed-loop, nonhydrocarbon-refrigerant refrigeration system.

## 2.11 Materials

### 2.11.1 GENERAL

**2.11.1.1** Materials of construction shall be the manufacturer's standard for the specified operating conditions, except as required or prohibited by the data sheets or by this standard. Appendix B lists material specifications that, when used with appropriate heat treatment or impact testing requirements, or both, are generally considered acceptable for major component parts. The metallurgy of all major

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components shall be clearly stated in the vendor's proposal. See 3.5 for requirements for auxiliary piping materials.

**2.11.1.2** Materials shall be identified in the proposal with their applicable ASTM, AISI[11], ASME, or SAE numbers, including the material grade (see Appendix B). When no such designation is available, the vendor's material specification, giving physical properties, chemical composition, and test requirements, shall be included in the proposal.

● **2.11.1.3** Copper and copper alloys (excluding Monel or its equivalent, bearing babbitt, and precipitation-hardened stainless steels) shall not be used for parts of compressors or auxiliaries in contact with corrosive gas or with gases capable of forming explosive copper compounds. It will be the responsibility of the purchaser to note such gas characteristics on the inquiry data sheets.

**2.11.1.4** The vendor shall specify ASTM optional tests and inspection procedures necessary to ensure that materials are satisfactory for the service. Such tests and inspections shall be listed in the proposal. The purchaser should consider specifying additional tests and inspections, especially for materials in critical service.

**2.11.1.5** Selection of casing material shall be restricted by the limits imposed by 2.2.7 and 2.2.8.

**2.11.1.6** ASTM A 515 steel can be notch sensitive and prone to brittle fracture at ambient temperatures. The use of ASTM A 515 steel is therefore prohibited.

● **2.11.1.7** Materials exposed to a sour environment (wet $H_2S$) as defined by NACE MRO175-93 shall be in accordance with the requirements of that standard. Ferrous materials not covered by NACE MRO175-93 shall be limited to a yield strength not exceeding 620 MPa (90,000 pounds per square inch), and a hardness not exceeding Rockwell C 22. Components that are fabricated by welding shall be stress relieved, if required, so that both welds and heat affected zones meet the yield strength and hardness requirements. The purchaser will specify on the data sheets the presence and concentration of $H_2S$ and water in the process gas.

Note: Shafts in compressors of between-bearing design may exceed the stated limits of yield strength and hardness because of requirements for higher strength at reduced shaft sections near couplings and because of the low levels of working stress in the portion of the shaft between bearings.

● **2.11.1.8** The purchaser will specify the presence of any corrosive agents in the process gas, the process stream, and the environment, including constituents that may cause stress corrosion cracking.

[11]American Iron and Steel Institute, 1101 17th Street, N.W., Washington, D.C. 20036-4700.

**2.11.1.9** If parts made from austenitic stainless steels are exposed to a process gas or environmental condition that promotes intergranular corrosion, and they could be fabricated, hard-surfaced, repaired or overlayed by welding, then stabilized or low-carbon grades are required.

**2.11.1.10** Austenitic steels shall not be used in services where stress-corrosion cracking is a possibility.

Note: Overlays or hard surfaces that contain more than 0.10 percent carbon can sensitize both low-carbon and stabilized grades of austenitic stainless steel unless a buffer layer that is not sensitive to intergranular corrosion is applied.

**2.11.1.11** In hydrogen gas service at partial pressures greater than 689 kPa (100 pounds per square inch gauge) or at concentrations greater than 90 molal percent at any pressure, impeller materials that have a yield strength in excess of 827 MPa (120,000 pounds per square inch) or hardness in excess of Rockwell C 34 are prohibited.

**2.11.1.12** Materials, casting factors, and the quality of any welding shall be equal or superior to those required by Section VIII, Division 1, of the ASME Code. The manufacturer's data report forms, as specified in the code, are not required.

**2.11.1.13** External parts that are subject to rotary or sliding motions (such as Control linkage joints and adjusting mechanisms) shall be of corrosion-resistant materials suitable for the site environment.

**2.11.1.14** Minor parts that are not identified (such as nuts, springs, washers, gaskets, and keys) shall have corrosion resistance at least equal or superior to that of specified parts in the same environment.

**2.11.1.15** The minimum quality bolting material for pressure joints shall be carbon steel (ASTM A 307, Grade B) for cast iron casings, and high-temperature alloy steel (ASTM A 193, Grade B7) for steel casings. Nuts shall conform to ASTM A 194, Grade 2H (or ASTM A 307, Grade B, case-hardened, where space is limited). For temperatures below −29°C (−20°F), low-temperature bolting material in accordance with ASTM A 320 shall be used.

**2.11.1.16** Where mating parts such as studs and nuts of AISI Standard Type 300 stainless steel or materials with similar galling tendencies are used, they shall be lubricated with a suitable anti-seizure compound compatible with the process conditions.

Note: Torque loading values will differ considerably with and without an anti-seizure compound.

**2.11.1.17** O-rings shall be compatible with all specified services. For high-pressure services, special consideration shall be given to the selection of O-rings to ensure that they will not be damaged upon rapid depressuring of the compressor.

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### 2.11.2 PRESSURE-CONTAINING PARTS

**2.11.2.1** The vendor shall specify on the data sheets the material grade of castings.

**2.11.2.2** Welding of piping and pressure-containing parts, as well as any dissimilar metal welds and weld repairs, shall be performed and inspected by operators and procedures qualified in accordance with Section IX of the ASME Code.

**2.11.2.3** The vendor shall be responsible for establishing weld repair procedures that are in compliance with the applicable requirements of the ASME Code, and for the implementation of repairs in accordance with these procedures, including post repair heat treatment, if required, and nondestructive examination of repairs. Such procedures are subject to review by the purchaser before any repair is made.

**2.11.2.4** Castings shall be sound and free from porosity, hot tears, shrink holes, blow holes, cracks, scales, blisters, and similar injurious defects. Surfaces of castings shall be cleaned by sandblasting, shotblasting, chemical cleaning, pickling, or any other standard method. Mold parting fins and remains of gates and risers shall be chipped, filed, or ground flush.

**2.11.2.5** The use of chaplets in pressure castings shall be held to a minimum. The chaplets shall be clean and corrosion free (plating is permitted) and of a composition compatible with the casting.

**2.11.2.6** Ferrous castings shall not be repaired by welding, peening, plugging, burning in, or impregnating, except as specified in 2.11.2.6.1 and 2.11.2.6.2.

**2.11.2.6.1** Weldable grades of steel castings may be repaired by welding, using a qualified welding procedure based on the requirements of Section VIII, Division 1, and Section IX of the ASME Code.

**2.11.2.6.2** Cast gray iron or nodular iron may be repaired by plugging within the limits specified in ASTM A 278, A 395, or A 536. The drilled holes for plugs shall be carefully examined, using liquid penetrant, to ensure that all defective material has been removed. All necessary repairs that are not covered by ASTM specifications shall be subject to the purchaser's approval.

**2.11.2.7** Fully enclosed cored voids, including voids closed by plugging, are prohibited.

**2.11.2.8** Nodular iron castings shall be produced in accordance with ASTM A 395.

**2.11.2.9** Pressure containing casings made of wrought materials or combinations of wrought and cast materials shall conform to the conditions specified in 2.11.2.9.1 through 2.11.2.9.5.

**2.11.2.9.1** Plate edges shall be inspected by magnetic particle or liquid penetrant examination as required by Section VIII, Division 1, UG-93 (d) (3), of the ASME Code.

**2.11.2.9.2** Accessible surfaces of welds shall be inspected by magnetic particle or liquid penetrant examination after back chipping or gouging and again after post weld heat treatment.

**2.11.2.9.3** Pressure containing welds including welds of the case to horizontal or vertical joint flanges, shall be full penetration (complete joint) welds unless otherwise approved by the purchaser prior to any fabrication. For exception, see 2.4.3.

**2.11.2.9.4** Fabricated casings, including main nozzles, shall be post weld heat treated, regardless of thickness. For exception, see 2.4.3.

**2.11.2.9.5** All pressure containing welds shall be examined as required by Section VIII, Division 1, of the ASME Code. (See 2.11.4.1 for required procedures and acceptance criteria.) Requirements for additional examination shall be mutually agreed upon by the vendor and purchaser.

● **2.11.2.10** When specified, proposed connection designs shall be submitted to the purchaser for review before fabrication. The drawings shall show weld designs, size, materials, and pre- and post-weld heat treatments.

### ● 2.11.3 LOW TEMPERATURE

For operating temperatures below −29°C (−20°F) or when specified for other low ambient temperatures, steels shall have, at the lowest specified temperature, an impact strength sufficient to qualify under the minimum Charpy V-notch impact energy requirements of Section VIII, Division 1, UG-84, of the ASME Code. For materials and thicknesses not covered by the code, the purchaser will specify the requirements on the data sheets.

### 2.11.4 MATERIAL INSPECTION OF PRESSURE-CONTAINING PARTS

**2.11.4.1** Regardless of the generalized limits presented in this section, it shall be the vendor's responsibility to review the design limits of all materials and welds in the event that more stringent requirements are specified. Defects that exceed the limits imposed in Section 2.11.4 shall be removed to meet the quality standards cited, as determined by additional magnetic particle or liquid penetrant inspection as applicable before repair welding.

**2.11.4.2** When radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds or materials is required or specified, the procedures and acceptance criteria in 2.11.4.2.1 through 2.11.4.2.4 shall apply, except as noted (see 4.2.3).

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**2.11.4.2.1** Radiographic examination shall be in accordance with Section VIII, Division 1, UW-51 (100 percent) and UW-52 (spot), of the ASME Code. Spot radiography shall consist of a minimum of one 150-millimeter (6-inch) spot radiograph for each 7.6 meters (25 feet) of weld on each casing. As a minimum, one spot radiograph is required for each welding procedure and welder used for pressure-containing welds.

**2.11.4.2.2** Ultrasonic examination shall be in accordance with Section VIII, Division 1, UW 53 and Appendix 12 of the ASME Code.

**2.11.4.2.3** Magnetic particle examination shall be in accordance with Section VIII, Division 1, Appendix 6, of the ASME Code and ASTM A 709. Linear indications shall be considered relevant only if the major dimension exceeds 1.6 millimeters ($^1/_{16}$ inch). Individual indications that are separated by less than 1.6 millimeters ($^1/_{16}$ inch) shall be considered continuous.

**2.11.4.2.4** Liquid penetrant examination shall be in accordance with Section VIII, Division 1, Appendix 8, of the ASME Code.

**2.11.4.3** Cast steel casing parts shall be examined by magnetic particle methods. Acceptability of defects shall be based on a comparison with the photographs in ASTM E 125. For each type of defect, the degree of severity shall not exceed the limits specified in Table 2.

### 2.11.5 IMPELLERS

**2.11.5.1** All accessible areas of welds on welded impellers shall be inspected by magnetic particle or liquid penetrant examination.

**2.11.5.2** Cast impellers shall be inspected by radiographic or ultrasonic means prior to finish machining. Details of inspection techniques and acceptance criteria shall be mutually agreed upon by the vendor and the purchaser.

**2.11.5.3** After the overspeed test described in 4.3.3, each impeller shall be examined all over by means of magnetic particle or liquid penetrant methods.

Table 2—Maximum Severity of Defects in Castings

| Type | Defect | Degree |
|---|---|---|
| I | Linear discontinuities | 1 |
| II | Shrinkage | 2 |
| III | Inclusions | 2 |
| IV | Chills and chaplets | 1 |
| V | Porosity | 1 |
| VI | Welds | 1 |

## 2.12 Nameplates and Rotation Arrows

**2.12.1** A nameplate shall be securely attached at a readily visible point on the equipment and on any other major piece of auxiliary equipment.

**2.12.2** The purchaser's item number, the vendor's name, the machine's serial number, and the machine's size and type, as well as its minimum and maximum allowable design limits and rating data (including pressures, temperatures, speeds, and power), maximum allowable working pressures and temperatures, hydrostatic test pressures, and critical speeds shall appear on the machine's nameplate.

Note: Any lateral critical speeds determined on the running tests shall be stamped on the nameplate followed by the word test. Critical speeds predicted by calculation up to and including the critical speed above trip speed and not identifiable by test shall be stamped on the nameplate and noted as a calculated value.

**2.12.3** Rotation arrows shall be cast in or attached to each major item of rotating equipment. Nameplates and rotation arrows (if attached) shall be of AISI Standard Series 300 stainless steel or of nickel-copper alloy (Monel or its equivalent). Attachment pins shall be of the same material.

## 2.13 Quality

Refer to API Technical Publication 683 (TP 683) for guidelines to improve the quality of the equipment.

## 2.14 Safety

Refer to ANSI B19.3 for Safety Standards.

# SECTION 3—ACCESSORIES

## 3.1 Drivers

● **3.1.1** The type of driver will be specified by the purchaser. The driver shall be sized to meet the maximum specified operating conditions, including external gear or coupling losses, and shall be in accordance with applicable specifications, as stated in the inquiry and order. The driver shall be suitable for satisfactory operation under the utility and site conditions specified by the purchaser.

● **3.1.2** Anticipated process variations that may affect the sizing of the driver (such as changes in pressure, temperature, or properties of the fluid handled, as well as special plant startup conditions), will be specified by the purchaser.

● **3.1.3** The starting conditions for the driven equipment will be specified by the purchaser, and the starting method shall be mutually agreed upon by the purchaser and the

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vendor. The driver's starting-torque capabilities shall exceed the speed-torque requirements of the driven equipment.

● **3.1.4** Steam turbine drivers shall conform to API Standard 612. Steam turbine drivers shall be sized to deliver continuously 110 percent of the maximum power required for the purchaser's specified conditions while operating at a corresponding speed with the specified steam conditions.

**3.1.5** Unless otherwise specified, compressor drive motors shall be in accordance with API Standard 541 or API Standard 546, as applicable. The motor nameplate rating (exclusive of the service factor) shall be at least 110 percent of the maximum power required (including gear and coupling losses) for any of the specified operating conditions. Consider starting the compressor at the normal suction pressure. Compressors driven by induction motors shall be rated at the actual motor speed for the rated load condition.

● **3.1.6** The motor's starting-torque requirements shall be met at a reduced voltage specified by the purchaser, and the motor shall accelerate to full speed within a period of time agreed upon by the purchaser and the vendor.

Note: For most applications, the starting voltage is typically 80 percent of the normal voltage, and the time required to accelerate to full speed is generally less than 30 seconds.

**3.1.7** Gas turbine drivers shall conform to API Standard 616 and shall be sized as mutually agreed upon by the purchaser and the vendor.

**3.1.8** Speed increasers and reducers shall be in accordance with API Standard 613. Epicyclic gears may be used with the purchaser's approval.

## 3.2 Couplings and Guards

**3.2.1** Unless otherwise specified, the compressor vendor shall furnish all couplings and guards for the entire compressor train, including gears when they are part of the train. The vendor shall arrange for complete machining of all coupling halves, mount the compressor coupling half, and arrange for mounting of the drive train half couplings.

**3.2.2** Couplings, coupling to shaft junctures, and guards shall conform to API Standard 671. The make, type, and mounting arrangement of the coupling shall be agreed upon by the purchaser and the vendors of the driver and driven equipment.

**3.2.3** When gear couplings with external teeth on the hub are furnished and the compressor is a through-drive design, the through-drive coupling shall include an idling adapter (solo plate) in accordance with API Standard 671. When fixed-hub couplings (marine gear or flexible element couplings) are used on through-drive designs, an idling adapter designed to impose a mass moment on the shaft equivalent to that produced by one-half the coupling weight, including the spacer, shall be furnished. Centering, fits, and tolerances shall be in accordance with API Standard 671.

**3.2.4** Plug and ring gauges shall be supplied in accordance with API Standard 671, Appendix D.

## 3.3 Mounting Plates

### 3.3.1 GENERAL

● **3.3.1.1** The equipment shall be furnished with soleplates or a baseplate, as specified.

**3.3.1.2** In 3.3.1.2.1 through 3.3.1.2.10, the term mounting plate refers to both baseplates and soleplates.

**3.3.1.2.1** Axial, lateral, and vertical jackscrews shall be provided for all equipment in the compressor train. If the equipment is too heavy to use jack-screws, other means shall be provided. Vertical jackscrews shall be arranged to prevent marring of shimming surfaces. Lugs holding these jackscrews shall be arranged so as not to interfere with the installation or removal of the shims.

**3.3.1.2.2** Equipment supports shall be provided with stainless steel (AISI Standard Type 300) shim packs, 3 millimeters to 13 millimeters ($^1/_8$ inch to $^1/_2$ inch) thick, with jackscrews for easy removal or addition of shims. All shims shall straddle hold down bolts and jackscrews.

**3.3.1.2.3** All machinery mounting surfaces on the mounting plates shall be machined parallel after fabrication and shall extend at least 25 millimeters (1 inch) beyond the outer three sides of the equipment feet.

**3.3.1.2.4** When centerline supports are provided, they shall be designed and manufactured to permit the machine to be moved using horizontal jackscrews.

**3.3.1.2.5** Anchor bolts shall not be used to fasten machinery to the mounting plates.

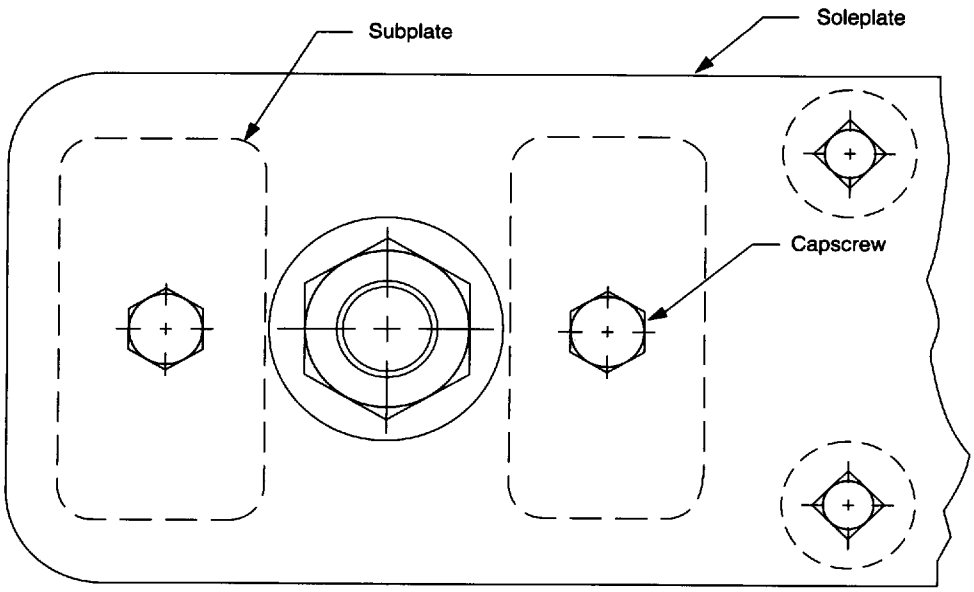
**3.3.1.2.6** Mounting plates shall not be drilled for equipment to be mounted by others. Mounting plates intended for installation on concrete shall be supplied with leveling screws. Mounting plate surfaces that are to be grouted shall have 50-millimeter-radiused (2-inch-radiused) outside corners (in the plan view). See Figures 9A , 9B, 9C, and 9D.

**3.3.1.2.7** Anchor bolts will be furnished by the purchaser.

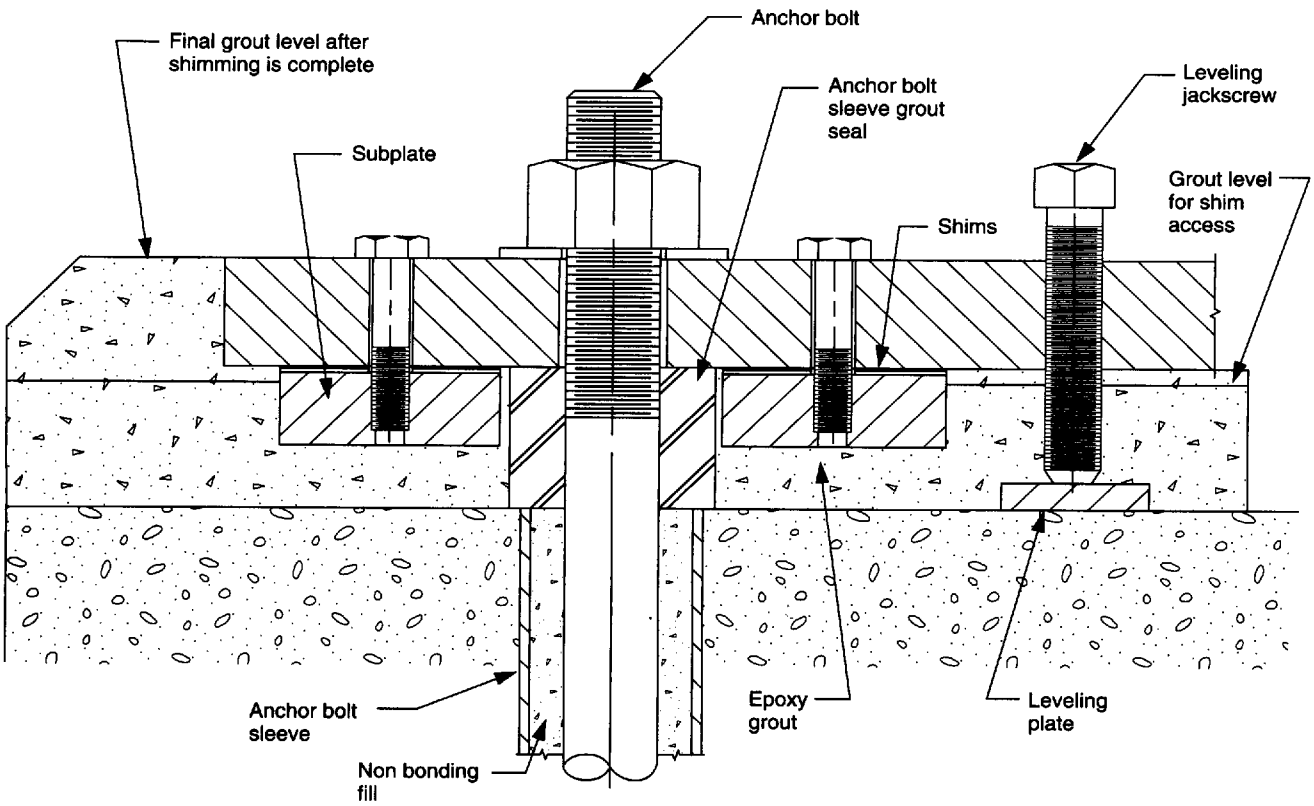
**3.3.1.2.8** Fasteners for attaching the components to the mounting plates and jackscrews for leveling the pedestal sole-plates shall be supplied by the vendor.

● **3.3.1.2.9** When epoxy grout is specified, the vendor shall commercially abrasive blast, in accordance with SSPC SP 6, all grouting surfaces of the mounting plates, and shall precoat these surfaces with a catalyzed epoxy primer. The

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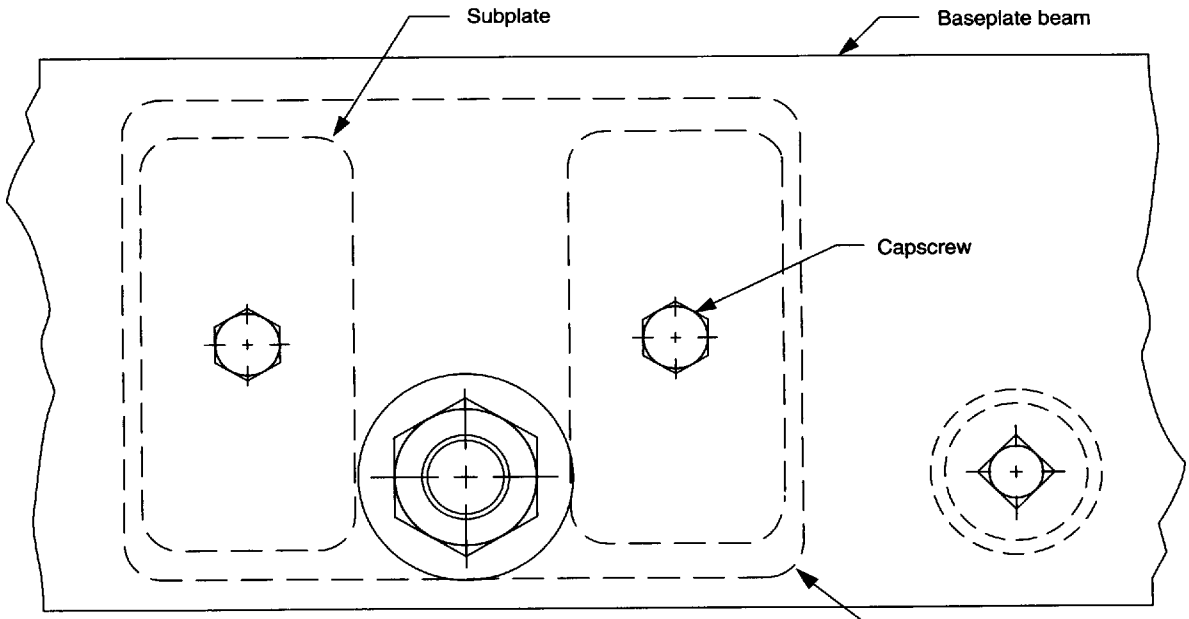

**TOP VIEW OF FOUNDATION AT FOUNDATION BOLT**



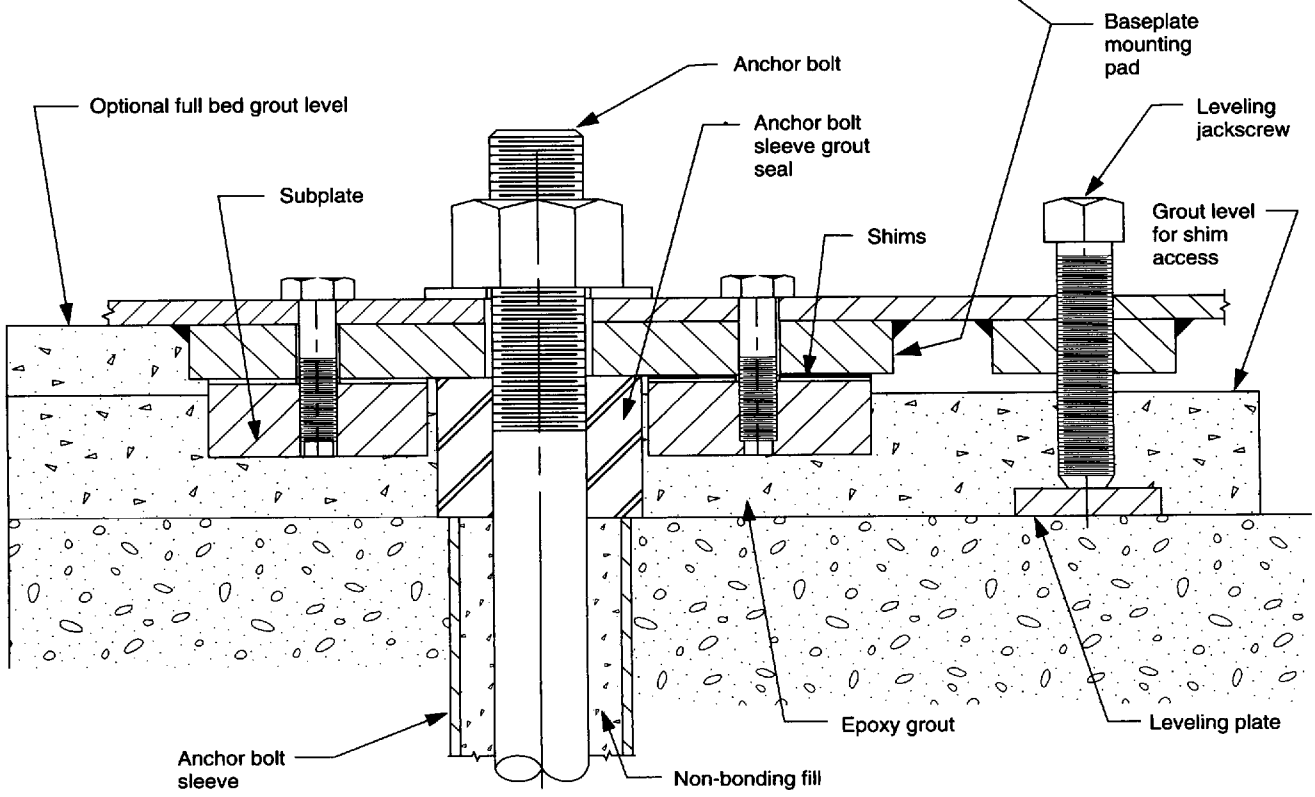

Note: Epoxy grout not to contact anchor bolt. Leveling jackscrew to be prevented from adhering to epoxy grout by applying wax, grease, or other protectant.

Figure 9A—Typical Mounting Plate Arrangement

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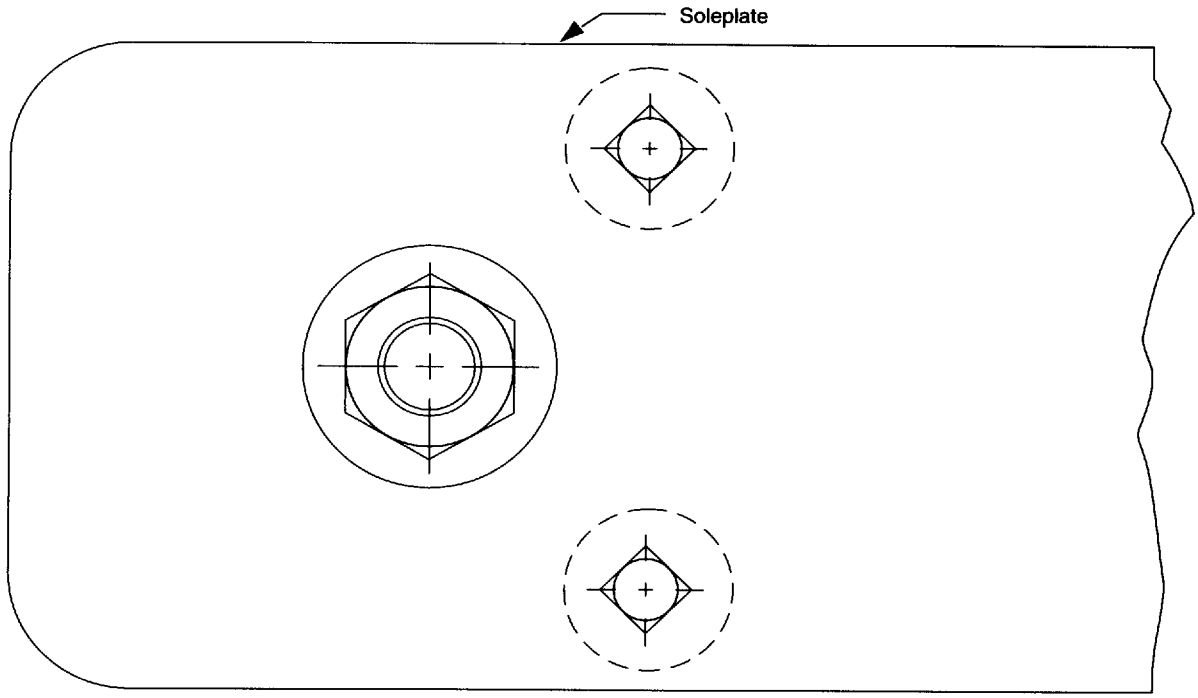

**TOP VIEW OF FOUNDATION AT FOUNDATION BOLT**



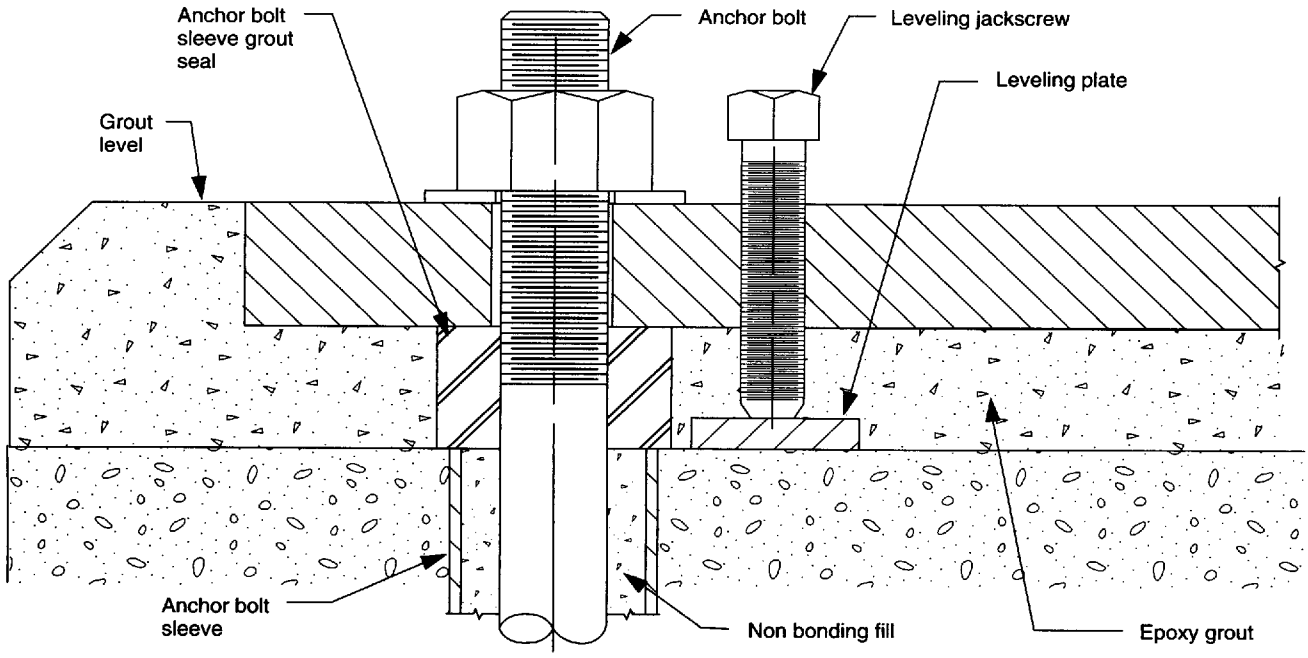

Notes:
1. This pour to be made after shimming is complete.
2. Epoxy grout not to contact anchor bolt. Leveling jackscrew to be prevented from adhering to epoxy grout by applying wax, grease, or other protectant.

Figure 9B—Typical Mounting Plate Arrangement

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**TOP VIEW OF FOUNDATION AT FOUNDATION BOLT**

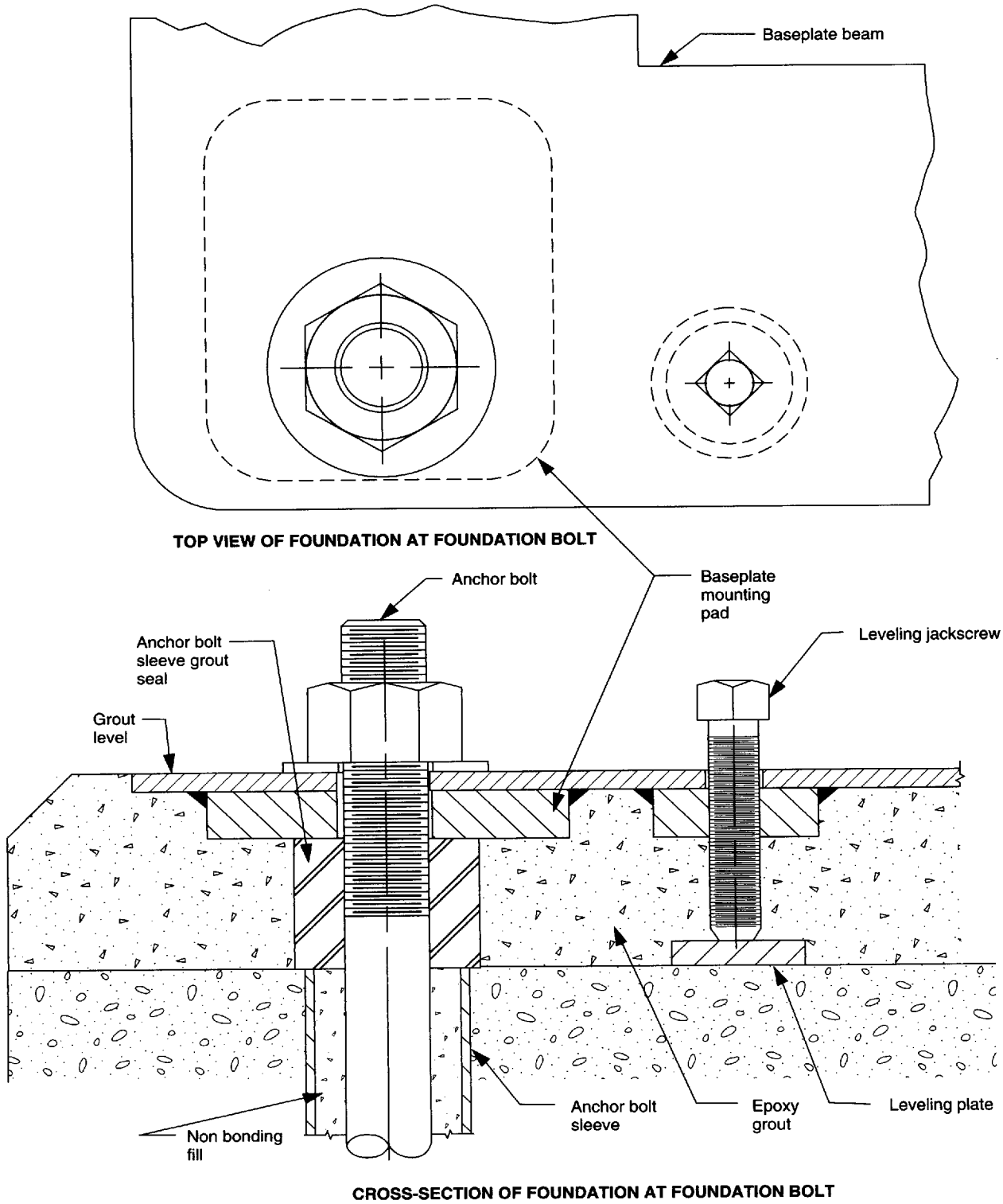



**CROSS-SECTION OF FOUNDATION AT FOUNDATION BOLT**

Note: Epoxy grout not to contact anchor bolt. Leveling jackscrew to be prevented from adhering to epoxy grout by applying wax, grease, or other protectant.

Figure 9C—Typical Mounting Plate Arrangement

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Note: Epoxy grout not to contact anchor bolt. Leveling jackscrew to be prevented from adhering to epoxy grout by applying wax, grease, or other protectant.

Figure 9D—Typical Mounting Plate Arrangement

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epoxy primer shall be compatible with epoxy grout. The vendor shall submit to the purchaser instructions for field preparation of the epoxy primer.

Note: Epoxy primers have a limited life after application. The grout manufacturer should be consulted to ensure proper field preparation of the mounting plate for satisfactory bonding of the grout.

**3.3.1.2.10** Mounting surfaces that are not to be grouted shall be coated with a rust preventive immediately after machining.

### 3.3.2 BASEPLATE

● **3.3.2.1** When a baseplate is specified, the data sheets will indicate the major equipment to be mounted on it. A baseplate shall be a single fabricated steel unit, unless the purchaser and the vendor mutually agree that it may be fabricated in multiple sections. Multiple-section baseplates shall have machined and doweled mating surfaces to ensure field reassembly.

Note: A baseplate with a nominal length of more than 12 meters (40 feet) or a nominal width of more than 3.6 meters (12 feet) may have to be fabricated in multiple sections because of shipping restrictions.

● **3.3.2.2** When specified, the baseplate shall be provided with leveling pads or targets protected with removable covers. The pads or targets shall be accessible for field leveling after installation, with the equipment mounted and the baseplate on the foundation.

● **3.3.2.3** When specified, the baseplate shall be suitable for column mounting (that is, of sufficient rigidity to be supported at specified points) without continuous grouting under structural members. The baseplate design shall be mutually agreed upon by the purchaser and the vendor.

**3.3.2.4** The baseplate shall be provided with lifting lugs for at least a four-point lift. Lifting the baseplate complete with all equipment mounted shall not permanently distort or otherwise damage the baseplate or the machinery mounted on it.

● **3.3.2.5** The bottom of the baseplate between structural members shall be open. When the baseplate is installed on a concrete foundation, accessibility shall be provided for grouting under all load-carrying structural members. The bottom of the baseplate shall be in one plane to permit use of a single-level foundation.

● **3.3.2.6** When specified, the bottom of the baseplate shall have machined mounting pads.

**3.3.2.7** The baseplate mounting pads shall be machined after the baseplate is fabricated.

**3.3.2.8** Unless otherwise specified, nonskid decking covering all walk and work areas shall be provided on the top of the baseplate.

**3.3.2.9** Oil reservoirs shall be separate from the baseplate (see API Standard 614) unless otherwise approved by the purchaser.

### 3.3.3 SOLEPLATES AND SUBSOLEPLATES

**3.3.3.1** When soleplates are specified, they shall meet the requirements of 3.3.3.1.1 through 3.3.3.1.2 in addition to those of 3.3.2.

**3.3.3.1.1** Adequate working clearance shall be provided at the bolting locations to allow the use of socket or box wrenches and to allow the equipment to be moved using the horizontal and vertical jackscrews.

**3.3.3.1.2** Soleplates shall be steel plates thick enough to transmit the expected loads from the equipment feet to the foundation, but in no case shall they be less than 38 millimeters (1.5 inches) thick.

**3.3.3.2** When subsoleplates are specified, they shall be steel plates at least 25 millimeters (1 inch) thick. The finish of the subsoleplates' mating surfaces shall match that of the soleplates.

## 3.4 Controls and Instrumentation

### 3.4.1 GENERAL

● **3.4.1.1** The vendor shall provide sufficient compressor performance data (in accordance with Section 5) to enable the purchaser to properly design a control system for startup operation, for all specified operating conditions, and for surge prevention. When specified by the purchaser, the vendor shall review the purchaser's overall compressor control system for compatibility with vendor-furnished control equipment.

**3.4.1.2** Instrumentation and installation shall conform to all detailed specifications. Unless otherwise specified, instrumentation shall conform to the requirements of API Standard 614, including supplied API 614 data sheets.

**3.4.1.3** Unless otherwise specified, controls and instrumentation shall be suitable for outdoor installation.

**3.4.1.4** All conduit on the compressor shall be designed so that it can be easily removed without damage, or be so located as not to hamper removal of bearings, seals, or the compressor internals.

### 3.4.2 CONTROL SYSTEMS

● **3.4.2.1** The compressor may be controlled on the basis of inlet pressure, discharge pressure, flow, or some combination of these parameters. This may be accomplished by suction throttling, variable inlet guide vanes, speed variation, discharge blowoff (when a constant-speed driver is used), or a cooled bypass from discharge to suction. The control system

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may be mechanical, pneumatic, hydraulic, electric, or any combination thereof. The system may be manual, or it may be automatic with a manual override. The purchaser will specify the source of the control signal, its sensitivity and range, and the equipment to be furnished by the vendor.

Note: A combination of control modes may be required on drives with a limited speed range and on multiservice or multistream applications.

● **3.4.2.2** When specified, an anti-surge system shall be provided.

Note: Anti-surge systems are generally required to prevent operating in unstable regions which may cause damage to the compressor. The scope of supply for these systems is process dependant and should be mutually determined.

● **3.4.2.3** When the driver is a motor and automatic capacity control with automatic override control specified, the override control shall prevent overloading the motor. The overriding control shall limit motor current to avoid overcurrent tripping of the motor. Such controls shall include the following:

a. A motor current indicator at the compressor control panel.
b. A current transformer/transducer in the motor switchgear to generate a milliampere signal for the control current indicator.

● **3.4.2.4** For a constant-speed drive, the control signal shall actuate either a purchaser-furnished control valve in the compressor inlet piping or the adjustable inlet guide vanes furnished by the vendor as an integral part of the compressor, as specified. In the latter case, the vendor shall also furnish a guide-vane positioner compatible with the type of control signal specified by the purchaser and shall include a handwheel or other means of local manual override during operation. A direct-driven vane position indicator shall be provided that will be visible during operation of the machine.

**3.4.2.5** For a variable-speed drive, the control signal shall act to adjust the set point of the driver's speed-control system. Unless otherwise specified, the control range limits shall be from maximum continuous speed to 95 percent of the minimum speed required for any specified operating case or 70 percent of the maximum continuous speed, whichever is lower.

**3.4.2.6** The full range of the purchaser's specified control signal shall correspond to the required operating range of the driven equipment. Unless otherwise specified, the maximum control signal shall correspond to the maximum continuous speed or the maximum flow.

### 3.4.3 INSTRUMENT AND CONTROL PANELS

● **3.4.3.1** When specified, a panel shall be provided and shall include all panel-mounted instruments for the driven equipment and the driver. Such panels shall be designed and fabricated in accordance with the purchaser's description. The purchaser will specify whether the panel is to be free-standing, located on the base of the unit, or in another location. The instruments on the panel shall be clearly visible to the operator from the driver control point. A lamp test push button shall be provided. The instruments to be mounted on the panel will be specified by the purchaser.

**3.4.3.2** Panels shall be completely assembled, requiring only connection to the purchaser's external piping and wiring circuits. When more than one wiring point is required on a unit for control or instrumentation, the wiring to each switch or instrument shall be provided from a single terminal box with terminal posts mounted on the unit (or its base if any). Wiring shall be installed in metal conduits or enclosures. All leads and posts on terminal strips, switches, and instruments shall be tagged for identification.

### 3.4.4 INSTRUMENTATION

#### 3.4.4.1 General

When specified, air purging shall be used to avoid moisture problems, even when weatherproof and watertight housings are used. Purge air will be clean and dry conforming to ISA S12.4 Type X or Y and NFPA 496, as specified.

#### 3.4.4.2 Temperature Gauges

**3.4.4.2.1** Dial-type temperature gauges shall be heavy duty and corrosion resistant. They shall be at least 125 millimeters (5 inches) in diameter and bimetallic type or liquid filled. Black printing on a white background is standard for gauges.

**3.4.4.2.2** The sensing elements of temperature gauges shall be in the flowing fluid.

#### 3.4.4.3 Thermowells

Temperature gauges that are in contact with flammable or toxic fluids or that are located in pressurized or flooded lines shall be furnished with $^3/_4$-inch NPT AISI Standard Type 300 stainless steel separable solid-bar thermowells.

#### 3.4.4.4 Thermocouples and Resistance Temperature Detectors

Where practical, the design and location of thermocouples and resistance temperature detectors shall permit replacement while the unit is operating. The lead wires of thermocouples and resistance temperature detectors shall be installed as continuous leads between the thermowell or detector and the terminal box. Conduit runs from thermocouple heads to a pull box or boxes located on the baseplate shall be provided.

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### 3.4.4.5 Pressure Gauges

Pressure gauges (not including built-in instrument air gauges) shall be furnished with AISI Standard Type 316 stainless steel bourdon tubes and stainless steel movements, 114-millimeter ($4^1/_2$-inch) dials [152-millimeter (6-inch) dials for the range over 5500 kPa (800 pounds per square inch)] and $^1/_2$-inch National Pipe Thread (NPT) male alloy steel connections. Black printing on a white background is standard for gauges. When specified, liquid-filled gauges shall be furnished in locations subject to vibration. Gauge ranges shall preferably be selected so that the normal operating pressure is at the middle of the gauge's range. In no case, however, shall the maximum reading on the dial be less than the applicable relief valve setting plus 10 percent. Each pressure gauge shall be provided with a device such as a disk insert or blowout back designed to relieve excess case pressure.

### 3.4.4.6 Solenoid Valves

**3.4.4.6.1** Direct solenoid-operated valves shall be used only in clean, dry instrument-air service, shall have Class F insulation or better, and shall have a continuous service rating. When required for other services, the solenoid shall act as a pilot valve to pneumatic or hydraulic valves.

**3.4.4.6.2** Solenoid valves shall not be used in continuous services that may affect normal operations, such as fuel controls. They may be used in intermittent instrument services, such as starting-cycle controls.

## 3.4.5 ALARMS AND SHUTDOWNS

### 3.4.5.1 General

**3.4.5.1.1** Each alarm switch and each shutdown switch shall be furnished in a separate housing located to facilitate inspection and maintenance. Hermetically sealed, single pole, double-throw switches with a minimum capacity of 5 amperes at 120 volts AC and $^1/_2$ ampere at 120 volts DC shall be used. Mercury switches shall not be used.

**3.4.5.1.2** Unless otherwise specified, electric switches that open (deenergize) to alarm and close (energize) to trip shall be furnished by the vendor.

**3.4.5.1.3** The sequence of alarm and shutdown operation shall be as described in 3.4.5.1.3.1 through 3.4.5.1.3.5.

**3.4.5.1.3.1** Any alarm or shutdown condition shall be initiated by the action of field contacts.

**3.4.5.1.3.2** Alarm and shutdown indications shall consist of corresponding panel-light illumination and sounding of a horn.

**3.4.5.1.3.3** Acknowledgment of the alarm or shutdown condition shall be accomplished by operating a common horn silence push button suitably located on the panel. This shall silence the horn.

**3.4.5.1.3.4** After the alarmed field condition has been corrected, the corresponding panel alarm light shall remain lit until a common panel-mounted reset push button is operated (a manual reset). In the interim, any additional field alarm condition or conditions will again sound the horn and illuminate the corresponding additional panel light or lights.

**3.4.5.1.3.5** Shutdown panel lights shall contain a first-out feature (that is, only the first shutdown light shall be illuminated). After a shutdown occurs, operation of the reset push button shall be required before the unit can be restarted.

**3.4.5.1.4** Connections shall be provided to actuate a remote alarm when any of the locally displayed compressor alarms or shutdowns operate.

● **3.4.5.1.5** When specified, the vendor shall furnish a first out annunciator in accordance with ANSI/ISA-S18.1-1979. The sequence configuration shall be as specified.

### 3.4.5.2 Alarm and Trip Switches

**3.4.5.2.1** Alarm and trip-switch settings shall not be adjustable from outside the housing. Alarm and trip switches shall be arranged to permit testing of the control circuit, including, when possible, the actuating element, without interfering with normal operation of the equipment. The vendor shall provide a clearly visible light on the panel to indicate when trip circuits are in a test bypass mode. Unless otherwise specified, shutdown systems shall be provided with switches or another suitable means to permit testing without shutting down the unit.

**3.4.5.2.2** Pressure-sensing elements shall be of AISI Standard Type 300 stainless steel. Low-pressure alarms shall be equipped with a valved bleed or vent connection to allow controlled depressurizing so that the operator can note the alarm set pressure on the associated pressure gauge. High-pressure alarms shall be equipped with valved test connections so that a portable test pump can be used to raise the pressure.

### 3.4.5.3 Removal of Instruments and Controls

All instruments and controls other than shutdown sensing devices shall be installed with sufficient valving to permit their removal while the system is in operation.

### 3.4.5.4 Housings for Arcing Switches

Particular attention is called to the requirements of 2.1.15 concerning the characteristics of housings for arcing switches outlined in the applicable codes.

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### 3.4.6 ELECTRICAL SYSTEMS

● **3.4.6.1** The characteristics of electrical power supplies for motors, heaters, and instrumentation will be specified. A pilot light shall be provided on the incoming side of each supply circuit to indicate that the circuit is energized. The pilot lights shall be installed on the control panels.

● **3.4.6.2** Electrical equipment located on the unit or on any separate panel shall be suitable for the hazard classification specified (see 2.1.14). Electrical starting and supervisory controls may be either AC or DC.

**3.4.6.3** Power and control wiring within the confines of the baseplate shall be resistant to heat, moisture, and abrasion. Stranded conductors shall be used within the confines of the baseplate and in other areas subject to vibration. Measurement and remote-control panel wiring may be solid conductor. Where elastomer insulation is used, a Neoprene or equivalent high-temperature thermo-plastic sheath shall be provided for insulation protection. Wiring shall be suitable for environmental temperatures.

**3.4.6.4** Unless otherwise specified, all leads on terminal strips, switches, and instruments shall be permanently tagged for identification. All terminal boards in junction boxes and control panels shall have at least 20 percent spare terminal points.

**3.4.6.5** To facilitate maintenance, liberal clearances shall be provided for all energized parts (such as terminal blocks and relays) on equipment. The clearances required for 600 volt service shall also be provided for lower voltages. To guard against accidental contact, enclosures shall be provided for all energized parts.

● **3.4.6.6** Electrical materials including insulation shall be corrosion resistant and nonhygroscopic insofar as is possible. When specified for tropical location, materials shall be given the treatments specified in 3.4.6.6.1 and 3.4.6.6.2.

**3.4.6.6.1** Parts (such as coils and windings) shall be protected from fungus attack.

**3.4.6.6.2** Unpainted surfaces shall be protected from corrosion by plating or another suitable coating.

**3.4.6.7** Control, instrumentation, and power wiring (including temperature element leads) within the limits of the baseplate shall be installed in rigid metallic conduits and boxes, properly bracketed to minimize vibration and isolated or shielded to prevent interference between voltage levels. Conduits may terminate (and in the case of temperature element heads, shall terminate) with a flexible metallic conduit long enough to permit access to the unit for maintenance without removal of the conduit. If temperature element heads will be exposed to temperatures above 60°C (140°F), a 19-millimeter ($^3/_4$-inch) bronze hose with four-wall interlocking construction and joints with packed on (heat-proof) couplings shall be used.

**3.4.6.8** For Division 2 locations, flexible metallic conduits shall have a liquidtight thermosetting or thermoplastic outer jacket. For Division 1 locations, an NFPA-approved connector shall be provided.

### 3.4.7 VIBRATION, POSITION, AND BEARING TEMPERATURE DETECTORS

**3.4.7.1** Unless otherwise specified, radial shaft vibration and axial-position transducers and bearing temperature sensors shall be supplied, installed, and calibrated in accordance with API Standard 670.

● **3.4.7.2** When specified, radial shaft vibration and axial-position monitors shall be supplied and calibrated in accordance with API Standard 670.

● **3.4.7.3** Purchaser will specify type of temperature detector required per 2.7.1.2.

● **3.4.7.4** When specified, a bearing-temperature monitor shall be supplied and calibrated according to API Standard 670 (see 2.7.1.2).

● **3.4.7.5** When specified, casing vibration transducers shall be supplied, installed, and calibrated in accordance with API Standard 670.

● **3.4.7.6** When specified, casing vibration monitors shall be supplied, installed and calibrated in accordance with API Standard 670.

## 3.5 Piping and Appurtences

### 3.5.1 GENERAL

**3.5.1.1** API Standard 614 shall apply to all auxiliary piping provided by the vendor.

**3.5.1.2** Auxiliary systems are defined as piping systems that are in the following services:

a. Instrument and control air.
b. Lubricating oil.
c. Control oil.
d. Seal fluid.
e. Cooling water.
f. Balance gas.
g. Reference gas.
h. Buffer gas.
i. Drains.
j. Liquid injection.

Note: Casing connections are discussed in 2.4.

● **3.5.1.3** When a baseplate has been specified, the vendor shall furnish all piping systems, including mounted appurtenances, located within its confines. The piping shall

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terminate with flanged connections at the edge of the baseplate. When soleplates are specified, the extent of the piping system supplied by the vendor shall be defined by the purchaser. The purchaser will furnish only interconnecting piping between equipment groupings and off base facilities.

**3.5.1.4** Instrument lines shall be provided with shutoff valves at the points of measurement.

● **3.5.1.5** When specified, a liquid injection manifold shall be supplied. It shall include a throttle valve, an armored flow meter, a check valve, a pressure indicator, and a block valve for each injection point.

● **3.5.1.6** When a buffer-gas manifold is specified, the required components such as valves, flow meters, check valves, pressure indicators, throttle valves, differential pressure indicators, controllers, and control valves shall be furnished by the vendor.

**3.5.1.7** Provision for bypassing the bearings (and seals if applicable) of compressors and drivers during oil system flushing operations shall be provided.

### 3.5.2 INSTRUMENT PIPING

Connections on equipment and piping for pressure instruments and test points shall conform to API Standard 614. Beyond the initial $^3/_4$-inch valve, $^1/_2$-inch piping, valves, and fittings may be used. Where convenient, a common connection may be used for remotely mounted instruments that measure the same pressure. Separate secondary $^1/_2$-inch isolating valves are required for each instrument on a common connection. Where a pressure gauge is to be used for testing pressure alarm or shutdown switches, common connections are required for the pressure gauge and switches.

### 3.5.3 PROCESS PIPING

● **3.5.3.1** The extent and requirements for process piping to be supplied by the vendor will be specified.

● **3.5.3.2** When specified, the vendor shall review all piping, appurtenances (intercoolers, aftercoolers, separators, knockout drums, air intake filters, and expansion joints), and vessels immediately upstream and downstream of the equipment and supports. The purchaser and the vendor shall mutually agree on the scope of this review.

Note: Piping arrangements may affect stable operation.

## 3.6 Special Tools

**3.6.1** When special tools and fixtures are required to disassemble, assemble, or maintain the unit, they shall be included in the quotation and furnished as part of the initial supply of the machine. For multiunit installations, the requirements for quantities of special tools and fixtures shall be mutually agreed upon by the purchaser and the vendor. These or similar special tools shall be used during shop assembly and post-test disassembly of the equipment.

**3.6.2** When special tools are provided, they shall be packaged in separate, rugged boxes and marked "Special Tools for (tag/item number)." Each tool shall be stamped or tagged to indicate its intended use.

# SECTION 4—INSPECTION, TESTING, AND PREPARATION FOR SHIPMENT

## 4.1 General

**4.1.1** After advance notification of the vendor by the purchaser, the purchaser's representative shall have entry to all vendor and subvendor plants where manufacturing, testing, or inspection of the equipment is in progress.

**4.1.2** The vendor shall notify subvendors of the purchaser's inspection and testing requirements.

**4.1.3** The vendor shall provide sufficient advance notice to the purchaser before conducting any inspection or test that the purchaser has specified to be witnessed or observed.

● **4.1.4** The purchaser will specify the extent of his participation in the inspection and testing.

**4.1.4.1** The purchaser and the vendor shall meet to coordinate manufacturing hold points and inspectors' visits.

**4.1.4.2** *Witnessed* means that a hold shall be applied to the production schedule and that the inspection or test shall be carried out with the purchaser or his representative in attendance. For mechanical running or performance tests, this requires written notification of a successful preliminary test.

**4.1.4.3** *Observed* means that the purchaser shall be notified of the timing of the inspection or test; however, the inspection or test shall be performed as scheduled, and if the purchaser or his representative is not present, the vendor shall proceed to the next step. (The purchaser should expect to be in the factory longer than for a witnessed test.)

**4.1.5** Equipment for the specified inspection and tests shall be provided by the vendor.

● **4.1.6** When specified, the purchaser's representative, the vendor's representative or both shall indicate compliance in accordance with the inspector's checklist (Appendix H) by initialing, dating, and submitting the completed form before shipment.

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**4.1.7** The purchaser's representative shall have access to the vendor's quality control program for review.

## 4.2 Inspection

### 4.2.1 GENERAL

**4.2.1.1** The vendor shall keep the following data available for at least five years for examination by the purchaser or his representative upon request:

a. Necessary certification of materials, such as mill test reports.
b. Purchase specifications for all items on bills of materials.
c. Test data to verify that the requirements of the specification have been met.
d. Fully identified records of all heat treatment, whether performed in the normal course of manufacture or as part of a repair procedure.
e. Results of quality-control tests and inspections.
f. Running test data (see 4.3.4).
g. Final-assembly maintenance and running clearances.

**4.2.1.2** Pressure-containing parts shall not be painted until the specified inspection of the parts is completed.

● **4.2.1.3** The purchaser will specify the following:

a. Parts that shall be subjected to surface and subsurface examination.
b. The type of examination required, such as magnetic particle, liquid penetrant, radiographic, or ultrasonic examination.

**4.2.1.4** During assembly of the compressor and before testing, each component (including cast in passages of these components) and all piping and appurtenances shall be cleaned by pickling or by another appropriate method to remove foreign materials, corrosion products, and mill scale.

● **4.2.1.5** When specified, the purchaser may inspect for cleanliness of the equipment and all piping and appurtenances furnished by or through the vendor before heads are welded to vessels, openings in vessels or exchangers are closed, or piping is finally assembled.

● **4.2.1.6** When specified, the hardness of parts, welds, and heat-affected zones shall be verified as being within the allowable values by testing of the parts, welds, or heat affected zones. The method, extent, documentation, and witnessing of the testing shall be mutually agreed upon by the purchaser and the vendor.

### ● 4.2.2 MATERIAL INSPECTION

#### 4.2.2.1 General

**4.2.2.1.1** When radiographic, ultrasonic, magnetic particle, or liquid penetrant inspection of welds or materials is specified, the recommended practices in 4.2.2.2 through 4.2.2.5 shall apply unless other procedures are specified by the purchaser. Cast iron may be inspected in accordance with 4.2.2.4 and 4.2.2.5. Welds, cast steel, and wrought material may be inspected in accordance with 4.2.2.2 through 4.2.2.5. The material inspection of pressure-containing parts is covered in 2.11.2 and 2.11.4; impellers, in 2.11.5.

Note: These recommended practices describe examination techniques that are applicable to great varieties of sizes and shapes of materials and widely varying examination requirements. Since the specification for the actual component being inspected depends on metallurgy, component configuration, and method of manufacture, specific procedures and acceptance standards for the application should be covered by written standards, developed by the manufacturer for the specific application.

**4.2.2.1.2** Acceptance standards for 4.2.2.2 through 4.2.2.5 shall be mutually agreed upon between the purchaser and vendor.

#### 4.2.2.2 Radiographic Inspections

Radiography shall be based upon the procedures of ASTM Standard Practices E94.

#### 4.2.2.3 Ultrasonic Inspection

Ultrasonic inspection shall be based upon the procedures ASTM A609 (castings), ASTM A388 (forgings), or ASTM A578 (plate).

#### 4.2.2.4 Magnetic Particle Inspection

Both wet and dry methods of magnetic particle inspection shall be based upon the procedures of ASTM Standard Practice E 709.

#### 4.2.2.5 Liquid Penetrant Inspection

Liquid penetrant inspection shall be based upon the procedures of ASTM E 165.

## 4.3 Testing

### 4.3.1 GENERAL

**4.3.1.1** Equipment shall be tested in accordance with 4.3.2 through 4.3.5. Other tests that may be specified by the purchaser are described in 4.3.6.

Note: See Appendix I for typical gas seal testing considerations.

**4.3.1.2** At least six weeks before the first scheduled test, the vendor shall submit to the purchaser, for his review and comment, detailed procedures for all running tests, including acceptance criteria for all monitored parameters.

**4.3.1.3** The vendor shall notify the purchaser not less than five working days before the date the equipment will be ready for testing. If the testing is rescheduled, the vendor shall notify the purchaser not less than five working days before the new test date.

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### 4.3.2 HYDROSTATIC TEST

**4.3.2.1** Pressure-containing parts (including auxiliaries) shall be tested hydrostatically with liquid at a minimum of 1.5 times the maximum allowable working pressure but not less than 140 kPa gauge (20 pounds per square inch gauge).

**4.3.2.2** Tests shall be maintained for a sufficient period of time to permit complete examination of parts under pressure. The hydrostatic test shall be considered satisfactory when neither leaks nor seepage through the casing or casing joint is observed for a minimum of 30 minutes. Large, heavy castings may require a longer testing period to be agreed upon by the purchaser and the vendor. Seepage past internal closures required for testing of segmented cases and operation of a test pump to maintain pressure are acceptable.

**4.3.2.3** The chloride content of liquids used to test austenitic stainless steel materials shall not exceed 50 parts per million. To prevent deposition of chlorides as a result of evaporative drying, all residual liquid shall be removed from tested parts at the conclusion of the test.

**4.3.2.4** If the part tested is to operate at a temperature at which the strength of a material is below the strength of that material at room temperature, the hydrostatic test pressure shall be multiplied by a factor obtained by dividing the allowable working stress for the material at room temperature by that at operating temperature. The stress values used shall conform to those given in ASME B31.3. For compressor casings and pressure vessels the stress values shall conform to those given in Section VIII, Division 1 or Division 2, as applicable, of the ASME Code. The pressure thus obtained shall then be the minimum pressure at which the hydrostatic test shall be performed. The data sheets shall list actual hydrostatic test pressures.

### 4.3.3 IMPELLER OVERSPEED TEST

Each impeller shall be subjected to an overspeed test at not less than 115 percent of maximum continuous speed for a minimum duration of 1 minute. Impeller dimensions identified by the manufacturer as critical (such as bore, eye seal, and outside diameter) shall be measured before and after each over-speed test. All such measurements and the test speeds shall be recorded and submitted for the purchaser's review following the test. Any permanent deformation of the bore or other critical dimension outside drawing tolerances might be cause for rejection.

### 4.3.4 MECHANICAL RUNNING TEST

**4.3.4.1** The requirements of 4.3.4.1.1 through 4.3.4.1.10 shall be met before the mechanical running test is performed.

**4.3.4.1.1** The contract shaft seals and bearings shall be used in the machine for the mechanical running test. The contract inner seal shall be subjected to design differential pressure during the test. When permitted by the purchaser, the atmospheric breakdown bushing or bushings on an oil-buffered seal may be replaced with a test bushing. (Low-pressure mechanical testing may require greater than design clearance or fewer elements for proper heat removal.)

**4.3.4.1.2** All oil pressures, viscosities, and temperatures shall be within the range of operating values recommended in the vendor's operating instructions for the specific unit being tested. Oil flow rates for each oil supply line shall be determined.

**4.3.4.1.3** Test-stand oil filtration shall be 10 micro-meters (microns) nominal or better. Oil system components downstream of the filters shall meet the cleanliness requirements of API Standard 614 before any test is started.

**4.3.4.1.4** All joints and connections shall be checked for tightness, and any leaks shall be corrected.

**4.3.4.1.5** All warning, protective, and control devices used during the test shall be checked, and adjustments shall be made as required.

**4.3.4.1.6** Facilities shall be installed to prevent the entrance of oil into the compressor during the mechanical running test. These facilities shall be in operation throughout the test.

**4.3.4.1.7** Testing with the contract coupling is preferred. If this is not practical, the mechanical running test shall be performed with a coupling-hub idling-adapter(s) in place, resulting in a moment equal to plus or minus 10 percent that of the contract coupling hub plus one-half that of the coupling spacer. When all testing is completed, idling adapters if used with contract couplings, shall be furnished to the purchaser as part of the special tools.

**4.3.4.1.8** All purchased vibration probes, transducers, oscillator-demodulators, and accelerometers shall be in use during the test. If vibration probes are not furnished by the compressor vendor, or if the purchased probes are not compatible with shop readout facilities, then shop probes and readouts that meet the accuracy requirements of API Standard 670 shall be used.

**4.3.4.1.9** Shop test facilities shall include instrumentation with the capability of continuously monitoring and plotting revolutions per minute, peak-to-peak displacement, and phase angle (x-y-y'). Presentation of vibration displacement and phase marker shall also be by oscilloscope.

**4.3.4.1.10** The vibration characteristics determined by the use of the instrumentation specified in 4.3.4.1.8 and 4.3.4.1.9 shall serve as the basis for acceptance or rejection of the machine (see 2.9.5.5).

**4.3.4.2** The mechanical running test of the equipment shall be conducted as specified in 4.3.4.2.1 through 4.3.4.2.5.

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**4.3.4.2.1** The equipment shall be operated at speed increments of approximately 10 percent from zero to the maximum continuous speed and run at the maximum continuous speed until bearings, lube-oil temperatures, and shaft vibrations have stabilized.

**4.3.4.2.2** The speed shall be increased to trip speed and the equipment shall be run for a minimum of 15 minutes.

**4.3.4.2.3** The speed shall be reduced to the maximum continuous speed, and the equipment shall be run for 4 hours.

Note: Exercise caution when operating equipment at or near critical speeds.

**4.3.4.2.4** The inner seal oil leakage rate shall be measured at each seal. Where, because of design considerations or test-stand conditions, contract seal performance cannot be duplicated during the four-hour test, an additional reduced-speed test or some other acceptable means of demonstrating contract seal performance shall be substituted (see 4.3.4.1.1). Any such alternate procedures shall be described in the vendor's proposed test plan and must be approved in advance by the purchaser.

● **4.3.4.2.5** When specified, lube oil and seal oil inlet pressures and temperatures shall be varied through the range permitted in the compressor operating manual. This shall be done during the four-hour test. This option, when specified, does not constitute a waiver of the other specified test requirements.

**4.3.4.3** During the mechanical running test, the requirements of 4.3.4.3.1 through 4.3.4.3.7 shall be met.

**4.3.4.3.1** During the mechanical running test, the mechanical operation of all equipment being tested and the operation of the test instrumentation shall be satisfactory. The equipment measured unfiltered vibration shall not exceed the limits of 2.9.5.5 and shall be recorded throughout the operating speed range.

**4.3.4.3.2** While the equipment is operating at maximum continuous speed and at other speeds that may have been specified in the test agenda, sweeps shall be made for vibration amplitudes at frequencies other than synchronous. As a minimum, these sweeps shall cover a frequency range from 0.25 to 8 times the maximum continuous speed but not more than 1500 hertz (90,000 cycles per minute). If the amplitude of any discrete, nonsynchronous vibration exceeds 20 percent of the allowable vibration as defined in 2.9.5.5, the purchaser and the vendor shall mutually agree on requirements for any additional testing and on the equipment's suitability for shipment.

● **4.3.4.3.3** Plots showing synchronous vibration amplitude and phase angle versus speed for acceleration and deceleration shall be made before and after the four-hour run. Plots shall be made of both the filtered (one per revolution) and unfiltered vibration levels. When specified, these data shall also be furnished in polar form. The speed range covered by these plots shall be zero to the specified driver trip speed.

**4.3.4.3.4** The mechanical running test shall verify that lateral critical speeds conform to the requirements of 2.9.2 and 2.9.3.

**4.3.4.3.5** Shop verification of the unbalanced response analysis shall be performed in accordance with 2.9.3.

● **4.3.4.3.6** When specified, tape recordings shall be made of all real-time vibration data as mutually agreed upon by the purchaser and the vendor.

● **4.3.4.3.7** When specified, the tape recordings of real-time vibration data shall be given to the purchaser.

**4.3.4.4** Unless otherwise specified, the requirements of 4.3.4.4.1 through 4.3.4.4.4 shall be met after the mechanical running test is completed.

● **4.3.4.4.1** All bearings shall be removed, inspected, and reassembled after the mechanical running test is completed. When specified, shaft end seals shall be removed for inspection following a successful running test.

**4.3.4.4.2** If replacement or modification of bearings or seals or dismantling of the case to replace or modify other parts is required to correct mechanical or performance deficiencies, the initial test will not be acceptable, and the final shop tests shall be run after these replacements or corrections are made. A mechanical retest is not required when the compressor case must be dismantled to comply with the requirements of 4.3.5.

**4.3.4.4.3** If minor scuffs and scratches occur on bearings or shaft end seal surfaces, minor cosmetic repairs of these parts is not a cause for rerunning the test. Shaft end seal contacts that result in a measurable change in dimension or metal transfer may be cause for rejection.

**4.3.4.4.4** When spare rotors are ordered to permit concurrent manufacture, each spare rotor shall also be given a mechanical running test in accordance with the requirements of this standard.

### 4.3.5 ASSEMBLED COMPRESSOR GAS LEAKAGE TEST

**4.3.5.1** After the mechanical running test is completed, each completely assembled compressor casing intended for toxic, hazardous, flammable, or hydrogen-rich service shall be tested as specified in 4.3.5.2 through 4.3.5.4 and subject to a soap-bubble test or another approved test to check for gas leaks.

Note: This test is intended to verify the integrity of the casing joint. Certain shaft seal designs will not be gas tight during this test, and therefore a small leakage from seals may be acceptable.

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**4.3.5.2** The casing (including end seals) shall be pressurized to the maximum sealing pressure or the maximum seal design pressure, with an inert gas as agreed upon by the purchaser and the vendor.

Note: Nitrogen is normally specified as the inert gas used; however, in the case of low mole weight applications, a lighter gas such as helium may be specified.

● **4.3.5.3** When specified, the casing (with or without end seals installed) shall be pressurized to the rated discharge pressure and held at this pressure for a minimum of 30 minutes. The test shall be considered satisfactory when no casing or casing joint leaks are observed.

**4.3.5.4** The requirements of 4.3.5.2 and 4.3.5.3 may necessitate two separate tests.

**4.3.5.5** Immediately upon completion of each witnessed mechanical or performance test, copies of the log data recorded during the test shall be given to the witnesses.

### ● 4.3.6 OPTIONAL TESTS

The purchaser will specify whether any of the following shop tests shall be performed. Test details shall be mutually agreed upon by the purchaser and the vendor.

#### 4.3.6.1 Performance Test

**4.3.6.1.1** The compressor shall be performance tested in accordance with ASME PTC 10. A minimum of five points, including surge and overload, shall be taken at normal speed. For variable-speed machines, additional points may be specified.

Note: Refer to ASME PTC 1 for general instructions.

**4.3.6.1.2** Head and capacity shall have zero negative tolerance at the normal operating point (or other point as specified), and the horsepower at this point shall not exceed 104 percent of the specified value. Surge shall comply with provisions of 2.1.4.

**4.3.6.1.3** For variable-speed compressors, a speed other than the normal speed may be used if necessary to achieve the specified performance and performance tolerances, provided that this adjusted speed meets the criteria specified in 2.9.

**4.3.6.1.4** For constant-speed compressors, the capacity shall be as specified in 4.3.6.1.1. The head shall be within the range of 100 percent to 105 percent of the normal head. The horsepower, based on measured head at normal capacity, shall not exceed 107 percent of the value at the specified normal operating point. If the power required at this point exceeds 107 percent, excess head may be removed by trimming impellers at the purchaser's option.

**4.3.6.1.5** The performance test shall be conducted using only one contract rotor, unless otherwise specified.

● **4.3.6.1.6** Multicase compressor trains and compressors with intermediate process pressures shall have individual sectional head (pressure) tolerances as mutually agreed.

#### 4.3.6.2 Complete–Unit Test

Such components as compressors, gears, drivers, and auxiliaries that make up a complete unit shall be tested together during the mechanical running test. A separate auxiliary test may be performed with the purchaser's approval. When specified, torsional vibration measurements shall be made to verify the vendor's analysis. The complete-unit test shall be performed in place of or in addition to separate tests of individual components specified by the purchaser.

#### 4.3.6.3 Tandem Test

Casings arranged for tandem drive shall be tested as a unit during the mechanical running test by using the shop driver and oil systems.

#### 4.3.6.4 Gear Test

The gear shall be tested with the machine unit during the mechanical running test.

#### 4.3.6.5 Helium Test

The compressor casing shall be tested for gas leakage with helium at the maximum allowable working pressure. The test shall be conducted with the casing submerged in water. The maximum allowable working pressure shall be maintained for a minimum of 30 minutes, with no bubbles permitted. As an alternative, a nonsubmerged soap-bubble test may be performed if approved by the purchaser.

#### 4.3.6.6 Sound–Level Test

The sound-level test shall be performed in accordance with purchaser requirements.

#### 4.3.6.7 Auxiliary–Equipment Test

Auxiliary equipment such as oil systems and control systems shall be tested in the vendor's shop. Details of the auxiliary-equipment tests shall be developed jointly by the purchaser and the vendor.

#### 4.3.6.8 Post–Test Inspection of Compressor Internals

The compressor shall be dismantled, inspected, and reassembled after satisfactory completion of the mechanical running test. The purchaser will specify whether the gas test required by 4.3.5 shall be performed before or after the post-test inspection.

Note: The merits of post-test inspection of compressor internals should be evaluated against the benefits of shipping a unit with proven mechanical assembly and casing joint integrity.

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#### 4.3.6.9 Full–Pressure/Full–Load/Full–Speed Test

The details of the full-pressure/full-load/full-speed test shall be developed jointly by the purchaser and the vendor. This test may be substituted for the mechanical running test.

#### 4.3.6.10 Post–Test Inspection of the Hydraulic Coupling Fit

After the running tests, the shrink fit of hydraulically mounted couplings shall be inspected by comparing hub/shaft match marks to ensure that the coupling hub has not moved on the shaft during the tests.

#### 4.3.6.11 Spare–Parts Test

Spare parts such as rotors, gears, diaphragms, bearings, and seals shall be tested as specified.

## 4.4 Preparation for Shipment

**4.4.1** Equipment shall be suitably prepared for the type of shipment specified, including blocking of the rotor when necessary. Blocked rotors shall be identified by means of corrosion-resistant tags attached with stainless steel wire. The preparation shall make the equipment suitable for six months of outdoor storage from the time of shipment, with no disassembly required before operation, except for inspection of bearings and seals. If storage for a longer period is contemplated, the purchaser will consult with the vendor regarding the recommended procedures to be followed.

**4.4.2** The vendor shall provide the purchaser with the instructions necessary to preserve the integrity of the storage preparation after the equipment arrives at the job site and before startup.

**4.4.3** The equipment shall be prepared for shipment after all testing and inspection have been completed and the equipment has been released by the purchaser. The preparation shall include that specified in 4.4.3.1 through 4.4.3.12. (See Appendix C, Item 44.)

**4.4.3.1** Exterior surfaces, except for machined surfaces, shall be given at least one coat of the manufacturer's standard finish paint. The paint shall not contain lead or chromates.

**4.4.3.2** Exterior machined surfaces shall be coated with a suitable rust preventive.

**4.4.3.3** The interior of the equipment shall be clean and free from scale, welding spatter, and foreign objects. The selection and application of preservatives or rust preventives shall be mutually agreed upon by the purchaser and the vendor.

**4.4.3.4** Internal steel areas of bearing housings and carbon steel oil systems' auxiliary equipment such as reservoirs, vessels, and piping shall be coated with a suitable oil-soluble rust preventive.

**4.4.3.5** Flanged openings shall be provided with metal closures at least 4.8 millimeters ($^3/_{16}$-inch) thick, with elastomer gaskets and at least four full-diameter bolts. For studded openings, all nuts needed for the intended service shall be used to secure closures.

**4.4.3.6** Threaded openings shall be provided with steel caps or solid-shank steel plugs. In no case shall nonmetallic (such as plastic) plugs or caps be used.

**4.4.3.7** Openings that have been beveled for welding shall be provided with closures designed to prevent entrance of foreign materials and damage to the bevel.

**4.4.3.8** Lifting points and lifting lugs shall be clearly identified on the equipment or equipment package. The recommended lifting arrangement shall be identified on boxed equipment.

**4.4.3.9** The equipment shall be identified with item and serial numbers. Material shipped separately shall be identified with securely affixed, corrosion-resistant metal tags indicating the item and serial number of the equipment for which it is intended. In addition, crated equipment shall be shipped with duplicate packing lists, one inside and one on the outside of the shipping container.

**4.4.3.10** When a spare rotor is purchased, the rotor shall be prepared for unheated indoor storage for a period of at least three years. The rotor shall be treated with a rust preventative and shall be housed in a vapor-barrier envelope with a slow release vapor-phase inhibitor. The rotor shall be crated for domestic or export shipment, as specified. Lead sheeting at least 3.2 millimeters ($^1/_8$-inch) thick, or a purchaser-approved equivalent shall be used between the rotor and the cradle at the support areas. The rotor shall not be supported at journals.

● **4.4.3.11** When specified, the fit-up and assembly of machine mounted piping and intercoolers shall be completed in the vendor's shop prior to shipment.

**4.4.3.12** Exposed shafts and shaft couplings shall be wrapped with waterproof, moldable waxed cloth or volatile-corrosion-inhibitor paper. The seams shall be sealed with oil proof adhesive tape.

**4.4.4** Auxiliary piping connections furnished on the purchased equipment shall be impression stamped or permanently tagged to agree with the vendor's connection table or general arrangement drawing.

**4.4.5** Connections on auxiliary piping removed for shipment shall be match marked clearly for reassembly.

**4.4.6** One copy of the manufacturer's standard installation instructions shall be packed and shipped with the equipment.

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# SECTION 5—VENDOR'S DATA

## 5.1 General

**5.1.1** The information to be furnished by the vendor is specified in 5.2. and 5.3. The vendor shall complete and forward the Vendor Drawing and Data Requirements form (see Appendix C), to the address or addresses noted on the inquiry or order. This form shall detail the schedule for transmission of drawings, curves, and data as agreed to at the time of the order, as well as the number and type of copies required by the purchaser.

**5.1.2** The data shall be identified on transmittal (cover) letters and in title blocks or title pages with the following information (see 5.2.1, 5.3.1.1, 5.3.5, and 5.3.6.1):

a. The purchaser/user's corporate name.
b. The job/project number.
c. The equipment item number and service name.
d. The inquiry or purchase order number.
e. Any other identification specified in the inquiry or purchase order.
f. The vendor's identifying proposal number, serial number, or other reference required to identify return correspondence completely.

**5.1.3** A coordination meeting shall be held, preferably at the vendor's plant, within 4 to 6 weeks after the purchase commitment. Unless otherwise specified, the vendor will prepare and distribute an agenda prior to this meeting, which, as a minimum shall include review of the following items:

a. The purchase order, scope of supply, unit responsibility, and sub-vendor items.
b. The data sheets.
c. Applicable specifications and previously agreed upon exceptions.
d. Schedules for transmittal of data, production, and testing.
e. The quality assurance program and procedures.
f. Inspection, expediting, and testing.
g. Schematics and bills of material for auxiliary systems.
h. The physical orientation of the equipment piping, and auxiliary systems.
i. Coupling selections.
j. Thrust-bearing sizing and estimated loadings.
k. The rotor dynamics analysis.
l. Other technical items.

## 5.2 Proposals

### 5.2.1 GENERAL

The vendor shall forward the original proposal and the specified number of copies to the addresses specified in the inquiry documents. As a minimum, the proposal shall include the data specified in 5.2.2 through 5.2.4, as well as a specific statement that the system and all its components are in strict accordance with this standard. If the system and components are not in strict accordance, the vendor shall include a list that details and explains each deviation. The vendor shall provide details to enable the purchaser to evaluate any proposed alternative designs. All correspondence shall be clearly identified in accordance with 5.1.2.

### 5.2.2 DRAWINGS

**5.2.2.1** The drawings indicated on the Vendor Drawing and Data Requirements form (see Appendix C) shall be included in the proposal. As a minimum, the following data shall be furnished:

a. A general arrangement or outline drawing for each major skid or system, showing overall dimensions, maintenance clearance dimensions, overall weights, erection weights, and maximum maintenance weights (indicated for each piece). The direction of rotation and the size and location of major purchaser connections shall also be indicated.
b. Cross-sectional drawings showing the details of the proposed equipment.
c. Schematics of all auxiliary systems, including the fuel, lube-oil, seal-oil, or self-acting gas, control, and electrical systems. Bills of material shall be included.
d. Sketches that show methods of lifting the assembled machine or machines and major components. (This information may be included on the drawings specified in 5.2.2.1, Item a.)

**5.2.2.2** If typical drawings, schematics, and bills of material are used, they shall be marked up to show the correct weight and dimension data and to reflect the actual equipment and scope proposed.

### 5.2.3 TECHNICAL DATA

The following data shall be included in the proposal:

a. The purchaser's data sheets, with complete vendor's information entered thereon and literature to fully describe details of the offering.
b. The purchaser's noise data sheet.
c. The Vendor Drawing and Data Requirements form (see Appendix C), indicating the schedule according to which the vendor agrees to transmit all the data specified as part of the contract.
d. A schedule for shipment of the equipment, in weeks after receipt of the order.
e. A list of major wearing components, showing interchangeability with the purchaser's other units.

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f. A list of recommended start-up spares, including any items that the vendor's experience indicates are likely to be required.
g. A list of the special tools furnished for maintenance. The vendor shall identify any metric items included in the offering.
h. A statement of any special protection required for startup, operation, and periods of idleness under the site conditions specified on the data sheets. The list shall show the protection to be furnished by the purchaser, as well as that included in the vendor's scope of supply.
i. A complete tabulation of utility requirements, such as those for steam, water, electricity, air, gas, and lube oil, including the quantity of lube oil required and the supply pressure, the heat load to be removed by the oil, and the nameplate power rating and operating power requirements of auxiliary drivers. (Approximate data shall be defined and clearly identified as such).
j. A description of the tests and inspection procedures for materials, as required by 2.11.1.4.
k. A description of any special requirements specified in the purchaser's inquiry and as outlined in 2.10.3 and 2.11.1.2.
l. A list of similar machines installed and operating under conditions analogous to those specified in the proposal.
m. Any start-up, shutdown, or operating restrictions required to protect the integrity of the equipment.
n. A summary of the materials of construction for the compressor, including hardness for materials exposed to $H_2S$ (see 2.11.1.7) and a detailed description of the impeller (type of construction, materials, and method of attachment to the shaft).
o. The maximum seal-gas rates (injection or eduction) and rated or expected inner seal-oil leakage rates, if applicable. The inner seal oil leakage shall be given on the basis of volume per day per machine at design gas or oil differential pressures and normal compressor speed. When self-acting gas seals are supplied, expected seal gas consumption, minimum seal gas supply flow, and primary vent flow should be given at the maximum sealing pressure, and at the conditions over the operating envelope of the compressor.
p. When interstage coolers are furnished by the vendor, data for the purchaser's heat and material balances and details of provisions for separating and withdrawing the condensate, including drawings that show cooling system details, and the vendor's recommendations regarding provision for support and piping expansion.
q. Drawings, details, and descriptions of the operations of instrumentation and controls, as well as the makes, materials, and types of auxiliary equipment. The vendor shall also include a complete description of the alarm and shutdown facilities to be provided.
r. The minimum length of straight pipe required for proper flow characteristics at the inlet and at any side connection.
s. Typical sound pressure levels for each item of equipment in the train covered by the proposal.
t. A description of the buffer gas system for oil seals or the barrier gas system for dry seals, when required.
u. Maximum and minimum allowable seal pressures for each compressor.
v. A statement of the manufacturer's capability regarding testing (including performance testing) of the compressor and any other specified items on the train. Details of each optional test specified shall be included.

### 5.2.4 CURVES

The vendor shall provide complete performance curves to fully define the envelope of operations and the point at which the manufacturer has illustrated the equipment. Performance curves shall be submitted for each section (between process nozzles) of each casing as well as an overall curve for the train. All curves shall be marked "PREDICTED."

**5.2.4.1** Curves for variable-speed compressors shall include the following:

a. Discharge pressure; brake horsepower; polytropic head; and polytropic efficiency versus inlet capacity (from predicted surge point to 115 percent rated capacity) at minimum operating speed and 80 percent, 90 percent, 100 percent, and 105 percent speed, and indicating the effect of specified inlet pressures, temperatures, and molecular weights. Any specified operating points shall be noted within the envelope of the performance curve predicted.
b. Steam flow of steam-turbine driven units, first under the conditions specified in 5.2.4.1, Item a, and at normal steam conditions, and then at maximum horsepower under the most adverse steam conditions.
c. Fuel flow for gas-turbine driven units, first under the conditions specified in 5.2.4.1, Item a, and based upon normal fuel and site-ambient conditions, and then at maximum horsepower under the most adverse fuel and site-ambient conditions.

**5.2.4.2** Curves and data for fixed speed compressors shall include the following:

a. Discharge pressure; brake horsepower; polytropic head; and polytropic efficiency versus capacity (from surge point to 115 percent rated capacity) at normal speed, indicating the effect of specified molecular weights, suction pressures, and temperatures. Alternate operating conditions requiring throttling shall be shown.
b. Speed versus torque to overcome friction and windage for the compressor and gear unit, if any, for normal starting conditions (throttled suction) and also with specified suction conditions for emergency starting (open suction).
c. Motor torque versus speed at rated voltage and at 80 percent of rated voltage.

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d. Motor current versus speed at rated voltage and at 80 percent of rated voltage.
e. $WK^2$ of the compressor, the gear, the motor, and the coupling(s) referred to motor speed.
f. Estimated times for acceleration to rated speed for throttled suction and for open suction at 80 percent of the nameplate motor voltage unless otherwise specified.

#### 5.2.5 OPTIONS

When specified, the vendor shall furnish a list of the procedures for any special or optional tests that have been specified by the purchaser or proposed by the vendor.

## 5.3 Contract Data

### 5.3.1 GENERAL

**5.3.1.1** The contract data to be furnished by the vendor is specified in Appendix C. Each drawing, bill of material, and data sheet shall have a title block in its lower right-hand corner that shows the date of certification, a reference to all identification data specified in 5.1.2, the revision number and date, and the title (see 5.3.2 and 5.3.3).

**5.3.1.2** The purchaser will promptly review the vendor's data when he receives them; however, this review shall not constitute permission to deviate from any requirements in the order unless specifically agreed upon in writing. After the data has been reviewed, the vendor shall furnish certified copies in the quantity specified.

**5.3.1.3** A complete list of vendor data shall be included with the first issue of the major drawings. This list shall contain titles, drawing numbers, and a schedule for transmission of all the data the vendor will furnish (see Appendix C).

### 5.3.2 DRAWINGS

The drawings furnished shall contain sufficient information so that with the drawings and the manuals specified in 5.3.6, the purchaser can properly install, operate, and maintain the ordered equipment. Drawings shall be clearly legible, and identified in accordance with 5.3.1.1. As a minimum, the drawings shall include the details listed in Appendix C.

### 5.3.3 TECHNICAL DATA

The data shall be submitted in accordance with Appendix C and identified in accordance with 5.3.1.1. Any comments on the drawings or revisions of specifications that necessitate a change in the data shall be noted by the vendor. These notations will result in the purchaser's issue of completed, corrected data sheets as part of the order specifications.

#### 5.3.3.1 Curves

**5.3.3.1.1** In accordance with the schedule set at the coordination meeting, the vendor shall provide complete performance curves to encompass the map of operations, with any limitations indicated. The curves shall comply with the requirements of 5.3.3.1.2 through 5.3.3.1.6.

**5.3.3.1.2** The compressor serial number shall be shown on all curves.

**5.3.3.1.3** All curves submitted prior to final performance testing (if required) shall be marked "PREDICTED." Any set of curves resulting from a test shall be marked "TESTED."

**5.3.3.1.4** If a performance test is specified, the vendor shall provide test data and curves when the test has been completed. The surge points shall be shown on the performance curves.

● **5.3.3.1.5** When specified, the vendor shall supply a nomograph (quadrant-type performance curve) for each compressor section. These curves typically shall show the influence of variation in speed, inlet volume, inlet pressure, inlet temperature, and molecular weight or composition of the gas on discharge pressure, discharge temperature, gas horsepower, and efficiency. If a performance test is requested in the vendor's plant, the quadrant curves shall be based on the test results. The normal or rated point shall be plotted through each curve.

**5.3.3.1.6** For compressors that have a back-to-back impeller arrangement, the vendor shall furnish a curve showing the expected loading on the active or inactive side of the thrust bearing versus any combination of the differential pressures across the low-pressure and high-pressure sections of the casing.

#### 5.3.3.2 Data Sheets

The vendor shall provide full information to enable completion of the data sheets for the train and auxiliary equipment, first for "as purchased," and then for "as built." This should be done by the vendor correcting and filling out the data sheets and submitting copies to the purchaser.

### ● 5.3.4 PROGRESS REPORTS

The vendor shall submit progress reports to the purchaser at the intervals specified on the Vendor Drawing and Data Requirements form (see Appendix C). The reports shall include engineering, purchasing, and manufacturing schedules for all major components. Planned and actual dates and the percentage completed shall be indicated for each milestone in the schedule.

### 5.3.5 RECOMMENDED SPARE PARTS

The vendor shall submit a complete list of spare parts, including those shown in the original proposal. The list

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shall include spare parts for all equipment and accessories supplied, with cross-sectional or assembly drawings for identification, parts numbers, and delivery times. Parts numbers shall identify each part for purposes of interchangeability. Standard purchased items shall be identified by the original manufacturer's numbers. The vendor shall forward the list to the purchaser promptly after receipt of the reviewed drawings and in time to permit order and delivery of the parts before field start-up. The transmittal letter shall be identified with the data specified in 5.1.2.

### 5.3.6 INSTALLATION, OPERATION, MAINTENANCE, AND TECHNICAL DATA MANUALS

#### 5.3.6.1 General

The vendor shall provide sufficient written instructions, including a list of all drawings, to enable the purchaser to correctly install, operate, and maintain the complete equipment ordered. This information shall be compiled in a manual (or manuals) with a cover sheet containing all identifying data required in 5.1.2 and an index sheet containing section titles and a complete list of referenced and enclosed drawings by title and drawing number. The manual shall be prepared for the specified installation; a typical manual is not acceptable.

#### 5.3.6.2 Installation Manual

All special information required for proper installation design that is not on the drawings shall be compiled in a manual that is separate from the operating and maintenance instructions. This manual shall be forwarded at a time that is mutually agreed upon in the order, but not later than the final issue of drawings. The manual shall contain information such as special alignment or grouting procedures, utility specifications (including quantities), and all installation design data, including the drawings and data specified in Appendix C, Item 36.

#### 5.3.6.3 Operating and Maintenance Manual

The manual containing operating and maintenance data shall be forwarded no more than two weeks after all of the specified tests have been successfully completed. This manual shall include a section that provides special instructions for operation at specified extreme environmental conditions (such as temperature). As a minimum, the manual shall also include all of the data listed in Appendix C, Item 37.

#### ● 5.3.6.4 Technical Data Manual

When specified, the vendor shall provide the purchaser with a technical data manual within 30 days of completion of shop testing (see Appendix C, Item 43).

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# APPENDIX A—TYPICAL DATA SHEETS



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PAGE _____ OF _____

# CENTRIFUGAL COMPRESSOR DATA SHEET CUSTOMARY UNITS

JOB NO. _____ ITEM NO. _____
PURCH. ORDER NO. _____ DATE _____
INQUIRY NO. _____ BY _____
REVISION _____ DATE _____

APPLICABLE TO: ○ PROPOSAL ○ PURCHASE ○ AS BUILT

FOR _____ UNIT _____
SITE _____ SERIAL NO. _____
SERVICE _____ NO. REQUIRED _____
○ CONTINUOUS ○ INTERMITTENT ○ STANDBY DRIVER TYPE (3.1.1) _____
MANUFACTURER _____ MODEL _____ DRIVER ITEM NO. _____
NOTE: INFORMATION TO BE COMPLETED: ○ BY PURCHASER ☐ BY MANUFACTURER

**OPERATING CONDITIONS**

| (ALL DATA ON PER UNIT BASIS) | NORMAL | OTHER CONDITIONS (3.1.2) | | | | |
|---|---|---|---|---|---|---|
| | | A | B | C | D | E |
| ○ GAS HANDLED (ALSO SEE PAGE _____) | | | | | | |
| ○ MMSCFD/SCFM (14.7 PSIA & 60° F DRY) | | | | | | |
| ○ WEIGHT FLOW, #/MIN (WET) (DRY) | | | | | | |
| **INLET CONDITIONS** | | | | | | |
| ○ PRESSURE (PSIA) | | | | | | |
| ○ TEMPERATURE (°F) | | | | | | |
| ○ RELATIVE HUMIDITY % | | | | | | |
| ○ MOLECULAR WEIGHT (%) | | | | | | |
| ☐ Cp/Cv($K_1$) OR ($K_{AVG}$) (NOTE 1) | | | | | | |
| ☐ COMPRESSIBILITY ($Z_1$) OR ($Z_{AVG}$) (NOTE 1) | | | | | | |
| ☐ INLET VOLUME, (CFM) (WET/DRY) | | | | | | |
| **DISCHARGE CONDITIONS** | | | | | | |
| ○ PRESSURE (PSIA) | | | | | | |
| ☐ TEMPERATURE (°F) | | | | | | |
| ☐ Cp/Cv ($K_2$) OR ($K_{AVG}$) (NOTE 1) | | | | | | |
| ☐ COMPRESSIBILITY ($Z_2$) OR ($Z_{AVG}$) (NOTE 1) | | | | | | |
| ☐ BHP REQUIRED (ALL LOSSES INCLUDED) | | | | | | |
| ☐ SPEED (RPM) | | | | | | |
| ☐ ESTIMATED SURGE, ICFM (AT SPEED ABOVE) | | | | | | |
| ☐ POLYTROPIC HEAD (FT-LBS/LB) | | | | | | |
| ☐ POLYTROPIC EFFICIENCY (%) | | | | | | |
| ○ CERTIFIED POINT | | | | | | |
| ☐ PERFORMANCE CURVE NUMBER | | | | | | |

**PROCESS CONTROL**

METHOD ○ SUCTION THROTTLING FROM _____ PSIA TO _____ PSIA
○ VARIABLE INLET GUIDE VANES (3.4.2.4)
○ SPEED VARIATION FROM _____ % TO _____ %
○ DISCHARGE BLOWOFF TO _____
○ COOLED BYPASS FROM _____ TO _____

SIGNAL SOURCE (3.4.2.1) _____
TYPE ○ ELECTRONIC ○ PNEUMATIC ○ OTHER _____
RANGE _____ MA _____ PSIG _____

ANTI-SURGE BYPASS ○ MANUAL ○ AUTOMATIC ○ NONE

REMARKS: _____

NOTE 1: IF GAS ANALYSIS IS GIVEN, MANUFACTURER SHALL SUPPLY DATA, OTHERWISE DATA SHALL BE SUPPLIED BY USER.



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# CENTRIFUGAL COMPRESSOR DATA SHEET CUSTOMARY UNITS

PAGE _____ OF _____
JOB NO. _____ ITEM NO. _____
REVISION _____ DATE _____
BY _____

**OPERATING CONDITIONS (Continued) (3.1.2) (3.1.3)**

| GAS ANALYSIS: ○ MOL % ○ | | NORMAL | OTHER CONDITIONS A | B | C | D | E | REMARKS: (2.11.1.3) (2.11.1.8) |
|---|---|---|---|---|---|---|---|---|
| | MW | | | | | | | |
| AIR | 28.966 | | | | | | | |
| OXYGEN | 32.000 | | | | | | | |
| NITROGEN | 28.016 | | | | | | | |
| WATER VAPOR | 18.016 | | | | | | | |
| CARBON MONOXIDE | 28.010 | | | | | | | |
| CARBON DIOXIDE | 44.010 | | | | | | | |
| HYDROGEN SULFIDE | 34.076 | | | | | | | (2.11.1.7) |
| HYDROGEN | 2.016 | | | | | | | |
| METHANE | 16.042 | | | | | | | |
| ETHYLENE | 28.052 | | | | | | | |
| ETHANE | 30.068 | | | | | | | |
| PROPYLENE | 42.078 | | | | | | | |
| PROPANE | 44.094 | | | | | | | |
| I-BUTANE | 58.120 | | | | | | | |
| n-BUTANE | 58.120 | | | | | | | |
| I-PENTANE | 72.146 | | | | | | | |
| n-PENTANE | 72.146 | | | | | | | |
| HEXANE PLUS | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| TOTAL | | | | | | | | |
| AVG. MOL. WT. | | | | | | | | |

**LOCATION:** (2.1.9)
○ INDOOR ○ OUTDOOR ○ GRADE
○ HEATED ○ UNDER ROOF ○ MEZZANINE
○ UNHEATED ○ PARTIAL SIDES ○ _____
○ ELEC. AREA CLASSIFICATION (2.1.15) CL.____ GR.____ DIV.____
○ WINTERIZATION REQD. (2.1.9) ○ TROPICALIZATION REQD. (3.4.6.6)

**SITE DATA:**
○ ELEVATION _____ FT BAROMETER _____ PSIA
○ RANGE OF AMBIENT TEMPS:

| | DRY BULB | WET BULB |
|---|---|---|
| NORMAL °F | | |
| MAXIMUM °F | | |
| MINIMUM °F | | |
| _____ °F | | |

**UNUSUAL CONDITIONS:** ○ DUST ○ FUMES
OTHER (2.1.9) _____

REMARKS: _____

**NOISE SPECIFICATIONS:** (2.1.10)
○ APPLICABLE TO MACHINE:
SEE SPECIFICATION _____
○ APPLICABLE TO NEIGHBORHOOD
SEE SPECIFICATION _____
ACOUSTIC HOUSING ○ YES ○ NO

**APPLICABLE SPECIFICATIONS:**
API 617, CENTRIFUGAL COMPR. FOR GEN. REFINERY SERV.
○ VENDOR HAVING UNIT RESPONSIBILITY (2.9.1.7)
_____
○ GOVERNING SPECIFICATION (IF DIFFERENT)
_____

PAINTING:
○ MANUFACTURER'S STD.
○ OTHER _____

**SHIPMENT:** (4.4.1)
○ DOMESTIC ○ EXPORT ○ EXPORT BOXING REQD.
○ OUTDOOR STORAGE MORE THAN 6 MONTHS (4.4.1)
SPARE ROTOR ASSEMBLY PACKAGED FOR (4.4.3.10)
○ HORIZONTAL STORAGE ○ VERTICAL STORAGE

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# CENTRIFUGAL COMPRESSOR DATA SHEET CUSTOMARY UNITS

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## CONSTRUCTION FEATURES

☐ **SPEEDS:**
MAX. CONT. _____ RPM TRIP _____ RPM
MAX. TIP SPEEDS: _____ FPS @ 100% SPEED
_____ FPS @ MAX. CONT. SPEED

☐ **LATERAL CRITICAL SPEEDS (DAMPED)**
FIRST CRITICAL _____ RPM _____ MODE
SECOND CRITICAL _____ RPM _____ MODE
THIRD CRITICAL _____ RPM _____ MODE
FOURTH CRITICAL _____ RPM _____ MODE

○ TRAIN LATERAL ANALYSIS REQUIRED (2.9.2.3)
○ TRAIN TORSIONAL ANALYSIS REQUIRED
(TURBINE DRIVEN TRAIN) (2.9.4.5)

**TORSIONAL CRITICAL SPEEDS:**
☐ FIRST CRITICAL _____ RPM
SECOND CRITICAL _____ RPM
THIRD CRITICAL _____ RPM
FOURTH CRITICAL _____ RPM

☐ **VIBRATION:** _____
ALLOWABLE TEST LEVEL _____ MILS
(PEAK TO PEAK)

☐ **ROTATION, VIEWED FROM DRIVEN END** ○ **CW** ○ **CCW**

○ **MATERIALS INSPECTION REQUIREMENTS (4.2.2)**
○ SPECIAL CHARPY TESTING (2.11.3)
○ RADIOGRAPHY REQUIRED FOR _____
○ ULTRASONIC REQUIRED FOR _____
○ MAGNETIC PARTICLE REQUIRED FOR _____
○ LIQUID PENETRANT REQUIRED FOR _____

☐ **CASING:**
MODEL _____
CASING SPLIT _____
MATERIAL _____
THICKNESS (IN.) _____ CORR. ALLOW. (IN.) _____
MAX. WORKING PRESS _____ PSIG
MAX DESIGN PRESS _____ PSIG
TEST PRESS (PSIG): HELIUM _____ HYDRO _____
MAX. OPER. TEMP. _____ °F MIN. OPER. TEMP. _____ °F
MAX. NO. OF IMPELLERS FOR CASING _____
MAX. CASING CAPACITY (ICFM) _____
CASING SPLIT SEALING (2.2.10) _____
○ SYSTEM RELIEF VALVE SET PT. (2.2.4) _____ PSIG

☐ **DIAPHRAGMS:**
MATERIAL _____

☐ **IMPELLERS:**
NO. _____ DIAMETERS _____
NO. VANES EA. IMPELLER _____

TYPE (OPEN, ENCLOSED, ETC.) _____
TYPE FABRICATION _____
MATERIAL _____
MAX. YIELD STRENGTH (PSI) _____
BRINNEL HARDNESS: MAX _____ MIN. _____
SMALLEST TIP INTERNAL WIDTH (IN.) _____
MAX. MACH. NO. @ IMPELLER EYE _____
MAX. IMPELLER HEAD @ 100% SPD (FT-LBS/LB) _____

☐ **SHAFT:**
MATERIAL _____
DIA @ IMPELLERS (IN) _____ DIA @ COUPLING (IN.) _____
SHAFT END: ○ TAPERED ○ CYLINDRICAL
MAX. YIELD STRENGTH (PSI) _____
SHAFT HARDNESS (BNH) (Rc) _____
MAX TORQUE CAPABILITY (FT-LBS) _____

☐ **BALANCE PISTON:**
MATERIAL _____ AREA _____ ($IN^2$)
FIXATION METHOD _____

☐ **SHAFT SLEEVES :**
AT INTERSTG. CLOSE MATL _____
CLEARANCE POINTS
AT SHAFT SEALS _____ MATL _____

○ ACCESSIBLE (2.8.2)

☐ **LABYRINTHS:**
INTERSTAGE
TYPE _____ MATERIAL _____
BALANCE PSITON
TYPE _____ MATERIAL _____

**SHAFT SEALS:**
○ SEAL TYPE: (2.8.3) _____
○ SETTLING OUT PRESSURE (PSIG) _____
○ SPECIAL OPERATION (2.8.1) _____
○ SUPPLEMENTAL DEVICE REQUIRED FOR CONTACT
SEALS (2.8.3.2) TYPE _____
○ BUFFER GAS SYSTEM REQUIRED (2.8.7) ○ MANIFOLD (3.5.1.6)
○ TYPE BUFFER GAS _____
○ BUFFER GAS CONTROL SYSTEM SCHEMATIC BY VENDOR
○ PRESSURIZING GAS FOR SUBATMOSPHERIC SEALS (2.8.8)
☐ TYPE SEAL _____
☐ INNER OIL LEAKAGE GUAR. (GAL/DAY/SEAL) _____
BUFFER GAS REQUIRED FOR:
☐ AIR RUN-IN ☐ OTHER _____

☐ BUFFER GAS FLOW (PER SEAL):
NORM: _____ LBS/MIN @ _____ PSI Δ P
MAX: _____ LBS/MIN @ _____ PSI Δ P

☐ **BEARING HOUSING CONSTRUCTION:**
TYPE (SEPARATE, INTERGRAL) _____ SPLIT _____
MATERIAL _____

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# CENTRIFUGAL COMPRESSOR DATA SHEET CUSTOMARY UNITS

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## CONSTRUCTION FEATURES (CONTINUED)

### BEARINGS AND BEARING HOUSINGS

| RADIAL | INLET | EXHAUST |
|---|---|---|
| ☐ TYPE | | |
| ☐ MANUFACTURER | | |
| ☐ LENGTH (IN) | | |
| ☐ SHAFT DIA (IN) | | |
| ☐ UNIT LOAD (ACT/ALLOW) | | |
| ☐ BASE MATERIAL | | |
| ☐ BABBITT THICKNESS (IN) | | |
| ☐ No. PADS | | |
| ☐ LOAD: B'TWN/ON PAD | | |
| ☐ PIVOT: CTR/OFFSET, % | | |
| ☐ _____ | | |
| ☐ BEARING SPAN _____ INCHES | | |

| THRUST | ACTIVE | INACTIVE |
|---|---|---|
| ☐ TYPE | | |
| ☐ MANUFACTURER | | |
| ☐ UNIT LOADING (MAX PSI) | | |
| ☐ UNIT LOAD (ULT.) (PSI) | | |
| ☐ AREA ($IN^2$) | | |
| ☐ NO. PADS | | |
| ☐ PIVOT: CENTER/OFFSET, % | | |
| ☐ PAD BASE MATL | | |

LUBRICATION: ○ FLOODED ○ DIRECTED
THRUST COLLAR: ○ INTEGRAL ○ REPLACEABLE
MATERIAL _____

**BEARING TEMPERATURE DEVICES** ○ SEE ATTACHED API-670 DATA SHT

○ THERMISTORS
- ○ TYPE _____ POS TEMP COEFF _____ NEG TEMP COEFF
- ○ TEMP SWITCH & INDICATOR BY: _____ PURCH _____ MFR

○ THERMOCOUPLES
- ○ SELECTOR SWITCH & INDIC BY: _____ PURCH _____ MFR

○ RESISTANCE TEMP DETECTORS
- ○ RESISTANCE MAT'L _____ ○ _____ OHMS
- ○ SELECTOR SWITCH & INDICATOR BY: _____ PURCH _____ MFR

○ LOCATION-JOURNAL BRG
No. _____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____

○ LOCATION-THRUST BRG
No. _____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____
No. (INACT) _____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____

○ MONITOR SUPPLIED BY (3.4.7.4) _____
- ○ LOCATION _____ ENCLOSURE _____
- ○ MFR. _____ ☐ MODEL _____
- ☐ SCALE RGE _____ ALARM ☐ SET @ _____ °F
- ○ SHTDWN ☐ SET @ _____ °F ○ TIME DELAY _____ SEC

**VIBRATION DETECTORS:** ○ SEE ATTACHED API-670 DATA SHEET

- ○ TYPE _____ ☐ MODEL _____
- ○ MFR _____
- ○ No. AT EA SHAFT BEARING _____ TOTAL No. _____
- ○ OSCILLATOR-DETECTORS SUPPLIED BY _____
  - ○ MFR _____ ☐ MODEL _____
  - MONITOR SUPPLIED BY (3.4.7.2) _____
  - ○ LOCATION _____ ENCLOSURE _____
  - ○ MFR. _____ ☐ MODEL _____
  - ☐ SCALE RGE ____ ALARM SET @ _____ MILS
  - ○ SHUTDOWN: ☐ SET @ ____ MILS ○ TIME DELAY ___ SEC

**AXIAL POSITION DETECTOR:** ○ SEE ATTACHED API-670 DATA SHEET

- ○ TYPE _____ ☐ MODEL _____
- ○ MFR _____ ○ NO. REQUIRED _____
- ○ OSCILLATOR-DEMODULATOR SUPPLIED BY _____
  - ○ MFR _____ ☐ MODEL _____
  - MONITOR SUPPLIED BY (3.4.7.2) _____
  - ○ LOCATION _____ ENCLOSURE _____
  - ○ MFR. _____ ☐ MODEL _____
  - ☐ SCALE RGE ____ ○ ALARM ☐ SET @ _____ MILS
  - ○ SHTDWN: ☐ SET @ _____ MILS ○ TIME DELAY _____ SEC

### CASING CONNECTIONS

| CONNECTION | ○ DESIGN APPROVAL REQ'D (2.11.2.10) | ☐ ASME ☐ API 605 ☐ OTHER _____ SIZE | ☐ FACING | ○ POSITION (2.4.2.1) | ☐ FLANGED OR STUDDED (2.4.2.1) | ○ MATING FLG & GASKET BY VENDOR (2.4.2.3) | ☐ GAS VELOCITY FT/SEC |
|---|---|---|---|---|---|---|---|
| INLET | | | | | | | |
| DISCHARGE | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

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☐ **OTHER CONNECTIONS**

| SERVICE: | NO. | SIZE | TYPE |
|---|---|---|---|
| LUBE-OIL INLET | | | |
| LUBE-OIL OUTLET | | | |
| SEAL-OIL INLET | | | |
| SEAL-OIL OUTLET | | | |
| SEAL-GAS INLET | | | |
| SEAL-GAS OUTLET | | | |
| CASING DRAINS | | | |
| STAGE DRAINS | | | |
| VENTS | | | |
| COOLING WATER | | | |
| PRESSURE | | | |
| TEMPERATURE | | | |
| SOLVENT INJECTION | | | |
| PURGE FOR: | | | |
| BRG HOUSING | | | |
| BTWN BRG & SEAL | | | |
| BTWN SEAL & GAS | | | |

**ALLOWABLE PIPING FORCES AND MOMENTS:**

| | INLET | | DISCHARGE | | | |
|---|---|---|---|---|---|---|
| | FORCE LB | MOMT FT.LB | FORCE LB | MOMT FT.LB | FORCE LB | MOMT FT.LB |
| AXIAL | | | | | | |
| VERTICAL | | | | | | |
| HORIZ. 90° | | | | | | |
| | FORCE LB | MOMT FT.LB | FORCE LB | MOMT FT.LB | FORCE LB | MOMT FT.LB |
| AXIAL | | | | | | |
| VERTICAL | | | | | | |
| HORIZ. 90° | | | | | | |

○ CASING VIBRATION TRANSDUCER (3.4.7.5)
○ SEE ATTACHED API-670 DATA SHEET
○ TYPE _____ MODEL _____
○ MFR _____ No. REQUIRED _____
○ LOCATION _____
○ OSCILATOR-DEMODULATORS SUPPLIED BY _____
○ MFR _____ ☐ MODEL _____
○ MONITOR SUPPLIED BY (3.4.7.6) _____
○ LOCATION _____ ENCLOSURE _____
○ MFR _____ ☐ MODEL _____
☐ SCALE RANGE _____ ○ ALARM: ☐SET @ ____ IN/SEC²
○ SHTDWN ☐ SET @ ____ IN/SEC² ○ TIME DELAY ____ SEC

○ INDIVIDUAL STAGE DRAINS REQUIRED (2.4.3.2)
○ VALVED & BLINDED
○ VALVED & BLINDED & MANIFOLD

KEY PHASOR REQUIRED
○ COMPRESSOR ○ GEAR ○ DRIVER

## ACCESSORIES

**COUPLINGS AND GUARDS** (3.2)
NOTE: SEE ROTATING ELEMENTS - SHAFT ENDS
○ SEE ATTACHED API-671 DATA SHEET ○ KEYLESS HYDRAULIC ○ KEYED
COUPLING FURNISHED BY _____
MANUFACTURER _____ TYPE _____ MODEL _____
COUPLING GUARD FURNISHED BY: _____
TYPE: ○ FULLY ENCLOSED ○ SEMI-OPEN ○ OTHER

**COUPLING DETAILS**
☐ MAX O.D. _____ IN.
☐ HUB WEIGHT _____ LBS.
☐ SPACER LENGTH _____ IN.
☐ SPACER WEIGHT _____ LBS.

○ VENDOR MOUNT HALF COUPLING
LUBRICATING REQUIREMENTS:
○ NON-LUBE ○ GREASE ○ CONT. OIL LUBE ○ OTHER
QUANTITY PER HUB _____ LBS OR GPM

**MOUNTING PLATES**
○ BASEPLATES: FURNISHED BY (3.3.1.1)
○ COMPRESSOR ONLY (3.3.2.1) ○ DRIVER ○ GEAR
○ OTHER _____
○ DRIP RIM ○ LEVELING PADS (3.3.3.2)
○ COLUMN MOUNTING (3.3.2.3)
○ SUB-SOLE PLATES REQ'D (3.3.3.2)
☐ STAINLESS STEEL SHIM THICKNESS _____ INCHES
PRIMER FOR EPOXY GROUT REQ'D (3.3.1.2.9)
TYPE _____
○ BASE PLATE WILL BE ON CONCRETE FOUNDATION (3.3.2.5)
○ MACHINED MOUNTING PADS REQ'D (3.3.2.6)

○ SOLEPLATES: FURNISHED BY:
☐ THICKNESS _____ IN
○ SUBSOLE PLATES REQ'D (3.3.3.2)
☐ STAINLESS STEEL SHIM THICKNESS - INCHES
○ DRIVER ____ ○ GEAR ____ ○ COMPRESSOR ____
○ PRIMER FOR EPOXY GROUT REQ'D (3.3.1.2.9)
TYPE _____

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## UTILITIES

○ **UTILITY CONDITIONS**

| **STEAM:** | DRIVERS | | HEATING | |
|---|---|---|---|---|
| INLET MIN | _____ PSIG | _____ °F | _____ PSIG | _____ °F |
| NORM | _____ PSIG | _____ °F | _____ PSIG | _____ °F |
| MAX | _____ PSIG | _____ °F | _____ PSIG | _____ °F |
| EXHAUST MIN. | _____ PSIG | _____ °F | _____ PSIG | _____ °F |
| NORM | _____ PSIG | _____ °F | _____ PSIG | _____ °F |
| MAX | _____ PSIG | _____ °F | _____ PSIG | _____ °F |

**ELECTRICITY: (3.4.6.1)**

| | DRIVERS | HEATING | CONTROL | SHUTDOWN |
|---|---|---|---|---|
| VOLTAGE | | | | |
| HERTZ | | | | |
| PHASE | | | | |

**COOLING WATER:**

TEMP INLET _____ °F MAX RETURN _____ °F
PRESS NORM _____ PSIG DESIGN _____ PSIG
MIN RETURN _____ PSIG MAX ALLOW ΔP _____ PSI
WATER SOURCE _____

**INSTRUMENT AIR:**

MAX PRESS _____ PSIG MIN PRESS _____ PSIG

**SHOP INSPECTION AND TESTS: (4.1.4)**

| | REQ'D. |
|---|---|
| CLEANLINESS (4.2.1.5) | ○ |
| HYDROSTATIC | ● |
| IMPELLER OVERSPEED | ● |
| MECHANICAL RUN | ● |
| ○ CONTRACT COUPLING ○ IDLING ADAPTOR(S) | |
| ○ CONTRACT PROBES ○ SHOP PROBES | |
| VARY LUBE & SEAL OIL PRESSURES AND TEMPERATURES (4.3.4.2.5) | ○ |
| POLAR FORM VIB DATA (4.3.4.3.3) | ○ |
| TAPE RECORD VIB DATA (4.3.4.3.6) | ○ |
| TAPE DATA TO PURCHASER (4.3.4.3.7) | ○ |
| SHAFT END SEAL INSP (4.3.4.4.1) | ○ |
| GAS LEAK TEST DISCH PRESS (4.3.5.3) | ○ |
| ○ BEFORE ○ AFTER POST TEST INSPECTION (4.3.6.8) | |
| PERFORMANCE TEST (GAS) (AIR) (4.3.6.1) | ○ |
| COMPLETE UNIT TEST (4.3.6.2) | ○ |
| TORSIONAL VIB MEAS (4.3.6.2) | ○ |
| TANDEM TEST (4.3.6.3) | ○ |
| GEAR TEST (4.3.6.4) | ○ |
| HELIUM LEAK TEST (4.3.6.5) | ○ |
| SOUND LEVEL TEST (4.3.6.6) | ○ |
| FULL LOAD/SPEED/PRESS TEST (4.3.6.9) | ○ |
| HYDRAULIC COUPLING INSP (4.3.6.10) | ○ |
| _____ | ○ |
| _____ | ○ |
| _____ | ○ |
| _____ | ○ |

☐ **TOTAL UTILITY CONSUMPTION:**

| | |
|---|---|
| COOLING WATER | _____ GPM |
| STEAM, NORMAL | _____ LBS/HR |
| STEAM, MAX | _____ LBS/HR |
| INSTRUMENT AIR | _____ SCFM |
| HP (DRIVER) | _____ HP |
| HP (AUXILIARIES) | _____ HP |
| HEATERS | _____ KW |
| PURGE (AIR OR $N_2$) | _____ SCFM |

**MISCELLANEOUS:**

☐ RECOMMENDED STRAIGHT RUN OF PIPE DIAMETERS BEFORE SUCTION _____
○ NOMOGRAPHS REQUIRED FOR EACH SECTION (5.3.3.1.5)
○ VENDOR'S REVIEW & COMMENTS ON PURCHASER'S PIPING & FOUNDATION (3.5.3.2)
○ COMPRESSOR TO BE SUITABLE FOR FIELD RUN IN ON AIR (2.1.17)
○ PROVISION FOR LIQUID INJECTION (2.1.11) _____
○ VENDOR'S REVIEW & COMMENTS ON PURCHASER'S CONTROL SYSTEMS (3.4.1.1)
○ EXTENT OF PROCESS PIPING BY VENDOR (3.5.3.1) _____
○ SHOP FITUP OF VENDOR PROCESS PIPING (4.4.3.11)
○ WELDING HARDNESS TESTING (4.2.1.6)
○ _____

VENDOR'S REPRESENTATIVE SHALL (2.1.14)
○ OBSERVE FLANGE PARTING
○ CHECK ALIGNMENT AT TEMPERATURE
○ BE PRESENT AT INITIAL ALIGNMENT

☐ **WEIGHTS (LB):**

COMPR. _____ GEAR _____ DRIVER _____ BASE _____
ROTORS: COMPR. _____ DRIVER _____ GEAR _____
COMPRESSOR UPPER CASE _____
SOUR SEAL OIL TRAPS _____
LO. CONSOLE _____ S.O. CONSOLE _____
OVERHEAD SEAL OIL TANKS _____
MAX FOR MAINTENANCE (IDENTIFY) _____
TOTAL SHIPPING WEIGHT _____

☐ **SPACE REQUIREMENTS (FT & IN):**

COMPLETE UNIT: L _____ W _____ H _____
LO. CONSOLE: L _____ W _____ H _____
S.O. CONSOLE: L _____ W _____ H _____
SOUR SEAL OIL TRAPS _____
OVERHEAD SEAL OIL TANKS _____

REMARKS:

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APPLICABLE TO: ○ PROPOSAL ○ PURCHASE ○ AS BUILT
FOR _____ UNIT _____
SITE _____ SERIAL NO. _____
SERVICE _____ NO. REQUIRED _____
○ CONTINUOUS ○ INTERMITTENT ○ STANDBY DRIVER TYPE (3.1.1) _____
MANUFACTURER _____ MODEL _____ DRIVER ITEM NO. _____
NOTE: INFORMATION TO BE COMPLETED: ○ BY PURCHASER ☐ BY MANUFACTURER

**OPERATING CONDITIONS**

(ALL DATA ON PER UNIT BASIS)

| | NORMAL | OTHER CONDITIONS (3.1.2) | | | | |
|---|---|---|---|---|---|---|
| | | A | B | C | D | E |
| ○ GAS HANDLED (ALSO SEE PAGE _____) | | | | | | |
| ○ m³/h (1.013 BAR & 0°C DRY) | | | | | | |
| ○ WEIGHT FLOW, kg/h (WET) (DRY) | | | | | | |
| **INLET CONDITIONS** | | | | | | |
| ○ PRESSURE (BAR)(kPa abs) | | | | | | |
| ○ TEMPERATURE (°C) | | | | | | |
| ○ RELATIVE HUMIDITY % | | | | | | |
| ○ MOLECULAR WEIGHT (%) | | | | | | |
| ☐ Cp/Cv($K_1$) OR ($K_{AVG}$) (NOTE 1) | | | | | | |
| ☐ COMPRESSIBILITY ($Z_1$) OR ($Z_{AVG}$) (NOTE 1) | | | | | | |
| ☐ INLET VOLUME, (m³/h) (WET/DRY) | | | | | | |
| **DISCHARGE CONDITIONS** | | | | | | |
| ○ PRESSURE (BAR)(kPa abs) | | | | | | |
| ☐ TEMPERATURE (°C) | | | | | | |
| ☐ Cp/Cv ($K_2$) OR ($K_{AVG}$) (NOTE 1) | | | | | | |
| ☐ COMPRESSIBILITY ($Z_2$) OR ($Z_{AVG}$) (NOTE 1) | | | | | | |
| ☐ kW REQUIRED (ALL LOSSES INCLUDED) | | | | | | |
| ☐ SPEED (RPM) | | | | | | |
| ☐ ESTIMATED SURGE, m³/h (AT SPEED ABOVE) | | | | | | |
| ☐ POLYTROPIC HEAD (N-m/kg) | | | | | | |
| ☐ POLYTROPIC EFFICIENCY (%) | | | | | | |
| ○ CERTIFIED POINT | | | | | | |
| ☐ PERFORMANCE CURVE NUMBER | | | | | | |

**PROCESS CONTROL**

METHOD ○ SUCTION THROTTLING FROM _____ (BAR)(kPa abs) TO _____ (BAR)(kPa abs)
○ VARIABLE INLET GUIDE VANES (3.4.2.4) FROM _____ % TO _____ %
○ SPEED VARIATION
○ DISCHARGE BLOWOFF TO _____
○ COOLED BYPASS FROM _____ TO _____

SIGNAL SOURCE (3.4.2.1) _____
TYPE ○ ELECTRONIC ○ PNEUMATIC ○ OTHER _____
RANGE _____ MA _____ (BAR)(kPa abs) _____

ANTI-SURGE BYPASS ○ MANUAL ○ AUTOMATIC ○ NONE

REMARKS: _____

NOTE 1: IF GAS ANALYSIS IS GIVEN, MANUFACTURER SHALL SUPPLY DATA, OTHERWISE DATA SHALL BE SUPPLIED BY USER.



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**OPERATING CONDITIONS (Continued) (3.1.2) (3.1.3)**

| GAS ANALYSIS: ○ MOL % ○ | | NORMAL | OTHER CONDITIONS A | B | C | D | E | REMARKS: (2.11.1.3) (2.11.1.8) |
|---|---|---|---|---|---|---|---|---|
| | MW | | | | | | | |
| AIR | 28.966 | | | | | | | |
| OXYGEN | 32.000 | | | | | | | |
| NITROGEN | 28.016 | | | | | | | |
| WATER VAPOR | 18.016 | | | | | | | |
| CARBON MONOXIDE | 28.010 | | | | | | | |
| CARBON DIOXIDE | 44.010 | | | | | | | |
| HYDROGEN SULFIDE | 34.076 | | | | | | | (2.11.1.7) |
| HYDROGEN | 2.016 | | | | | | | |
| METHANE | 16.042 | | | | | | | |
| ETHYLENE | 28.052 | | | | | | | |
| ETHANE | 30.068 | | | | | | | |
| PROPYLENE | 42.078 | | | | | | | |
| PROPANE | 44.094 | | | | | | | |
| I-BUTANE | 58.120 | | | | | | | |
| n-BUTANE | 58.120 | | | | | | | |
| I-PENTANE | 72.146 | | | | | | | |
| n-PENTANE | 72.146 | | | | | | | |
| HEXANE PLUS | | | | | | | | |
| | | | | | | | | |
| | | | | | | | | |
| TOTAL | | | | | | | | |
| AVG. MOL. WT. | | | | | | | | |

**LOCATION:** (2.1.9)
○ INDOOR ○ OUTDOOR ○ GRADE
○ HEATED ○ UNDER ROOF ○ MEZZANINE
○ UNHEATED ○ PARTIAL SIDES ○ _____
○ ELEC. AREA CLASSIFICATION (2.1.15) CL.____ GR.____ ZONE ____
○ WINTERIZATION REQD. (2.1.9) ○ TROPICALIZATION REQD. (3.4.6.6)

**SITE DATA:**
○ ELEVATION _____ m BAROMETER _____ BAR
○ RANGE OF AMBIENT TEMPS:

| | DRY BULB | WET BULB |
|---|---|---|
| NORMAL °C | | |
| MAXIMUM °C | | |
| MINIMUM °C | | |
| _____ °C | | |

**UNUSUAL CONDITIONS:** ○ DUST ○ FUMES
OTHER (2.1.9) _____

REMARKS: _____

**NOISE SPECIFICATIONS:** (2.1.10)
○ APPLICABLE TO MACHINE:
SEE SPECIFICATION _____
○ APPLICABLE TO NEIGHBORHOOD
SEE SPECIFICATION _____
ACOUSTIC HOUSING ○ YES ○ NO

**APPLICABLE SPECIFICATIONS:**
API 617, CENTRIFUGAL COMPR. FOR GEN. REFINERY SERV.
○ VENDOR HAVING UNIT RESPONSIBILITY (2.9.1.7)
_____
○ GOVERNING SPECIFICATION (IF DIFFERENT)
_____

PAINTING:
○ MANUFACTURER'S STD.
○ OTHER _____

**SHIPMENT:** (4.4.1)
○ DOMESTIC ○ EXPORT ○ EXPORT BOXING REQD.
○ OUTDOOR STORAGE MORE THAN 6 MONTHS (4.4.1)
SPARE ROTOR ASSEMBLY PACKAGED FOR (4.4.3.10)
○ HORIZONTAL STORAGE ○ VERTICAL STORAGE

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## CONSTRUCTION FEATURES

☐ **SPEEDS:**
MAX. CONT. _____ RPM TRIP _____ RPM
MAX. TIP SPEEDS: _____ FPS @ 100% SPEED
_____ FPS @ MAX. CONT. SPEED

☐ **LATERAL CRITICAL SPEEDS (DAMPED)**
FIRST CRITICAL _____ RPM _____ MODE
SECOND CRITICAL _____ RPM _____ MODE
THIRD CRITICAL _____ RPM _____ MODE
FOURTH CRITICAL _____ RPM _____ MODE

○ TRAIN LATERAL ANALYSIS REQUIRED (2.9.2.3)
○ TRAIN TORSIONAL ANALYSIS REQUIRED
(TURBINE DRIVEN TRAIN) (2.9.4.5)

**TORSIONAL CRITICAL SPEEDS:**
☐ FIRST CRITICAL _____ RPM
SECOND CRITICAL _____ RPM
THIRD CRITICAL _____ RPM
FOURTH CRITICAL _____ RPM

☐ **VIBRATION:** _____
ALLOWABLE TEST LEVEL _____ μ
(PEAK TO PEAK)

☐ **ROTATION, VIEWED FROM DRIVEN END** ○ **CW** ○ **CCW**

○ **MATERIALS INSPECTION REQUIREMENTS (4.2.2)**
○ SPECIAL CHARPY TESTING (2.11.3)
○ RADIOGRAPHY REQUIRED FOR _____
○ ULTRASONIC REQUIRED FOR _____
○ MAGNETIC PARTICLE REQUIRED FOR _____
○ LIQUID PENETRANT REQUIRED FOR _____

☐ **CASING:**
MODEL _____
CASING SPLIT _____
MATERIAL _____
THICKNESS (mm) _____ CORR. ALLOW. (mm) _____
MAX. WORKING PRESS _____ (BARG)(kPaG)
MAX DESIGN PRESS _____ (BARG)(kPaG)
TEST PRESS (BAR)(kPaG): HELIUM _____ HYDRO _____
MAX. OPER. TEMP. _____ °C MIN. OPER. TEMP. _____ °C
MAX. NO. OF IMPELLERS FOR CASING _____
MAX. CASING CAPACITY ($m^3/h$) _____
CASING SPLIT SEALING (2.2.10) _____
○ SYSTEM RELIEF VALVE SET PT. (2.2.4) _____ (BAR)(kPaG)

☐ **DIAPHRAGMS:**
MATERIAL _____

☐ **IMPELLERS:**
NO. _____ DIAMETERS _____
NO. VANES EA. IMPELLER _____

TYPE (OPEN, ENCLOSED, ETC.) _____
TYPE FABRICATION _____
MATERIAL _____
MAX. YIELD STRENGTH (N) _____
BRINNEL HARDNESS: MAX _____ MIN. _____
SMALLEST TIP INTERNAL WIDTH (mm) _____
MAX. MACH. NO. @ IMPELLER EYE _____
MAX. IMPELLER HEAD @ 100% SPD (N-m/kg) _____

☐ **SHAFT:**
MATERIAL _____
DIA @ IMPELLERS (mm) _____ DIA @ COUPLING (mm) _____
SHAFT END: ○ TAPERED ○ CYLINDRICAL
MAX. YIELD STRENGTH (N) _____
SHAFT HARDNESS (BNH) (Rc) _____
MAX TORQUE CAPABILITY (N-m) _____

☐ **BALANCE PISTON:**
MATERIAL _____ AREA _____ ($mm^2$)
FIXATION METHOD _____

☐ **SHAFT SLEEVES :**
AT INTERSTG. CLOSE MATL _____
CLEARANCE POINTS
AT SHAFT SEALS _____ MATL _____

○ ACCESSIBLE (2.8.2)

☐ **LABYRINTHS:**
INTERSTAGE
TYPE _____ MATERIAL _____
BALANCE PSITON
TYPE _____ MATERIAL _____

**SHAFT SEALS:**
○ SEAL TYPE: (2.8.3) _____
○ SETTLING OUT PRESSURE (BAR)(kPaG) _____
○ SPECIAL OPERATION (2.8.1) _____
○ SUPPLEMENTAL DEVICE REQUIRED FOR CONTACT SEALS (2.8.3.2) TYPE _____
○ BUFFER GAS SYSTEM REQUIRED (2.8.7) ○ MANIFOLD (3.5.1.6)
○ TYPE BUFFER GAS _____
○ BUFFER GAS CONTROL SYSTEM SCHEMATIC BY VENDOR
○ PRESSURIZING GAS FOR SUBATMOSPHERIC SEALS (2.8.8)
☐ TYPE SEAL _____
☐ INNER OIL LEAKAGE GUAR. (λ/DAY/SEAL) _____
BUFFER GAS REQUIRED FOR:
☐ AIR RUN-IN ☐ OTHER _____
☐ BUFFER GAS FLOW (PER SEAL):
NORM: _____ kg/MIN @ _____ (BAR)(kPa) Δ P
MAX: _____ kg/MIN @ _____ (BAR)(kPa) Δ P

☐ **BEARING HOUSING CONSTRUCTION:**
TYPE (SEPARATE, INTERGRAL) _____ SPLIT _____
MATERIAL _____

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## CONSTRUCTION FEATURES (CONTINUED)

### BEARINGS AND BEARING HOUSINGS

| RADIAL | INLET | EXHAUST |
|---|---|---|
| ☐ TYPE | | |
| ☐ MANUFACTURER | | |
| ☐ LENGTH (mm) | | |
| ☐ SHAFT DIA (mm) | | |
| ☐ UNIT LOAD (ACT/ALLOW) | | |
| ☐ BASE MATERIAL | | |
| ☐ BABBITT THICKNESS (mm) | | |
| ☐ No. PADS | | |
| ☐ LOAD: B'TWN/ON PAD | | |
| ☐ PIVOT: CTR/OFFSET, % | | |
| ☐ _____ | | |

☐ BEARING SPAN _____ INCHES

| THRUST | ACTIVE | INACTIVE |
|---|---|---|
| ☐ TYPE | | |
| ☐ MANUFACTURER | | |
| ☐ UNIT LOADING (MAX)(BAR) | | |
| ☐ UNIT LOAD (ULT.) (BAR) | | |
| ☐ AREA (mm²) | | |
| ☐ NO. PADS | | |
| ☐ PIVOT: CENTER/OFFSET, % | | |
| ☐ PAD BASE MATL | | |

LUBRICATION: ○ FLOODED ○ DIRECTED
THRUST COLLAR: ○ INTEGRAL ○ REPLACEABLE
MATERIAL _____

**BEARING TEMPERATURE DEVICES** ○ SEE ATTACHED API-670 DATA SHT

○ THERMISTORS
- ○ TYPE ____ POS TEMP COEFF _____ NEG TEMP COEFF
- ○ TEMP SWITCH & INDICATOR BY: ____ PURCH ____ MFR

○ THERMOCOUPLES
- ○ SELECTOR SWITCH & INDIC BY: ____ PURCH ____ MFR

○ RESISTANCE TEMP DETECTORS
- ○ RESISTANCE MAT'L _____ ○ _____ OHMS
- ○ SELECTOR SWITCH & INDICATOR BY: ____ PURCH ____ MFR

○ LOCATION-JOURNAL BRG
No. ____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____

○ LOCATION-THRUST BRG
No. ____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____
No. (INACT) ____ EA PAD _____ EVERY OTH PAD _____ PER BRG
OTHER _____

○ MONITOR SUPPLIED BY (3.4.7.4) _____
- ○ LOCATION _____ ENCLOSURE _____
- ○ MFR. _____ ☐ MODEL _____
- ☐ SCALE RGE _____ ALARM ☐ SET @ _____ °C
- ○ SHTDWN ☐ SET @ _____ °C ○ TIME DELAY _____ SEC

**VIBRATION DETECTORS:** ○ SEE ATTACHED API-670 DATA SHEET
- ○ TYPE _____ ☐ MODEL _____
- ○ MFR _____
- ○ No. AT EA SHAFT BEARING _____ TOTAL No. _____
- ○ OSCILLATOR-DETECTORS SUPPLIED BY _____
  - ○ MFR _____ ☐ MODEL _____
  - MONITOR SUPPLIED BY (3.4.7.2) _____
  - ○ LOCATION _____ ENCLOSURE _____
  - ○ MFR. _____ ☐ MODEL _____
  - ☐ SCALE RGE ____ ALARM SET @ _____ μ
  - ○ SHUTDOWN: ☐ SET @ ____ μ ○ TIME DELAY ___ SEC

**AXIAL POSITION DETECTOR:** ○ SEE ATTACHED API-670 DATA SHEET
- ○ TYPE _____ ☐ MODEL _____
- ○ MFR _____ ○ NO. REQUIRED _____
- ○ OSCILLATOR-DEMODULATOR SUPPLIED BY _____
  - ○ MFR _____ ☐ MODEL _____
  - MONITOR SUPPLIED BY (3.4.7.2) _____
  - ○ LOCATION _____ ENCLOSURE _____
  - ○ MFR. _____ ☐ MODEL _____
  - ☐ SCALE RGE ____ ○ ALARM ☐ SET @ _____ μ
  - ○ SHTDWN: ☐ SET @ ____ μ ○ TIME DELAY ____ SEC

### CASING CONNECTIONS

| CONNECTION | ○ DESIGN APPROVAL REQ'D (2.11.2.10) | ☐ ASME ☐ API 605 ☐ OTHER _____ SIZE | ☐ FACING | ○ POSITION (2.4.2.1) | ☐ FLANGED OR STUDDED (2.4.2.1) | ○ MATING FLG & GASKET BY VENDOR (2.4.2.3) | ☐ GAS VELOCITY m/m |
|---|---|---|---|---|---|---|---|
| INLET | | | | | | | |
| DISCHARGE | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |
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☐ **OTHER CONNECTIONS**

| SERVICE: | NO. | SIZE | TYPE |
|---|---|---|---|
| LUBE-OIL INLET | | | |
| LUBE-OIL OUTLET | | | |
| SEAL-OIL INLET | | | |
| SEAL-OIL OUTLET | | | |
| SEAL-GAS INLET | | | |
| SEAL-GAS OUTLET | | | |
| CASING DRAINS | | | |
| STAGE DRAINS | | | |
| VENTS | | | |
| COOLING WATER | | | |
| PRESSURE | | | |
| TEMPERATURE | | | |
| SOLVENT INJECTION | | | |
| PURGE FOR: | | | |
| BRG HOUSING | | | |
| BTWN BRG & SEAL | | | |
| BTWN SEAL & GAS | | | |

**ALLOWABLE PIPING FORCES AND MOMENTS:**

| | INLET | | DISCHARGE | | | |
|---|---|---|---|---|---|---|
| | FORCE N | MOMT N-m | FORCE N | MOMT N-m | FORCE N | MOMT N-m |
| AXIAL | | | | | | |
| VERTICAL | | | | | | |
| HORIZ. 90° | | | | | | |
| | FORCE N | MOMT N-m | FORCE N | MOMT N-m | FORCE N | MOMT N-m |
| AXIAL | | | | | | |
| VERTICAL | | | | | | |
| HORIZ. 90° | | | | | | |

- ○ INDIVIDUAL STAGE DRAINS REQUIRED (2.4.3.2)
  - ○ VALVED & BLINDED
  - ○ VALVED & BLINDED & MANIFOLD
- KEY PHASOR REQUIRED
  ○ COMPRESSOR ○ GEAR ○ DRIVER
- ○ CASING VIBRATION TRANSDUCER (3.4.7.5)
- ○ SEE ATTACHED API-670 DATA SHEET
- ○ TYPE _____ MODEL _____
- ○ MFR _____ No. REQUIRED _____
- ○ LOCATION _____
- ○ OSCILATOR-DEMODULATORS SUPPLIED BY _____
  - ○ MFR _____ ☐ MODEL _____
- ○ MONITOR SUPPLIED BY (3.4.7.6) _____
  - ○ LOCATION _____ ENCLOSURE _____
  - ○ MFR _____ ☐ MODEL _____
  - ☐ SCALE RANGE _____ ○ ALARM: ☐SET @ ____ mm/SEC²
  - ○ SHTDWN ☐ SET @ ____ mm/SEC² ○ TIME DELAY ____ SEC

## ACCESSORIES

**COUPLINGS AND GUARDS** (3.2)

NOTE: SEE ROTATING ELEMENTS - SHAFT ENDS

○ SEE ATTACHED API-671 DATA SHEET ○ KEYLESS HYDRAULIC ○ KEYED

COUPLING FURNISHED BY _____

MANUFACTURER _____ TYPE _____ MODEL _____

COUPLING GUARD FURNISHED BY: _____

TYPE: ○ FULLY ENCLOSED ○ SEMI-OPEN ○ OTHER

**COUPLING DETAILS**

- ☐ MAX O.D. _____ mm
- ☐ HUB WEIGHT _____ kg
- ☐ SPACER LENGTH _____ mm
- ☐ SPACER WEIGHT _____ kg

○ VENDOR MOUNT HALF COUPLING

LUBRICATING REQUIREMENTS:

○ NON-LUBE ○ GREASE ○ CONT. OIL LUBE ○ OTHER

QUANTITY PER HUB _____ kgs OR l/MIN

**MOUNTING PLATES**

- ○ BASEPLATES: FURNISHED BY (3.3.1.1)
  - ○ COMPRESSOR ONLY (3.3.2.1) ○ DRIVER ○ GEAR
  - ○ OTHER _____
  - ○ DRIP RIM ○ LEVELING PADS (3.3.2.2)
  - ○ COLUMN MOUNTING (3.3.2.3)
  - ○ SUB-SOLE PLATES REQ'D (3.3.3.2)
  - ☐ STAINLESS STEEL SHIM THICKNESS _____ mm
  - PRIMER FOR EPOXY GROUT REQ'D (3.3.1.2.9)
    TYPE _____
  - ○ BASE PLATE WILL BE ON CONCRETE FOUNDATION (3.3.2.5)
  - ○ MACHINED MOUNTING PADS REQ'D (3.3.2.6)
- ○ SOLEPLATES: FURNISHED BY:
  - ☐ THICKNESS _____ mm
  - ○ SUBSOLE PLATES REQ'D (3.3.3.2)
- ☐ STAINLESS STEEL SHIM THICKNESS - mm
  - ○ DRIVER ____ ○ GEAR ____ ○ COMPRESSOR ____
- ○ PRIMER FOR EPOXY GROUT REQ'D (3.3.1.2.9)
  TYPE _____

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## UTILITIES

○ **UTILITY CONDITIONS**

| STEAM: | DRIVERS | HEATING |
|---|---|---|
| INLET MIN | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |
| NORM | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |
| MAX | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |
| EXHAUST MIN. | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |
| NORM | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |
| MAX | _____ BARG (kPaG) ____ °C | _____ BARG (kPaG ____ °C |

**ELECTRICITY: (3.4.6.1)**

| | DRIVERS | HEATING | CONTROL | SHUTDOWN |
|---|---|---|---|---|
| VOLTAGE | | | | |
| HERTZ | | | | |
| PHASE | | | | |

**COOLING WATER:**

TEMP INLET _____ °C MAX RETURN _____ °C

PRESS NORM _____ BARG(kPaG) DESIGN _____ BARG(kPaG)

MIN RETURN _____ BARG(kPaG) MAX ALLOW ΔP _____ BARG(kPa)

WATER SOURCE _____

**INSTRUMENT AIR:**

MAX PRESS _____ BARG(kPa) MIN PRESS _____ BARG(kPa)

**SHOP INSPECTION AND TESTS: (4.1.4)**

| | REQ'D. |
|---|---|
| CLEANLINESS (4.2.1.5) | ○ |
| HYDROSTATIC | ● |
| IMPELLER OVERSPEED | ● |
| MECHANICAL RUN | ● |
| ○ CONTRACT COUPLING | ○ IDLING ADAPTOR(S) |
| ○ CONTRACT PROBES | ○ SHOP PROBES |
| VARY LUBE & SEAL OIL PRESSURES AND TEMPERATURES (4.3.4.2.5) | ○ |
| POLAR FORM VIB DATA (4.3.4.3.3) | ○ |
| TAPE RECORD VIB DATA (4.3.4.3.6) | ○ |
| TAPE DATA TO PURCHASER (4.3.4.3.7) | ○ |
| SHAFT END SEAL INSP (4.3.4.4.1) | ○ |
| GAS LEAK TEST DISCH PRESS (4.3.5.3) | ○ |
| ○ BEFORE ○ AFTER POST TEST INSPECTION (4.3.6.8) | |
| PERFORMANCE TEST (GAS) (AIR) (4.3.6.1) | ○ |
| COMPLETE UNIT TEST (4.3.6.2) | ○ |
| TORSIONAL VIB MEAS (4.3.6.2) | ○ |
| TANDEM TEST (4.3.6.3) | ○ |
| GEAR TEST (4.3.6.4) | ○ |
| HELIUM LEAK TEST (4.3.6.5) | ○ |
| SOUND LEVEL TEST (4.3.6.6) | ○ |
| FULL LOAD/SPEED/PRESS TEST (4.3.6.9) | ○ |
| HYDRAULIC COUPLING INSP (4.3.6.10) | ○ |
| _____ | ○ |
| _____ | ○ |
| _____ | ○ |
| _____ | ○ |

☐ **TOTAL UTILITY CONSUMPTION:**

| | |
|---|---|
| COOLING WATER | _____ m³/h |
| STEAM, NORMAL | _____ kg/h |
| STEAM, MAX | _____ kg/h |
| INSTRUMENT AIR | _____ m³/h |
| HP (DRIVER) | _____ kW |
| HP (AUXILIARIES) | _____ kW |
| HEATERS | _____ kW |
| PURGE (AIR OR $N_2$) | _____ m³/h |

**MISCELLANEOUS:**

☐ RECOMMENDED STRAIGHT RUN OF PIPE DIAMETERS BEFORE SUCTION _____

○ NOMOGRAPHS REQUIRED FOR EACH SECTION (5.3.3.1.5)

○ VENDOR'S REVIEW & COMMENTS ON PURCHASER'S PIPING & FOUNDATION (3.5.3.2)

○ COMPRESSOR TO BE SUITABLE FOR FIELD RUN IN ON AIR (2.1.17)

○ PROVISION FOR LIQUID INJECTION (2.1.11) _____

_____

○ VENDOR'S REVIEW & COMMENTS ON PURCHASER'S CONTROL SYSTEMS (3.4.1.1)

○ EXTENT OF PROCESS PIPING BY VENDOR (3.5.3.1) _____

○ SHOP FITUP OF VENDOR PROCESS PIPING (4.4.3.11)

○ WELDING HARDNESS TESTING (4.2.1.6)

○ _____

VENDOR'S REPRESENTATIVE SHALL (2.1.14)

○ OBSERVE FLANGE PARTING

○ CHECK ALIGNMENT AT TEMPERATURE

○ BE PRESENT AT INITIAL ALIGNMENT

☐ **WEIGHTS (kg):**

COMPR. _____ GEAR _____ DRIVER _____ BASE _____

ROTORS: COMPR. _____ DRIVER _____ GEAR _____

COMPRESSOR UPPER CASE _____

SOUR SEAL OIL TRAPS _____

LO. CONSOLE _____ S.O. CONSOLE _____

OVERHEAD SEAL OIL TANKS _____

MAX FOR MAINTENANCE (IDENTIFY) _____

TOTAL SHIPPING WEIGHT _____

_____

☐ **SPACE REQUIREMENTS (mm):**

COMPLETE UNIT: L _____ W _____ H _____

LO. CONSOLE: L _____ W _____ H _____

S.O. CONSOLE: L _____ W _____ H _____

SOUR SEAL OIL TRAPS _____

OVERHEAD SEAL OIL TANKS _____

REMARKS:

_____

_____

_____

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# APPENDIX B—TYPICAL MATERIAL SPECIFICATIONS FOR MAJOR COMPONENT PARTS



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API STD*617 95 ■ 0732290 0543495 255 ■



## Table B-1—Typical Material Specifications for Major Component Parts

| Part | Material[a] | Specification[b] | Form | Temperature Limits[c] Minimum °C (°F) | Maximum °C (°F) |
|---|---|---|---|---|---|
| Casings | | | | | |
| Cast | Cast iron | ASTM A 278, Class 30 | Cast | –45 (–50) | 230 (450) |
| | | ASTM A 278, Class 40 | Cast | –28 (–20) | 260 (500) |
| | Austenitic cast iron | ASTM A 436, Type 2 | Cast | –45 (–50) | 260 (500) |
| | | ASTM A 571, Type D-2M Classes 1 & 2 | Cast | –195 (–320) | 260 (500) |
| | Ductile Iron | ASTM A 395 | Cast | –28 (–20) | 260 (500) |
| | Cast steel | ASTM A 216, Grade WCB[d] | Cast | –28 (–20) | 400 (750) |
| | | ASTM A 352, Grade LCB | Cast | –45 (–50) | 345 (650) |
| | | ASTM A 352, Grade LC2 | Cast | –75 (–100) | 345 (650) |
| | | ASTM A 352, Grade LC3 | Cast | –100 (–150) | 345 (650) |
| | | ASTM A 352, Grade LC4 | Cast | –115 (–175) | 345 (650) |
| | | ASTM A 217 | Cast | –28 (–20) | 400 (750) |
| | Cast stainless steel | ASTM A 743/744 or A 351, Grades CF3, CF3M, CF8 or CF8M | Cast | –195 (–320) | 345 (650) |
| | Cast Titanium | ASTM B 367, Grades C3 or C4 | Cast | –45 (–50) | 150 (300) |
| Fabricated | Steel | ASTM A 285, Grade C | Plate | –45 (–50) | 345 (650) |
| | | ASTM A 516, Grade 55, 60, 65, 70 | Plate | –45 (–50) | 345 (650) |
| | | ASTM A 203, Grade A or B | Plate | –60 (–75) | 345 (650) |
| | | ASTM A 203, Grade D or E | Plate | –105 (–160) | 345 (650) |
| | | ASTM A 537, Class 1 or 2 | Plate | –60 (–75) | 345 (650) |
| | | ASTM A 353 | Plate | –195 (–320) | 345 (650) |
| | | ASTM A 553, Type I | Plate | –195 (–320) | 345 (650) |
| | | ASTM A 553, Type II | Plate | –170 (–275) | 345 (650) |
| | | ASTM A 266, Class 1 or 4 | Forged | –28 (–20) | 345 (650) |
| | | ASTM A 336, Class F1 | Forged | –28 (–20) | 345 (650) |
| | | ASTM A 414 | Sheet | –28 (–20) | 345 (650) |
| | | ASTM A 508, Class 5a | Forged | –28 (–20) | 345 (650) |
| | Stainless steel | ASTM A 240, Type 304, 304L, 316, 316l or 321 | Plate | –195 (–320) | 345 (650) |
| Diaphragms and guide vanes | Cast iron | ASTM A 48 or A 278, Class 30 | Cast | –195 (–320) | 345 (650) |
| | Ductile iron | ASTM A 536 | Cast | –195 (–320) | 345 (650) |
| | Cast iron | ASTM A 216, Grade WCB | Cast | –195 (–320) | 345 (650) |
| | Steel | ASTM A 283, A284, A285 A516 or A543 | Plate | –195 (–320) | 345 (650) |
| | Stainless steel | ASTM A 743/744 or A351, Grades CA15, CF3, CF3M, CF8 or CF8M | Cast | –195 (–320) | 345 (650) |
| | Aluminum | ASTM B 26, Alloy 355 or C355 | Cast | –195 (–320) | 150 (300) |
| Shaft | Steel | ASTM A 470, Class 1 | Forged | –28 (–20) | 345 (650) |
| | | ASTM A 470, Class 7 | Forged | –115 (–175) | 400 (750) |
| | | AISI Types 1040-1050[e] | Bar or Forged | –28 (–20) | 345 (650) |
| | | AISI Types 4140-4150[e] | Bar or Forged | –28 (–20) | 400 (750) |
| | | AISI Types 2320[e] | Bar or Forged | –110 (–170) | 345 (650) |
| | | ASTM A522, Type 1 | Forged | –195 (–320) | 345 (650) |
| | | ASTM 4340-4345 | Forged | –115 (–175) | 345 (650) |
| | Stainless steel | ASTM A 336, Grade F6 | Forged | –60 (–75) | 345 (650) |
| | | ASTM A 473, Type 410 | Forged | –60 (–75) | 345 (650) |
| | Precipitation hardening stainless steel | ASTM A 705, Types 630 or XM-12 | Forged | –75 (–100) | 345 (650) |

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## Table B-1—Typical Material Specifications for Major Component Parts—continued

| Part | Material[a] | Specification[b] | Temperature Limits[c] Form | Minimum °C (°F) | Maximum °C (°F) |
|---|---|---|---|---|---|
| Impellers | Aluminum | ASTM B 26, Alloy C355 | Cast | –195 (–320) | 150 (300) |
| Cast | Precipitation hardening stainless steel | ASTM A 747, Types CB7CU-1 or CB7CU-2 | Cast | –75 (–100) | 345 (650) |
| | Steel | ASTM A 148 | Cast | –28 (–20) | 345 (650) |
| | | ASTM A 487 Gs 4Q | Cast | –45 (–50) | 345 (650) |
| | Stainless steel | ASTM A 743/744 or A 351, Grade CA15 or CA6NM | Cast | –45 (–50) | 345 (650) |
| | | ASTM A 743/744 or A 351, Grade CF3, CF3m, CF8, CF8M | Cast | –195 (–320) | 345 (650) |
| | Titanium | ASTM B 367, Grade C3 or C4 | Cast | –45 (–50) | 345 (650) |
| Fabricated (covers, hubs, blades) | Steel | AISI Types 4130-4140[e] | Plate or Forged | –28 (–20) | 400 (750) |
| | | AISI Types 4320-4345[e] | Plate or Forged | –115 (–175) | 400 (750) |
| | | AISI Type 3140[e] | Forged | –45 (–50) | 400 (750) |
| | | ASTM A 543 | Plate | –115 (–175) | 400 (750) |
| | | ASTM A 522, Type I | Forged | –145 (–230) | 345 (650) |
| | | ASTM A 522, Type II | Forged | –170 (–275) | 345 (650) |
| | | ASTM A 353 or A583 Type I | Plate | –195 (–320) | 345 (650) |
| | | AISI Type 403[e] | Forged | –60 (–75) | 345 (650) |
| | | ASTM A 473 Type 410 | Forged | –60 (–75) | 345 (650) |
| | | ASTM A 240 Type 304, 304L, 316, 316L | Plate | –195 (–320) | 345 (650) |
| | | ASTM A 473, Type 304, 304L, 316, or 316L | Forged | –195 (–320) | 345 (650) |
| | Precipitation hardening stainless steel | ASTM A 705, Types 630 or XM-12 | Forged | –75 (–100) | 345 (650) |
| | | ASTM A 693, Types 630 or XM-12 | Plate | –75 (–100) | 345 (650) |
| | Ni-Cu | SAE AMS 4646 | Forged | –115 (–175) | 345 (650) |
| | | ASTM B 127 | Plate | –115 (–175) | 345 (650) |
| | | QQ-N-286 | Plate | –115 (–175) | 345 (650) |
| Labyrinths | | | | | |
| Impeller interstage shaft seal and balance piston | Aluminum | ASTM B 26, Alloys 443, 335, 850, A850 or B850 | Cast | –195 (–320) | 315 (600) |
| | | 6061-T6 or 1100 | Plate | –195 (–320) | 315 (600) |
| | Babbitt | ASTM B 23 | Cast | –195 (–320) | 175 (350) |
| | Stainless steel | AISI Type 403, 410, 416, 303, 304 or 316 | Wrought | –195 (–320) | 345 (650) |
| | Cr-Ni-Fe-Mo-Cu-Cb alloy | ASTM B 462 | Wrought | –195 (–320) | 345 (650) |
| | Stainless steel honeycomb | ASTM A 240, Types 304, 304L, 316 or 316L | Fabricated | –195 (–320) | 345 (650) |
| | Ni-Cu alloy | ASTM B 164 | Wrought | –115 (–175) | 345 (650) |
| | Nonmetallic TFE[f] | | Molded | –195 (–320) | 260 (500) |
| | Nonmetallic TFE[f] Carbon-filled | | Molded | –28 (–20) | 260 (500) |
| | Nonmetallic TFE[f] mica-filled | | Molded | –54 (–65) | 260 (500) |
| | Lead | ASTM B 29 | Cast | –101 (–150) | 205 (400) |

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## Table B-1—Typical Material Specifications for Major Component Parts—continued

| Part | Material[a] | Specification[b] | Temperature Limits[c] Form | Minimum °C (°F) | Maximum °C (°F) |
|---|---|---|---|---|---|
| Balance piston | Steel (new) | ASTM A 470, Class 1 | Forged | –28 (–20) | 345 (650) |
| | | ASTM A 470, Class 7 | Forged | –115 (–175) | 400 (750) |
| | | AISI Types, 1040-1050[e] | Forged | –28 (–20) | 345 (650) |
| | | AISI Types, 4130-4145[e] | Forged | –28 (–20) | 400 (750) |
| | | AISI Types, 4330, 4345[e] | Forged | –115 (–175) | 455 (850) |
| | | AISI Type 2320[e] | Forged | –112 (–170) | 345 (650) |
| | | ASTM A 522, Type 1 | Forged | –195 (–320) | 345 (650) |
| | Stainless steel (new) | ASTM A 336, Grade F6 | Forged | –60 (–75) | 345 (650) |
| | | ASTM A 473, Type 410 | Forged | –60 (–75) | 345 (650) |
| | | AISI Type 403 or 410[e] | Forged | –28 (–20) | 345 (650) |
| | Precipitation hardening stainless steel | ASTM A 705, Types 630 or XM-12 | Forged | –75 (–100) | 345 (650) |
| | Ni-Cu alloy | SAE AMS 4676 | Forged | –115 (–175) | 345 (650) |
| Shaft sleeves | Steel | AISI Types 4130-4150[e] | Forged | –45 (–50) | 345 (650) |
| | | AISI Types 4320-4345[e] | Forged | –115 (–175) | 400 (750) |
| | | ASTM A 470, Class 7 | Forged | –115 (–175) | 400 (750) |
| | | ASTM A 522, Type I | Forged | –195 (–320) | 345 (650) |
| | Stainless steel | AISI Types 403 or 410[e] | Forged | –75 (–100) | 400 (750) |
| | Ni-Cu alloy | ASTM B 164 or SAE AMS 4676 | Forged | –115 (–175) | 345 (650) |
| | Ni-Mo-Cr alloy | ASTM B 574, Alloy N10276 | Wrought | –115 (–175) | 345 (650) |
| | | ASTM A 494, Grade CW-12M-1 | Cast | –115 (–175) | 345 (650) |
| | Precipitation hardening stainless steel | A705 Type 630 or XM-12 | Forged | –75 (–100) | 345 (650) |

Notes:
[a]The materials shown in this table are those commonly used by compressor manufacturers, but the list is not all inclusive. Other suitable materials may exist and may be used as indicated by specific design considerations.
[b]Descriptions of AISI Types can be found in ASTM DS 56D. See Note (e).
[c]The temperature limits shown in this table are those commonly observed by compressor manufacturers and are not necessarily the same as any temperature limits specified in the applicable material specifications.
[d]Normalized or normalized and tempered.
[e]AISI designations are only a description of chemical analyses of types of steel; they are not procurement specifications. All materials should be purchased to a specification that adequately defines the required properties and controls.
[f]TFE = tetrafluoroethylene.

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# APPENDIX C—CENTRIFUGAL COMPRESSOR VENDOR DRAWING AND DATA REQUIREMENTS



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# CENTRIFUGAL COMPRESSOR VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PURCHASE ORDER NO. _____ DATE _____
REQUISITION NO. _____ DATE _____
INQUIRY NO. _____ DATE _____
PAGE 1 OF 2 BY _____
FOR _____ REVISION _____
SITE _____ UNIT _____
SERVICE _____ NO. REQUIRED _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

**DISTRIBUTION RECORD**

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

| | | | DESCRIPTION | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1. Certified dimensional outline drawing and list of connections. | | | | | | |
| | | | 2. Cross-sectional drawing and bill of materials. | | | | | | |
| | | | 3. Rotor assembly drawings and bills of materials. | | | | | | |
| | | | 4. Thrust-bearing assembly drawing and bill of materials. | | | | | | |
| | | | 5. Journal-bearing assembly drawings and bills of materials. | | | | | | |
| | | | 6. Shaft-coupling assembly drawings and bills of materials. | | | | | | |
| | | | 7. Seal-oil schematic and bill of materials. | | | | | | |
| | | | 8. Shaft-seal assembly drawing and list of connections. | | | | | | |
| | | | 9. Seal-oil component drawings and data. | | | | | | |
| | | | 10. Lube-oil/control-oil schematics and bills of materials. | | | | | | |
| | | | 11. Lube-oil-system assembly and arrangement drawings and list of connections. | | | | | | |
| | | | 12. Electrical and instrumentation schematics and bills of materials. | | | | | | |
| | | | 13. Electrical and instrumentation assembly drawings and lists of connections. | | | | | | |
| | | | 14. Tabulation of utility requirements. | | | | | | |
| | | | 15. Curves showing polytropic head and efficiency versus ICFM. | | | | | | |
| | | | 16. Curves showing discharge pressure and brake horsepower versus ICFM. | | | | | | |
| | | | 17. Balance line pressure versus thrust load. | | | | | | |
| | | | 18. Speed versus starting torque. | | | | | | |
| | | | 19. Vibration analysis data. | | | | | | |
| | | | 20. Lateral critical analysis report. | | | | | | |
| | | | 21. Torsional critical analysis report. | | | | | | |
| | | | 22. Transient torsional analysis report. | | | | | | |
| | | | 23. Allowable flange loading. | | | | | | |
| | | | 24. Alignment diagram. | | | | | | |
| | | | 25. Weld procedures. | | | | | | |
| | | | 26. Certified hydrostatic test logs. | | | | | | |
| | | | 27. Mechanical running test logs. | | | | | | |
| | | | 28. Performance test logs. | | | | | | |
| | | | 29. Nondestructive test procedures. | | | | | | |
| | | | 30. Procedures for special and optional tests (see 4.3.4). | | | | | | |
| | | | 31. Certified mill test reports. | | | | | | |

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.



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# CENTRIFUGAL COMPRESSOR VENDOR DRAWING AND DATA REQUIREMENTS

JOB NO. _____ ITEM NO. _____
PAGE 2 OF 2 BY _____
DATE _____ REVISION _____

Proposal[a] Bidder shall furnish _____ copies of data for all items indicated by an X.

Review[b] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.

Final[c] Vendor shall furnish _____ copies and _____ transparencies of drawings and data indicated.
Vendor shall furnish _____ operating and maintenance manuals.

**DISTRIBUTION RECORD**

Final—Received from vendor
Final—Due from vendor[c]
Review—Returned to vendor
Review—Received from vendor
Review—Due from vendor[c]

| | | | DESCRIPTION | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | 32. Rotor balancing logs. | | | | | |
| | | | 33. Rotor combined mechanical and electrical runout. | | | | | |
| | | | 34. As-built data sheets. | | | | | |
| | | | 35. As-built dimensions and data. | | | | | |
| | | | 36. Installation manual. | | | | | |
| | | | 37. Operating and maintenance manuals. | | | | | |
| | | | 38. Spare-parts recommendations. | | | | | |
| | | | 39. Engineering, fabrication, and delivery schedule (progress reports). | | | | | |
| | | | 40. List of drawings. | | | | | |
| | | | 41. Shipping list. | | | | | |
| | | | 42. List of special tools furnished for maintenance. | | | | | |
| | | | 43. Technical data manual. | | | | | |
| | | | 44. Material Safety Data Sheets. | | | | | |
| | | | 45. ASME Pressure Vessel Fabrication Data. | | | | | |
| | | | | | | | | |
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| | | | | | | | | |
| | | | | | | | | |

[a]Proposal drawings and data do not have to be certified or as built. Typical data shall be clearly identified as such.
[b]Purchaser will indicate in this column the time frame for submission of materials using the nomenclature given at the end of this form.
[c]Bidder shall complete these two columns to reflect his actual distribution schedule and include this form with his proposal.

Notes:

1. Send all drawings and data to _____
   _____
2. All drawings and data must show project, appropriation, purchase order, and item numbers in addition to the plant location and unit. In addition to the copies specified above, one set of the drawings/instructions necessary for field installation must be forwarded with the shipment.

Nomenclature:

_____ S—number of weeks prior to shipment.
_____ F—number of weeks after firm order.
_____ D—number of weeks after receipt of approved drawings.

Vendor _____
Date _____ Vendor Reference _____
Signature _____
(Signature acknowledges receipt of all instructions)



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## DESCRIPTION

1. Certified dimensional outline drawing and list of connections, including the following:
   a. Size, rating, and location of all customer connections.
   b. Approximate overall handling weights.
   c. Overall dimensions.
   d. Shaft centerline height.
   e. Dimensions of baseplates (if furnished), complete with diameter, number, and locations of bolt holes, thickness of the metal through which the bolts must pass, centers of gravity, and details for foundation design.
   f. The weight and center of gravity of the machine, of the heaviest piece of equipment that must be handled for erection, and of significant items to be handled for maintenance.
   g. Principal dimensions, including those required for the piping design, maintenance clearances, and dismantling clearances, and the maximum loading limit on the main process flanges (both forces and moments).
   h. The direction of rotation of each drive train component.
   i. The size, type, location, and identification of all the purchaser's connections, including vents, drains, lubricating oil, conduits, and instruments. (The vendor's plugged connections shall be identified.)
   j. When couplings are furnished, their make, size, and type; and the style of the coupling guards.
   k. A list of reference drawings.
   l. A list of special weather protection and climitization features.
   m. Cold alignment setting data for equipment furnished by the vendor. A composite diagram or data on expected thermal growth, including transient effects, shall be included.
   n. Completed information to permit adequate foundation design by the purchaser. This shall include, but shall not be limited to, the following—
      1. Grouting details.
      2. The size and location of foundation on bolts.
      3. The weight distribution for each bolt/subsoleplate location.
   o. The location of the center of gravity and rigging provisions to permit removal of the top half of the casing, the rotor, and any subassemblies that weigh more than 300 pounds (136 kilograms).
   p. Equipment furnished by the vendor for mounting by the purchaser.

2. Cross-sectional drawing and bill of materials, including the following:
   a. Journal-bearing clearances and tolerances.
   b. Axial rotor float for all rotors (compressor, gas generator, power turbine).
   c. Shaft end and internal labyrinth seal clearances and tolerances.
   d. Axial position of rotor disks, blades relative to inlet nozzles or vanes, and tolerance allowed.
   e. Outside diameter of all disks (impellers) at the blade tip.

3. Rotor assembly drawings and bills of materials, including the following:
   a. Axial position from the active thrust-collar face to—
      1. Each impeller or rotating disc, inlet side.
      2. Each radial probe.
      3. Each journal-bearing centerline.
      4. Phase-angle notch.
      5. Coupling face or end of shaft.

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b. Thrust-collar assembly details, including—
   1. Collar-shaft, with tolerance.
   2. Concentricity (or axial runout) tolerance.
   3. Required torque for locknut.
   4. Surface finish requirements for collar faces.
   5. Preheat method and temperature requirements for shrunk-on collar installation.
c. Dimensioned shaft ends for coupling mountings.

4. Thrust-bearing assembly drawing and bill of materials.

5. Journal-bearing assembly drawings and bills of materials for all field-maintainable rotors.

6. Shaft-coupling assembly drawings and bills of materials, including the following:
   a. Hydraulic mounting procedure.
   b. Shaft end gap and tolerance.
   c. Coupling guards.
   d. Thermal growth from a baseline of 15°C (59°F).

7. Seal-oil schematic, including the following:
   a. Steady-state and transient oil flows and pressures.
   b. Control, alarm, and trip settings.
   c. Heat loads.
   d. Utility requirements, including electrical, water, and air.
   e. Pipe and valve sizes.
   f. Bill of materials.

8. Shaft-seal assembly and arrangement drawings, including size, rating, and location of all customer connections.

9. Seal-oil component drawings and data, including the following:
   a. Pumps and drivers—
      1. Certified dimensional outline drawings.
      2. Cross-section and bill of materials.
      3. Mechanical seal drawing and bill of materials.
      4. List of recommended spare parts.
      5. Instruction and operating manuals.
      6. Completed data sheets for pumps and drivers.
   b. Overhead tank, reservoir, and drain tanks—
      1. Arrangement drawings.
      2. Maximum, minimum, and normal liquid levels.
      3. Design calculations.
   c. Coolers and filters—
      1. Arrangement drawings.
      2. List of recommended spare parts.
      3. Completed data sheets for coolers.
   d. Instrumentation—
      1. Controllers.
      2. Switches.
      3. Control valves.
      4. Gauges.

10. Lube-oil/control-oil schematics and bills of materials, including the following:
    a. Steady-state and transient oil flows and pressures at each use point.
    b. Control, alarm, and trip settings (pressures and recommended temperatures).

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c. Heat loads at each use point at maximum load.
d. Utility requirements, including electricity, water, and air.
e. Pipe and valve sizes.
f. Instrumentation, safety devices, and control schemes.
g. Lube oil component drawings and data.
h. The vendor shall supply a schematic of a control system for the seal-buffer system when oil seals are supplied; or schematics of a control system for the seal gas system and the barrier gas system when self-acting gas seals are supplied.

11. Lube-oil-system assembly and arrangement drawings, including size, rating, and location of all customer connections.

12. Electrical and instrumentation schematics and bills of materials for all systems. The schematics shall show all alarm and shutdown limits (set points).

13. Electrical and instrumentation arrangement drawings and list of connections.

14. Tabulation of utility requirements.

15. Curves showing polytropic head and polytropic efficiency versus ICFM curves for each section or casing on multiple-section or multiple-casing units, in addition to composite curves at 80 percent, 90 percent, 100 percent, and 105 percent of rated speed.

16. Curves showing discharge pressure and brake horsepower versus ICFM at rated conditions for each section or casing on multiple-section or multiple-casing units, in addition to composite curves at 80 percent, 90 percent, 100 percent, and 105 percent of rated speed. For service with gases of varying molecular weights (MW), curves shall also be furnished at maximum and minimum MW. For air compressors, curves shall also be furnished at three additional specified inlet temperatures.

17. Curve showing the pressure above suction pressure behind the balance drum versus unit loading of the thrust shoes, both in bar (pounds per square inch), using rated conditions as the curve basis. The curve shall extend from a pressure equal to suction pressure behind the drum to a pressure corresponding to at least 34.5 bar (500 pounds per square inch) unit loading on the thrust shoes. Balance drum outside diameter, effective balance drum areas, and expected and recommended maximum allowable pressure behind the balance drum shall be shown on the curve sheet.

18. Speed versus starting torque curve. Speed versus torque curves for the compressor superimposed over the motor-speed-versus-torque starting curve with the motor values based on the specified starting voltage.

19. Vibration analysis data, including the following:
    a. Number of vanes—each impeller.
    b. Number of vanes—each guide vane.

20. Lateral critical analysis report, including, but not limited to, the following:
    a. Complete description of the method used.
    b. Graphic display of critical speeds versus operating speeds.
    c. Graphic display of bearing and support stiffness and its effect on critical speeds.
    d. Graphic display of rotor response to unbalance (including damping).
    e. Journal static loads.
    f. Stiffness and damping coefficients.

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g. Tilting-pad bearing geometry and configuration, including—
   1. Pad angle (arc) and number of pads.
   2. Pivot offset.
   3. Pad clearance (with journal radius, pad bore radius, and bearing-set bore radius, and bearing-set bore radius.
   4. Preload.

21. Torsional critical analysis report, including, but not limited to, the following:
   a. Complete description of the method used.
   b. Graphic display of the mass elastic system.
   c. Tabulation identifying the mass moment and torsional stiffness of each component identified in the mass elastic system.
   d. Graphic display of exciting forces versus speed and frequency.
   e. Graphic display of torsional critical speeds and deflections (mode-shape diagram).
   f. Effects of alternative coupling on analysis.

22. Transient torsional analysis for all units using synchronous starter/helper motors (mandatory) or driving synchronous generators (optional).

23. Allowable flange loadings for all customer's connections, including anticipated thermal movements referenced to a defined point.

24. Coupling alignment diagram, including recommended coupling limits during operation. Note all shaft-end position changes and support growth from a reference ambient temperature of 15°C (59°F) or another temperature specified by the purchaser. Include the recommended alignment method and cold setting targets.

25. Welding procedures for fabrication and repair.

26. Certified hydrostatic test logs.

27. Mechanical running test logs, including, but not limited to, the following:
   a. Oil flows, pressures, and temperatures.
   b. Vibration, including an *x–y* plot of amplitude and phase angle versus revolutions per minute during start-up and coast-down.
   c. Bearing metal temperatures.
   d. Observed critical speeds (for flexible rotors).
   e. When specified, tape recordings of real-time vibration data.

28. Performance test logs and reports in accordance with ASME PTC 10.

29. Nondestructive test procedures as itemized on the purchase order data sheets or the Vendor Drawing and Data Requirements form.

30. Procedures for any special or optional tests (see 4.3.6).

31. Certified mill test reports of items as agreed upon in the precommitment or preinspection meetings.

32. Rotor balancing logs, including a residual unbalance report in accordance with Appendix B.

33. Rotor combined mechanical and electrical runout in accordance with 2.9.5.6.

34. As-built data sheets.

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35. As-built dimensions (including nominal dimensions with design tolerances) and data for the following listed parts:
    a. Shaft or sleeve diameters at—
        1. Thrust collar (for separate collars).
        2. Each seal component.
        3. Each wheel (for stacked rotors) or bladed disks.
        4. Each interstage labyrinth.
        5. Each journal bearing.
    b. Each wheel or disk bore (for stacked rotors) and outside diameter.
    c. Each labyrinth or seal-ring bore.
    d. Thrust-collar bore (for separate collars).
    e. Each journal-bearing inside diameter.
    f. Thrust-bearing concentricity (axial runout).
    g. Metallurgy and heat treatment for—
        1. Shaft.
        2. Impellers or bladed disks.
        3. Thrust collar.
        4. Blades, vanes, and nozzles.

36. Installation manual describing the following (see 5.3.6.2):
    a. Storage procedures.
    b. Foundation plan.
    c. Grouting details.
    d. Setting equipment, rigging procedures, component weights, and lifting diagrams.
    e. Coupling alignment diagram (as specified in Appendix C, Item 24).
    f. Piping recommendations, including allowable flange loads.
    g. Composite outline drawings for the driver/driven-equipment train, including anchor-bolt locations.
    h. Dismantling clearances.

37. Operating and maintenance manuals describing the following:
    a. Start-up.
    b. Normal shutdown.
    c. Emergency shutdown.
    d. List of undesirable speeds (see 2.9.1.7).
    e. Lube-oil recommendations.
    f. Routine operational procedures, including recommended inspection schedules and procedures.
    g. Instructions for—
        1. Disassembly and reassembly of rotor in casing.
        2. Rotor unstacking and restacking procedures.
        3. Disassembly and reassembly of journal bearings (for tilting-pad bearings, the instructions shall include "go/no-go" dimensions with tolerances for three-step plug gauges).
        4. Disassembly and reassembly of thrust bearing.
        5. Disassembly and reassembly of seals (including maximum and minimum clearances).
        6. Disassembly and reassembly of thrust collar.
    h. Performance data, including—
        1. Polytropic head and polytropic efficiency versus ICFM.
        2. Discharge pressure and brake horsepower versus ICFM.
        3. Balance Drum $\Delta P$ versus thrust loading.
        4. Speed versus starting torque.

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i. Vibration analysis data, as specified in Appendix C, Items 19 to 22.
j. As-built data, including—
   1. As-built data sheets.
   2. As-built dimensions or data, including assembly clearances.
   3. Hydrostatic test logs, as specified in Appendix C, Item 26.
   4. Mechanical running test logs, as specified in Appendix C, Item 27.
   5. Rotor balancing logs, as specified in Appendix C, Item 32.
   6. Rotor mechanical and electrical runout at each journal, as specified in Appendix C, Item 33.
k. Drawings and data, including—
   1. Certified dimensional outline drawing and list of connections.
   2. Cross-sectional drawing and bill of materials.
   3. Rotor assembly drawings and bills of materials.
   4. Thrust-bearing assembly drawing and bill of materials.
   5. Journal-bearing assembly drawings and bills of materials.
   6. Seal-component drawing and bill of materials.
   7. Lube-oil schematics and bills of materials.
   8. Lube-oil assembly drawing and list of connections.
   9. Lube-oil component drawings and data.
   10. Electrical and instrumentation schematics and bills of materials.
   11. Electrical and instrumentation assembly drawings and list of connections.
l. The following maintenance information—
   1. Maximum and minimum bearing, labyrinth, and seal clearances.
   2. Instructions for measuring and adjusting cold clearances.
   3. Rotor float allowance.
   4. Interference fits on parts that are required to be removed or replaced for maintenance or normally consumable spares.
   5. Runout and concentricity tolerances on parts of assembled rotors.
   6. Balancing tolerances.
m. The following reassembly information—
   1. Bolting sequence and torque values for such items as casing bolting and internal bolting.
   2. Reassembly sequences together with required inspection checks.
   3. Adjustment procedures to achieve required positions, clearances, and float.
   4. Detailed procedures for preoperational checks, including settings and adjustments.
   5. Coupling installation procedures.

38. a. Pattern, stock, or production numbers and materials or construction.
    b. Identification of each part to determine interchangeability.

39. Progress reports and delivery schedule, including vendor buy-outs and milestones.

40. List of drawings, including latest revision numbers and dates.

41. Shipping list, including all major components that will ship separately and the preservation, package, and shipping procedure.

42. List of special tools furnished for maintenance (see 3.6).

43. Technical data manual, including the following:
    a. As-built purchaser data sheets, as specified in Appendix C, Item 34.
    b. Certified performance curves, as specified in Appendix C, Items 15 and 16.
    c. Drawings, in accordance with 5.3.2.

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d. As-built assembly clearances.
e. Spare parts list, in accordance with 5.3.5.
f. Utility data, as specified in Appendix C, Item 14.
g. Vibration data, as specified in Appendix C, Item 19.
h. Reports, as specified in Appendix C, Items 20, 21, 22, 24, 27, 28, 32 and 33.

44. Material Safety Data Sheets (OSHA Form 20).

45. ASME pressure vessel fabrication data, including the manufacturer's data report, rubbings of the code stamp, stress relief charts, and mill test reports.

Note: Items 8, 12, 13, 14, 26(f), and 42(k.7–k.9) are required only for a turbine manufacturer's scope of supply.

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# APPENDIX D—PROCEDURE FOR DETERMINATION OF RESIDUAL UNBALANCE

## D.1 Scope

This appendix describes the procedure to be used to determine residual unbalance in machine rotors. Although some balancing machines may be set up to read out the exact amount of unbalance, the calibration can be in error. The only sure method of determining residual unbalance is to test the rotor with a known amount of unbalance.

## D.2 Definition

Residual unbalance is the amount of unbalance remaining in a rotor after balancing. Unless otherwise specified, it shall be expressed in gram-millimeters or ounces-inches.

## D.3 Maximum Allowable Residual Unbalance

**D.3.1** The maximum allowable residual unbalance per plane shall be calculated using Equation 5 in 2.9.5.2 of this standard.

**D.3.2** If the actual static weight load on each journal is not known, assume that the total rotor weight is equally supported by the bearings. For example, a two-bearing rotor weighting 6000 pounds would be assumed to impose a static weight load of 3000 pounds on each journal.

## D.4 Residual Unbalance Check

### D.4.1 GENERAL

**D.4.1.1** When the balancing machine readings indicate that the rotor has been balanced to within the specified tolerance, a residual unbalance check shall be performed before the rotor is removed from the balancing machine.

**D.4.1.2** To check residual unbalance, a known trial weight is attached to the rotor sequentially in six (or twelve, if specified by the purchaser) equally spaced radial positions, each at the same radius. The check is run in each correction plane, and the readings in each plane are plotted on a graph, using the procedure specified in D.4.2.

### D.4.2 PROCEDURE

**D.4.2.1** Select a trial weight and a radius that will be equivalent to between one and two times the maximum allowable residual unbalance (that is, if $U_{max}$ is 2 ounces-inches, the trial weight should cause 2 to 4 ounces-inches of unbalance).

**D.4.2.2** Starting at the last known heavy spot in each correction plane, mark off the specified number of radial positions (six or twelve) in equal (60- or 30-degree) increments around the rotor. Add the trial weight to the last known heavy spot in one plane. If the rotor has been balanced very precisely and the final heavy spot cannot be determined, add the trial weight to any one of the marked radial positions.

**D.4.2.3** To verify that an appropriate trial weight has been selected, operate the balancing machine and the note units of unbalance indicated on the meter. If the meter pegs, a smaller trial weight should be used. If little or no meter reading results, a larger trial weight should be used. Little or no meter reading generally indicates that the rotor was not balanced correctly, the balancing machine was not sensitive enough, or that a balancing machine fault exists (i.e., a faulty pickup). Whatever the error, it must be corrected before proceeding with the residual unbalance check.

**D.4.2.4** Locate the weight at each of the equally spaced positions in turn, and record the amount of unbalance indicated on the meter for each position. Repeat the initial position as a check. All verification shall be performed using only one sensitivity range on the balance machine.

**D.4.2.5** Plot the readings on the residual unbalance work sheet and calculate the amount of residual unbalance (see Figure D-1). The maximum meter reading occurs when the trial weight is added at the rotor's heavy spot; the minimum reading occurs when the trial weight is opposite the heavy spot. Thus, the plotted readings should form an approximate circle (see Figure D-2). An average of the maximum and minimum meter readings represents the effect of the trial weight. The distance of the circle's center from the origin of the polar plot represents the residual unbalance in that plane.

**D.4.2.6** Repeat steps described in D.4.2.1 through D.4.2.5 for each balance plane. If the specified maximum allowable residual unbalance has been exceeded in any balance plane, the rotor shall be balanced more precisely and checked again. If a correction is made in any balance plane, the residual unbalance check shall be repeated in all planes.

**D.4.2.7** For stack component balanced rotors, a residual unbalance check shall be performed after the addition and balancing of the first rotor component, and at the completion of balancing the entire rotor, as a minimum.

Note: This ensures that time is not wasted and rotor components are not subjected to unnecessary material removal in attempting to balance a multiple component rotor with a faulty balancing machine.



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Equipment (rotor) no.: _____

Purchase order no.: _____

Correction plane (inlet, drive-end, etc.—use sketch): _____

Balancing speed: _____ rpm

*N*—Maximum allowable rotor speed: _____ rpm

*W*—Weight of journal (closest to this correction plane): _____ lbs.

$U_{max}$ = Maximum allowable residual unbalance =
4 × *W*/*N* (6350 *W*/*N*)
4 × _____ lbs./ _____ rpm _____ oz.-in. (gm-mm)

Trial unbalance (2 × $U_{max}$) _____ oz.-in. (gm-mm)

*R*—Radius (at which weight will be placed): _____ inches

Trial unbalance weight = trial unbalance/*R*
_____ oz.-in./_____ inches = _____ oz. (gm)

Conversion information: 1 ounce = 28.375 grams

**TEST DATA**

| Position | Amplitude | Angular Position |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| Repeat 1 | | |

**ROTOR SKETCH**

**TEST DATA—GRAPHIC ANALYSIS**

Step 1: Plot data on the polar chart (Figure D-1—continued). Scale the chart so the largest and smallest amplitude will fit conveniently.

Step 2: With the compass, draw the best fit circle through the six points and mark the center of this circle.

Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. _____ units

Step 4: Record the trial unbalance from above. _____ oz.-in. (gm-mm)

Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). _____ oz.-in. (gm-mm)

Step 6: Divide the answer in Step 5 by the answer in Step 3. _____ Scale Factor

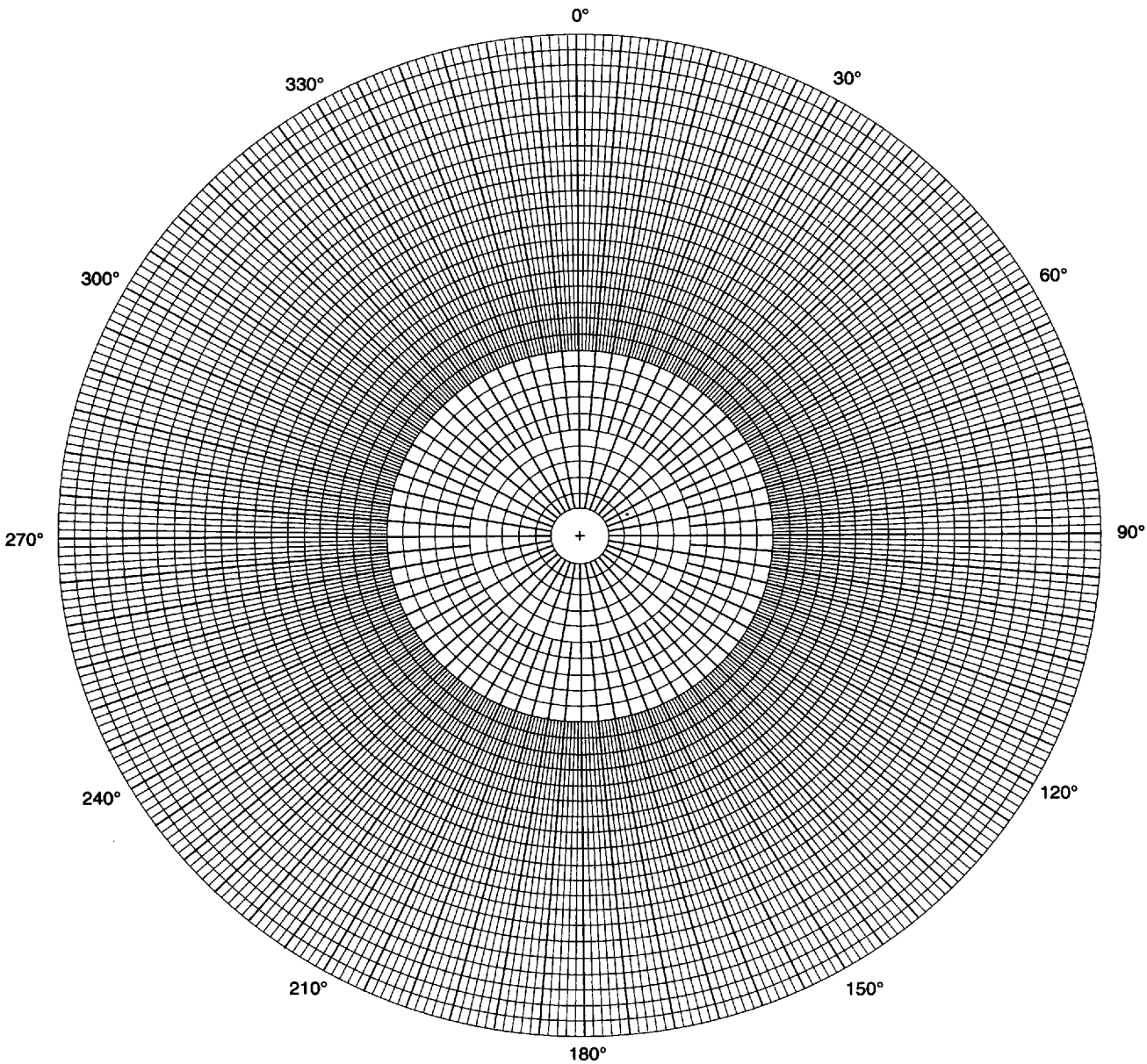
You now have a correlation between the units in the polar chart and the gm-in. of actual balance.

Figure D-1—Residual Unbalance Work Sheet

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance. Proceed with the balancing machine sensitivity check before rebalancing is attempted.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual balance.

Record actual residual unbalance _____ (oz.-in.)(gm-mm)

Record allowable residual unbalance (from Figure D-1) _____ (oz.-in.)(gm-mm)

Correction plane _____ for Rotor No. _____ (has/has not) passed.

By _____ Date _____

Figure D-1—continued

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*Equipment (rotor) no.:* C-101

Purchase order no.: _____

Correction plane (inlet, drive-end, etc.—use sketch): A

Balancing speed: 800 rpm

*N*—Maximum allowable rotor speed: 10,000 rpm

*W*—Weight of journal (closest to this correction plane): 908 lbs.

$U_{max}$ = Maximum allowable residual unbalance =
4 × *W*/*N* (6350 *W*/*N*)
4 × 908 lbs./ 10,000 rpm = 0.36 oz.-in. (gm-mm)

Trial unbalance (2 × $U_{max}$) = 0.72 oz.-in. (gm-mm)

*R*—Radius (at which weight will be placed): 6.875 inches

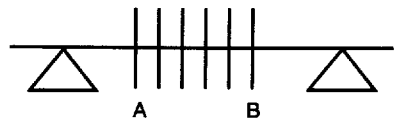
Trial unbalance weight = trial unbalance/*R*
0.72 oz.-in./ 6.875 inches = 0.10 oz. (gm)

Conversion information: 1 ounce = 28.375 grams

**TEST DATA**

| Position | Amplitude | Angular Position |
|---|---|---|
| 1 | 16.2 | 0° |
| 2 | 12.0 | 60° |
| 3 | 12.5 | 120° |
| 4 | 17.8 | 180° |
| 5 | 24.0 | 240° |
| 6 | 23.0 | 300° |
| Repeat 1 | 16.2 | 0° |

**ROTOR SKETCH**




C-101

**TEST DATA—GRAPHIC ANALYSIS**

Step 1: Plot data on the polar chart (Figure D-2—continued). Scale the chart so the largest and smallest amplitude will fit conveniently.

Step 2: With the compass, draw the best fit circle through the six points and mark the center of this circle.

Step 3: Measure the diameter of the circle in units of scale chosen in Step 1 and record. 35 units

Step 4: Record the trial unbalance from above. 0.72 oz.-in. (gm-mm)

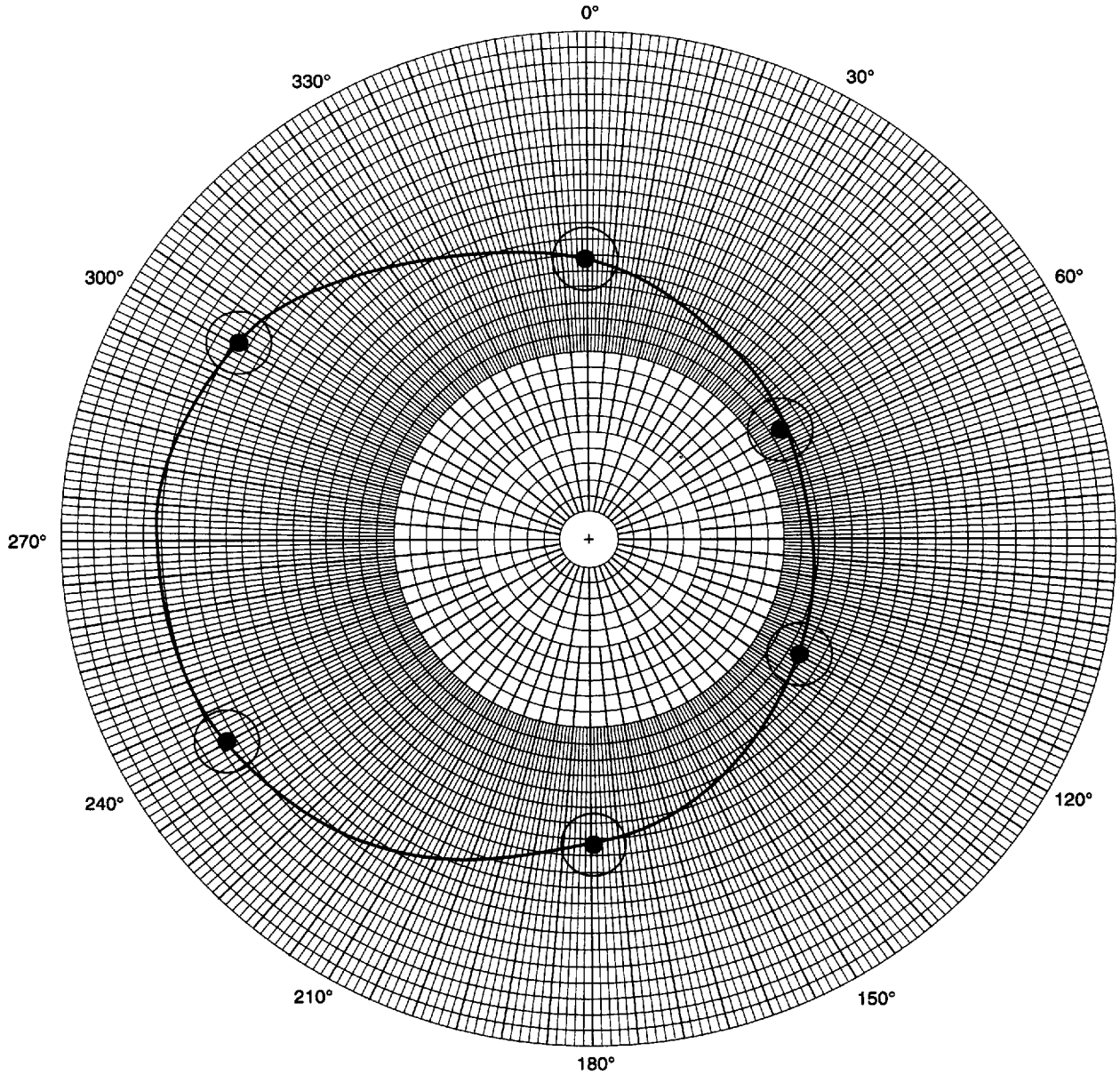
Step 5: Double the trial unbalance in Step 4 (may use twice the actual residual unbalance). 1.44 oz.-in. (gm-mm)

Step 6: Divide the answer in Step 5 by the answer in Step 3. 0.041 Scale Factor

You now have a correlation between the units in the polar chart and the gm-in. of actual balance.

Figure D-2—Sample Calculations for Residual Unbalance

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The circle you have drawn must contain the origin of the polar chart. If it doesn't, the residual unbalance of the rotor exceeds the applied test unbalance. Proceed with the balancing machine sensitivity check before rebalancing is attempted.

If the circle does contain the origin of the polar chart, the distance between origin of the chart and the center of your circle is the actual residual unbalance present on the rotor correction plane. Measure the distance in units of scale you choose in Step 1 and multiply this number by the scale factor determined in Step 6. Distance in units of scale between origin and center of the circle times scale factor equals actual residual balance.

Record actual residual unbalance _____6.5 (0.041) = 0.27_____ (oz.-in.)(gm-mm)

Record allowable residual unbalance (from Figure D-2) _____0.36_____ (oz.-in.)(gm-mm)

Correction plane _____A_____ for Rotor No. _____C-101_____ (has/has not) passed.

By _____John Inspector_____ Date _____11-16-92_____

Figure D-2—continued

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# APPENDIX E—ROTOR DYNAMICS LOGIC DIAGRAMS



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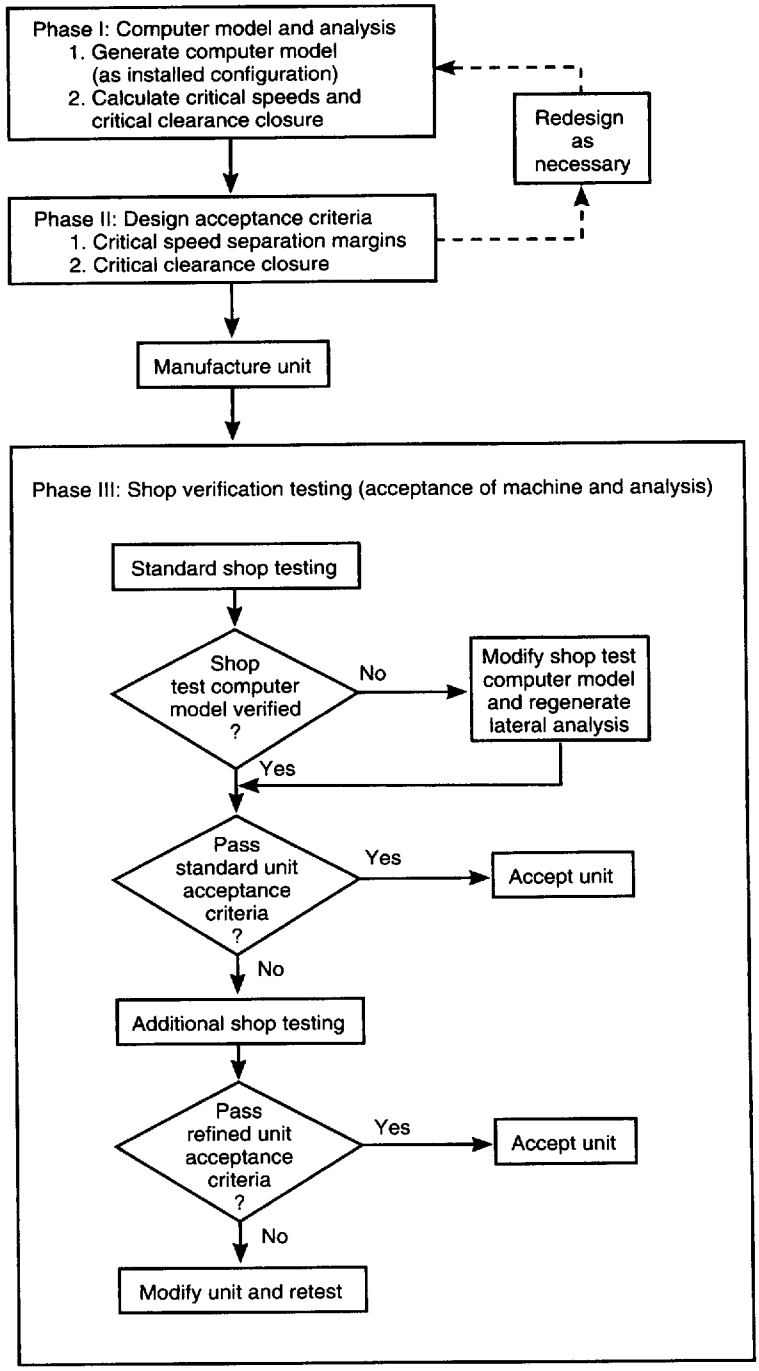



Figure E-1—API Three-Phase Vibration Acceptance Program

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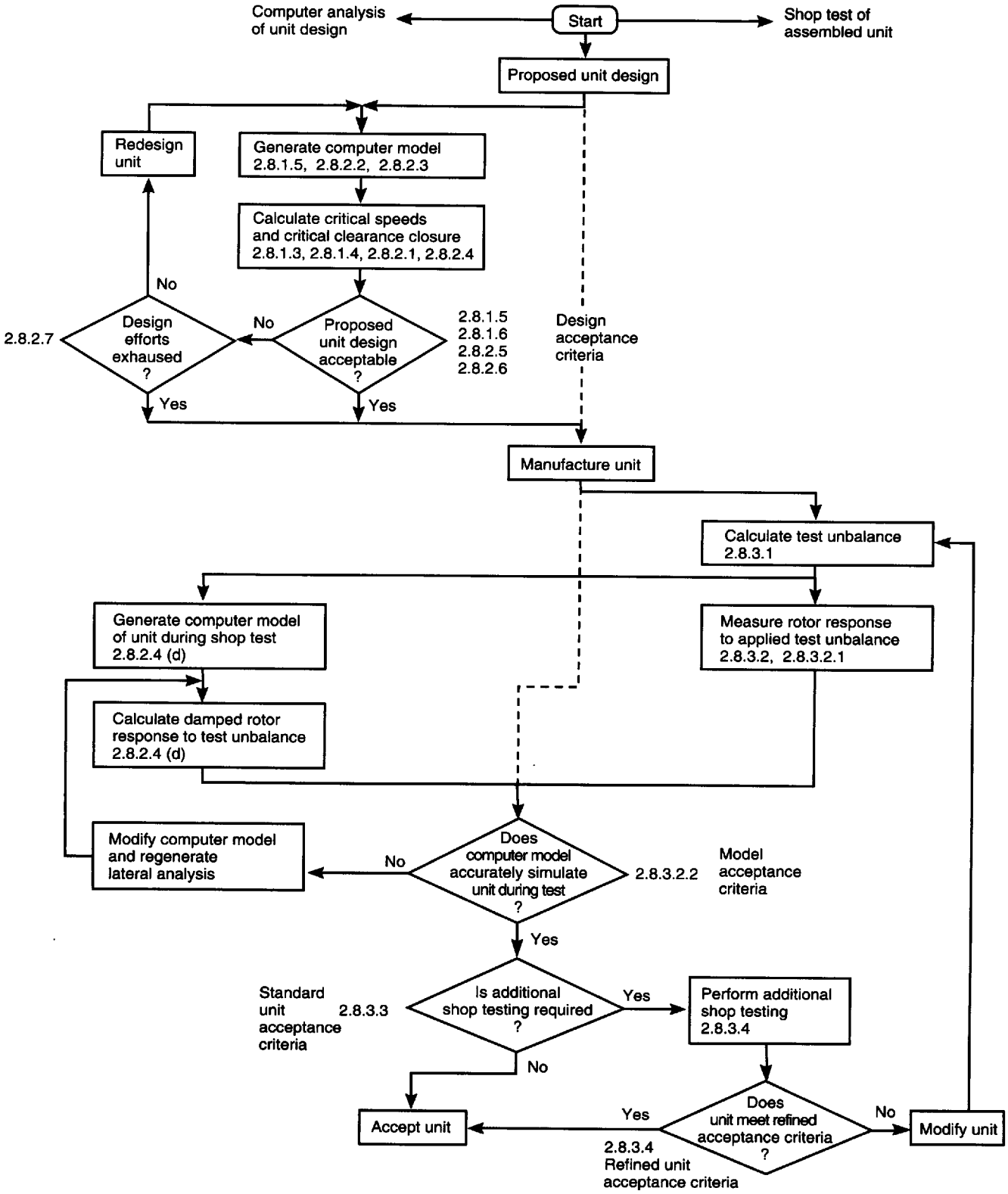



Figure E-2—Detailed Flow Chart of API Vibration Acceptance Program

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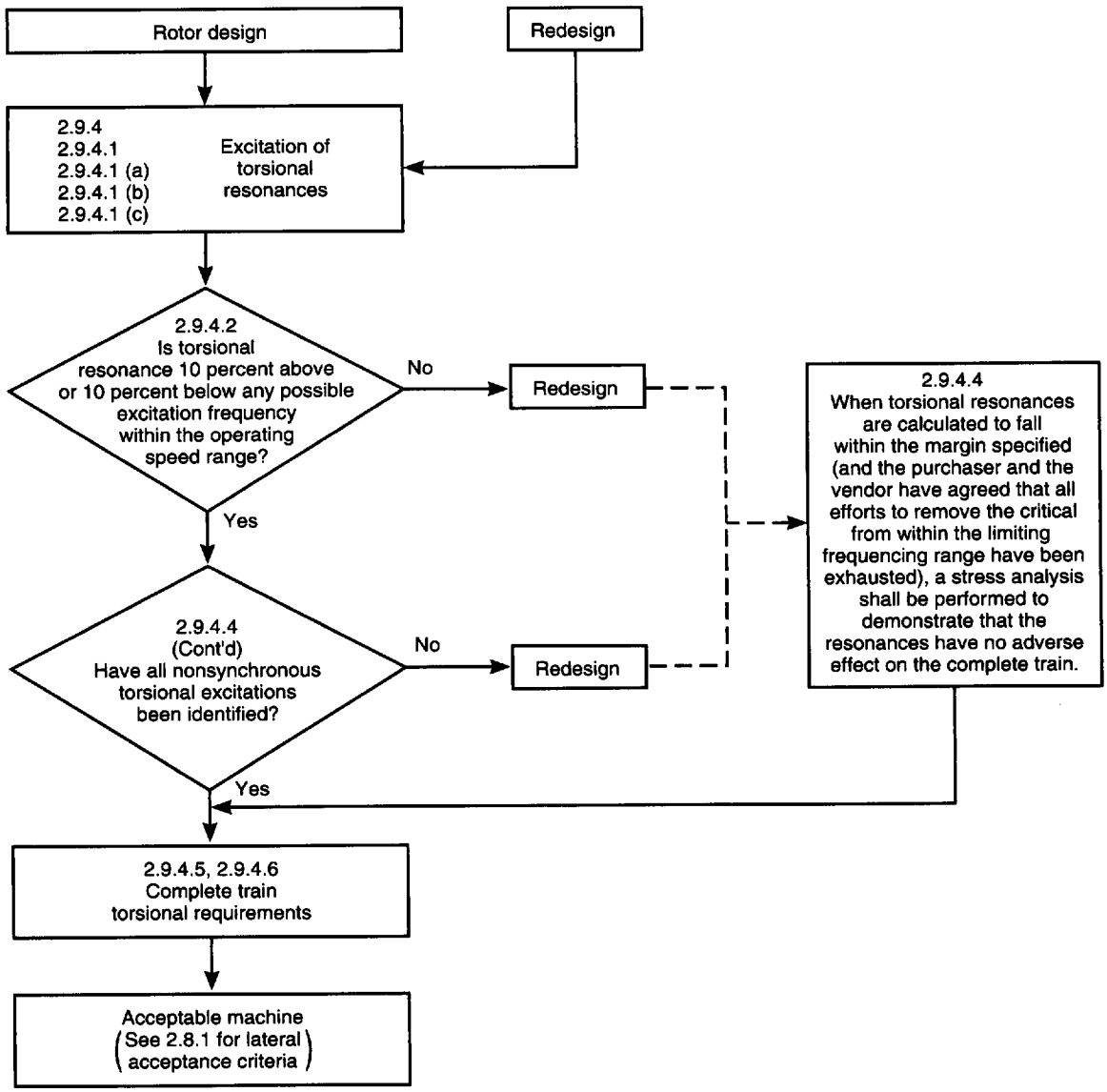



Figure E-3—Rotor Dynamics Logic Diagram (Torsional Analysis)

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# APPENDIX F—CENTRIFUGAL COMPRESSOR NOMENCLATURE



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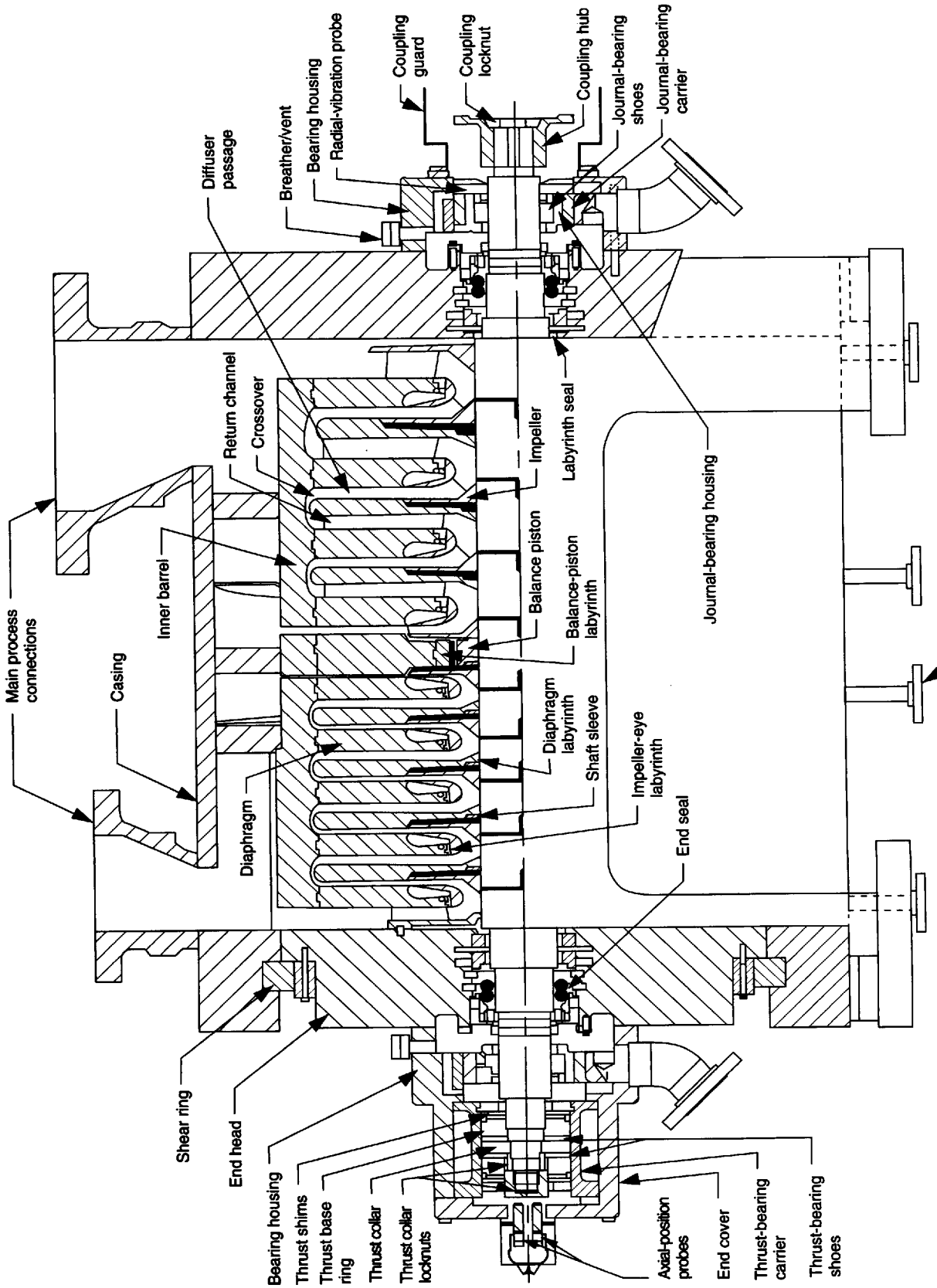



Figure F-1—Centrifugal Compressor Nomenclature

Note: Some compressors may use bolted-head construction.

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# APPENDIX G—FORCES AND MOMENTS

## G.1 General

The April 1988, November 1979, and October 1973 issues of this standard referred nozzle forces and moments calculations to appropriate NEMA documents with the stipulation that the constants in the equations be multiplied by 1.85. Experience has shown that there has not been a uniform interpretation of "1.85 times NEMA." Therefore, the equations have been adapted to compressors by identifying all the constants and clarifying that the equivalent of the exhaust nozzle in the NEMA calculation is the largest compressor nozzle. This is usually, but not necessarily, the inlet nozzle.

## G.2 Equations

The design of each compressor body must allow for limited piping loads on the various casing nozzles. For maximum system reliability, nozzle loads imposed by piping should be as low as possible regardless of the compressor's load-carrying capability. As a standard, the forces and moments acting on compressors due to the inlet, sidestream, and discharge connections should be limited by the following:

a. The total resultant force and total resultant moment imposed on the compressor at any connection should not exceed the values shown in Equation G-1.

$$F_r + 1.09\,M_r \le 54.1D_e \qquad \text{(G-1)}$$

In U.S. Customary units:

$$3F_r + M_r \le 927\,D_e$$

*Where:*

$F_r$ = Resultant force, in Newtons (pounds). (See Figure G-1.)

$$F_r = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

$M_r$ = Resultant moment, in Newton-meters (foot-pounds) from Figure G-1.

$$M_r = \sqrt{M_x^2 + M_y^2 + M_z^2}$$

For sizes up to 200 millimeters (8 inches) in diameter

$D_e$ = Nominal pipe diameter of the connection, in millimeters (inches).

For sizes greater than 200 millimeters (8 inches), use a value of

$$D_e = \frac{(400 + D_{nom})}{3}\text{(millimeters)}$$

In U.S. Customary units:

$$D_e = \frac{16 + D_{nom}}{3}\text{(inches)}$$

*Where:*

$D_e$ = Equivalent pipe diameter of the connection, in millimeters (inches).

$D_{nom}$ = Nominal pipe diameter, in millimeters (inches).

b. The combined resultants of the forces and moments of the inlet, sidestream, and discharge connections resolved at the centerlines of the largest connection should not exceed the following:

1. The resultants shall not exceed:

$$F_c + 1.64M_c \le 40.4D_c \qquad \text{(G-2)}$$

In U.S. Customary units,

$$2F_c + M_c \le 462D_c$$

*Where:*

$F_c$ = Combined resultant of inlet, sidestream, and discharge forces, in Newtons (pounds).

$M_c$ = Combined resultant of inlet, sidestream, and discharge moments, and moments resulting from forces, in Newton-meters (pound-feet).

$D_c$ = Diameter [in millimeters (inches)] of one circular opening equal to the total areas of the inlet, sidestream, and discharge openings. If the equivalent nozzle diameter is greater than 230 millimeters (9 inches), use a value of $D_c$ equal to:

$$= \frac{(460 + \textit{Equivalent diameter})}{3}\text{(millimeters)}$$

In Customary units,

$$= \frac{(18 + \textit{Equivalent diameter})}{3}\text{(inches)}$$

2. The individual components (Figure G-1) of these resultants should not exceed:

$$F_x = 16.1D_c \qquad M_x = 24.6D_c$$
$$F_y = 40.5D_c \qquad M_y = 12.3D_c$$
$$F_z = 32.4D_c \qquad M_z = 12.3D_c$$

In U.S. Customary units,

$$F_x = 92D_c \qquad M_x = 462D_c$$
$$F_y = 231D_c \qquad M_y = 231D_c$$
$$F_z = 185D_c \qquad M_z = 231D_c$$



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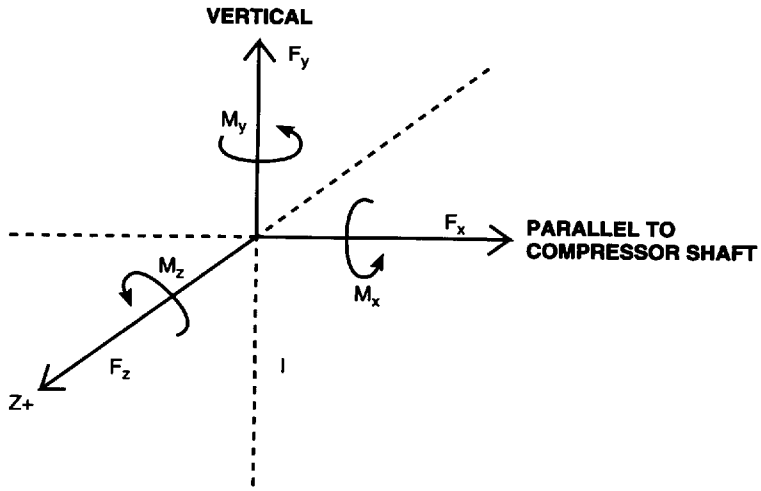



Figure G-1—Combined Resultants of the Forces and Moments of Corrections

*Where:*

$F_x$ = Horizontal component of $F_c$ parallel to the compressor shaft, in Newtons (pounds).
$F_y$ = Vertical component of $F_c$, in Newtons (pounds).
$F_z$ = Horizontal component of $F_c$ at right angles to be compressor shaft, in Newtons (pounds).
$M_x$ = Component of $M_c$ around the horizontal axis, in Newton-meters (foot-pounds).
$M_y$ = Component of $M_c$ around the vertical axis, in Newton-meters (foot-pounds).
$M_z$ = Component of $M_c$ around the horizontal axis at right angles to the compressor shaft, in Newton-meters (foot-pounds).

c. These values of allowable forces and moments pertain to the compressor structure only. They do not pertain to the forces and moments in the connecting pipes, flanges, and flange bolting, which should not exceed the allowable stress as defined by applicable codes and regulatory bodies.

Loads may be increased by mutual agreement between the purchaser and vendor. However, it is recommended that expected operating loads be minimized.

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# APPENDIX H—INSPECTOR'S CHECKLIST



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## INSPECTOR'S CHECKLIST

| Item | API Standard 617 Reference | Reviewed | Observed | Witnessed | Inspected by | Status |
|---|---|---|---|---|---|---|
| **General** | | | | | | |
| Vendor data records | 4.2.1.1 | | | | | |
| Final assembly maintenance & clearances | 4.2.1.1g | | | | | |
| Surface and subsurface inspection | 4.2.1.3 | | | | | |
| **Material Inspection** | | | | | | |
| Material inspection certification/testing | 4.2.2.1 | | | | | |
| **Mechanical Inspection** | | | | | | |
| Equipment feet (vertical & horizontal jackscrews | 3.3.1.2.1<br>3.3.1.2.2 | | | | | |
| Foot/baseplate shims | 3.3.1.2.2 | | | | | |
| Nozzle flange dimensions | | | | | | |
| Casing openings size/finish | 2.4.2.2 | | | | | |
| Rotor identification | 2.6.6 | | | | | |
| Shaft finishes | 2.6.5 | | | | | |
| Shaft electrical and mechanical runnout | 2.6.5<br>2.9.5.6 | | | | | |
| Shaft Gauss | 2.6.17 | | | | | |
| Coupling fit (API-671) | 2.6.2 | | | | | |
| Rotor balance (balance machine residual) | 2.9.5.2<br>2.9.5.3 | | | | | |
| Rotation arrow/nameplate data/units | 2.12 | | | | | |
| Mounting surfaces coated | 3.3.1.2.10 | | | | | |
| Mounting surfaces epoxy primed | 3.3.1.2.9 | | | | | |
| Oil system cleanliness | 4.3.4.1.3 | | | | | |
| Equipment cleanliness | 4.2.1.5 | | | | | |
| Material hardness | 4.2.1.6 | | | | | |
| Hydrostatic | 4.3.2 | | | | | |
| Impeller overspeed | 4.3.3 | | | | | |
| **Mechanical running test** | | | | | | |
| Contract shaft seals and bearings | 4.3.4.1.1 | | | | | |
| Oil flows, p, T as specified | 4.3.4.1.2 | | | | | |
| No leaks observed | 4.3.4.1.4 | | | | | |
| Protective devices operational | 4.3.4.1.5 | | | | | |
| Control devices operational | 4.3.4.1.5 | | | | | |



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## INSPECTOR'S CHECKLIST—continued

| Item | API Standard 617 Reference | Reviewed | Observed | Witnessed | Inspected by | Status |
|---|---|---|---|---|---|---|
| Contract instrumentation used | 4.4.4.1.8 | | | | | |
| Four-hour test complete | 4.3.4.2.3 | | | | | |
| Vary pressure and temperature | 4.3.4.2.5 | | | | | |
| Lateral critical speeds as predicted | 4.3.4.3.4 | | | | | |
| Tape recordings complete | 4.3.4.3.6 | | | | | |
| Bearing inspection after test satisfactory | 4.3.4.4.1 | | | | | |
| Spare rotor fit and run | 4.3.4.4.4 | | | | | |
| Gas leakage test | 4.3.5 | | | | | |
| **Optional tests** | | | | | | |
| Performance test | 4.3.6.1 | | | | | |
| Complete unit test | 4.3.6.2 | | | | | |
| Gear test | 4.3.6.4 | | | | | |
| Helium test | 4.3.6.5 | | | | | |
| Sound level test | 4.3.6.6 | | | | | |
| Aux. equipment test | 4.3.6.7 | | | | | |
| Post test casing internal inspection | 4.3.6.8 | | | | | |
| Full-pressure/full-load/ full-speed test | 4.3.6.9 | | | | | |
| Post-test inspection of hydraulic coupling fit | 4.3.6.10 | | | | | |
| Spare parts test | 4.3.6.11 | | | | | |
| **Preparation for shipment** | | | | | | |
| Preparation complete | 4.4.1 | | | | | |
| Paint | 4.4.3.1 | | | | | |
| Rust preventative (interior and exterior) | 4.4.3.2<br>4.4.3.3 | | | | | |
| Tags complete | 4.4.3.9<br>4.4.5 | | | | | |
| Installation instructions shipped | 4.4.6 | | | | | |
| Special tools complete | 3.6 | | | | | |
| Spare parts complete | | | | | | |



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# APPENDIX I—TYPICAL GAS SEAL TESTING CONSIDERATIONS

## I.1 Self-Acting Gas Seal Shop Test

**I.1.1** The self-acting gas seal shall be tested at the seal vendor's shop. Details of the test procedure and the performance on the test gas shall be mutually agreed upon by the purchaser, compressor vendor, and seal vendor.

**I.1.2** Test records shall include time, temperatures, pressures, speed, power, and flow across each sealing interface. The test record shall note any discontinuities in the measured parameters.

## I.2 Overspeed Test

Prior to assembly of the seal, the rotating face shall be subject to an overspeed test of 115 percent of maximum continuous speed for one minute.

## I.3 Static Test

Increase the pressure across each sealing interface to maximum sealing pressure in at least four increments, including normal sealing pressure. Hold each pressure to ensure steady state.

## I.4 Dynamic Test

**I.4.1** Maintain seal gas pressure(s) at normal sealing pressure, and increase from zero speed to maximum continuous speed in at least four increments. Record two sets of data at each speed increment. Ensure steady state.

**I.4.2** At normal sealing pressure, run for 15 minutes at 110 percent of maximum continuous speed.

**I.4.3** At normal sealing pressure, run for 60 minutes at maximum continuous speed.

**I.4.4** Reduce speed to zero, reduce the pressure to the minimum sealing pressure, then increase to maximum continuous speed. Increase pressure to maximum sealing pressure in at least four increments. Record two sets of data at each test point. Ensure steady state.

**I.4.5** Reduce pressure to normal sealing pressure. Run for 15 minutes.

**I.4.6** Reduce speed to zero. Maintain normal sealing pressure; record two sets of static data.

**I.4.7** For a tandem seal that is normally pressurized across only one sealing interface, repeat I.4.1 to I.4.6 inclusive for the second sealing interface.

**I.4.8** Pressurize seal for normal sealing at normal sealing pressure. Record static data.

**I.4.9** Gradually increase speed to maximum continuous speed, hold for 15 minutes. Record data every 5 minutes.

**I.4.10** Repeat I.4.8 and I.4.9 to steady or decreasing leakage trend.

## I.5 Visual Inspection

Following the test, disassemble the seal, ensuring critical parts are match-marked. Examine components for wear, buildup, and general condition. Record observances. Reassemble the seal.

## I.6 Confirmation Test

After reassembly, repeat I.4.8 and I.4.9 to verify proper reassembly of the seal.

## I.7 Special Optional Tests

The purchaser may specify additional optional tests such as temperature cycling, breakaway, and running torque. Test details shall be mutually agreed upon by the purchaser, compressor vendor, and seal vendor.



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# APPENDIX J—APPLICATION CONSIDERATIONS FOR ACTIVE MAGNETIC BEARINGS

## J.1 Scope

This appendix outlines the special considerations that are necessary when applying active magnetic bearings systems. These areas will require detailed interaction between the compressor vendor, the bearing vendor, and the end user.

## J.2 Definition

An *active magnetic bearing* is a bearing that generates load-bearing capacity by means of actively controlled electromagnets. These magnets are positioned radially around the circumference of a shaft to comprise a journal bearing, or are positioned on either side of a rotating disc to form a double acting thrust bearing. Materials used in both the stationary and rotating parts are magnetic iron or steel.

Shaft position sensors are used to detect the shaft axial and radial positions. The sensor signals are input to a control system which modulates the current flowing through power amplifiers to the electromagnets to stably maintain the shaft position in the center of the magnets.

## J.3 Bearing Loads

**J.3.1** It is difficult to predict all the axial and radial loads on bearings. Active magnetic bearings have a lower inherent transient overload capacity than hydrodynamic bearings.

**J.3.2** For accurate prediction of the bearing load requirements, it is critical that the purchaser and compressor vendor mutually define the operating envelope, including: start-up, shut-down, emergency, and unusual operating conditions.

**J.3.3** Bearings should be selected to have a load capacity higher than the calculated static and dynamic loads at the most adverse operating condition, including operation at minimum and maximum flows. (See 2.7.3.3 and 2.7.3.7.)

**J.3.4** The amount of additional bearing capacity should depend on the confidence in the accuracy of the load prediction.

**J.3.5** Magnetic bearings should be tested to verify their force generating capability. The test procedure should be mutually agreed upon.

**J.3.6** The allowable operating envelope for an active magnetic bearing machine may be different than for a hydrodynamic bearing machine.

**J.3.7** Special testing may be considered to simulate site bearing loading.

## J.4 Balance

Special rotor balancing setups may be required.

## J.5 Rotating Elements

The bearing area and the sensor viewing area on the shaft may be sleeved. (See 2.6.5.)

## J.6 Auxiliary Bearing Systems

**J.6.1** Auxiliary bearing systems are required to support the rotor when the active magnetic bearings are de-energized at rest, and to permit rundown if the active magnetic bearing system is de-energized at running speed.

**J.6.2** Considerations include:

a. Location of the auxiliary bearing relative to other components.
b. Axial loadings and location of the auxiliary bearing considering thermal expansion, and inertial landing loads.
c. Life, that is the number of uncontrolled run downs allowed before inspection of replacement of the auxiliary bearings.
d. Labyrinth clearances with the rotor resting on the auxiliary bearings.

## J.7 Rotor Dynamic Design

**J.7.1** Bearing stiffness and damping characteristics are adjustable electronically and can be adapted to optimize rotordynamic behavior.

**J.7.2** Rotordynamics must be evaluated with both normal magnetic bearing support and on auxiliary bearings under run down conditions, and should consider sensor location relative to rotor nodes and bearings.

**J.7.3** An estimate should be made of the maximum rotor unbalance which will not overload the bearings.

## J.8 Operator Interface

**J.8.1** Minimum instrumentation should include:

a. Vibration level per axis.
b. Bearing current per amplifier-controlled sector.
c. Bearing temperature per amplifier-controlled sector.
d. Bearing control panel diagnostics.

**J.8.2** Other information can be made available for monitoring and analyzing compressor and bearing system behavior. Vibration levels on each controlled axis are required to assess the relative health of the bearing system. Similarly, bearing currents and temperatures for each bearing segment or amplifier-specific portion are required.



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## J.9 Protection from the Process Environment

**J.9.1** Consider the compatibility of the magnet assembly with the process environment. Exposure of the magnetic bearing to the process requires the compatibility of the bearing windings, their insulation, the laminations, and any encapsulant.

**J.9.2** Electrical feeds through the pressure boundary should be above the shaft centreline if possible to avoid connection locations in areas subject to process liquid pooling.

**J.9.3** Bearing heating during shutdown must be considered.

**J.9.4** Provide low point drains in bearing cavities.

## J.10 Schedule

Allow time in the compressor vendor's shop and on-site for calibrating the control system.

## J.11 Bearing Cooling

Cooling is required to remove heat from electrical losses in the bearing windings and heat generated by windage losses.

## J.12 Electrical Design

**J.12.1** Electrical insulation should be specified with the NEMA terminology usually used to describe active magnetic bearing winding insulation systems.

**J.12.2** Provide insulation on bearing laminations as failures due to induction heating can be sudden and cause substantial damage.

**J.12.3** Consider the length of the cable run from the control system to the bearings.

**J.12.4** Provide an uninteruptable power supply or battery backup to maintain shaft levitation during electrical utility disruption. Backup systems are recommended for either continuous operation in the event of a power disturbance or to permit safe and orderly shutdown.

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